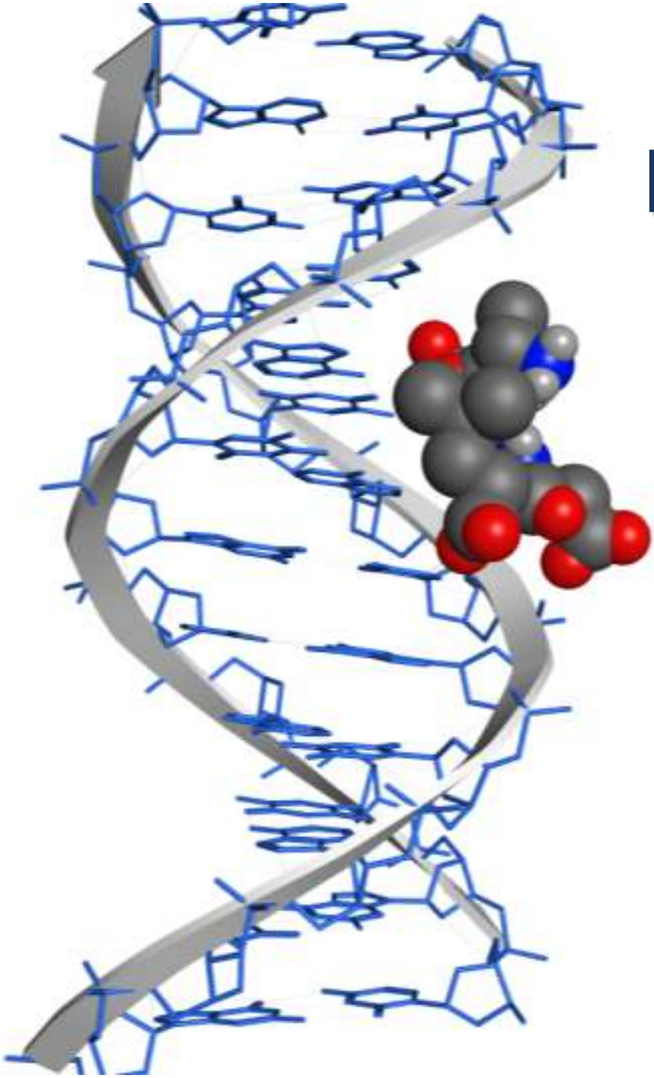
I.P. Pavlov Institute of Physiology of the Russian Academy of Sciences
Saint Petersburg Institute of Bioregulation and Gerontology



Peptides, Genome, Ageing

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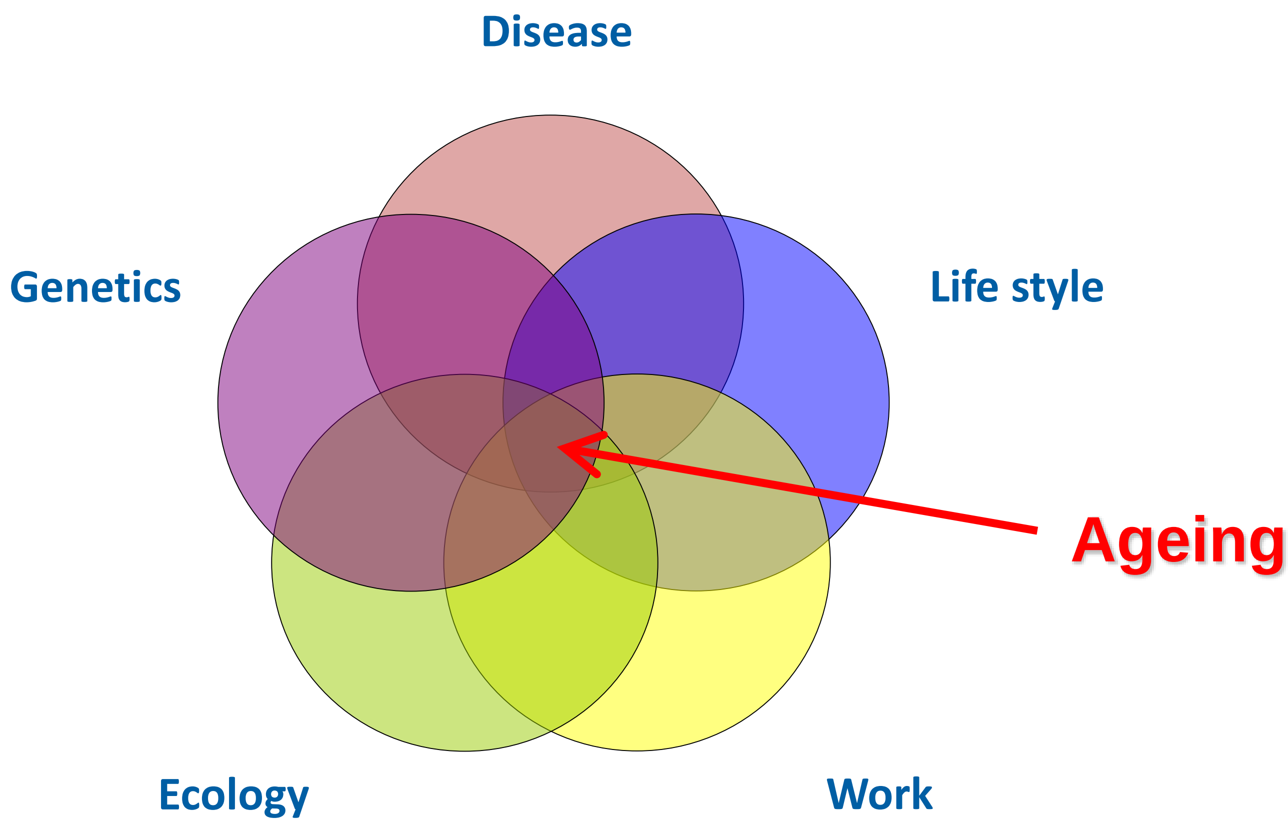
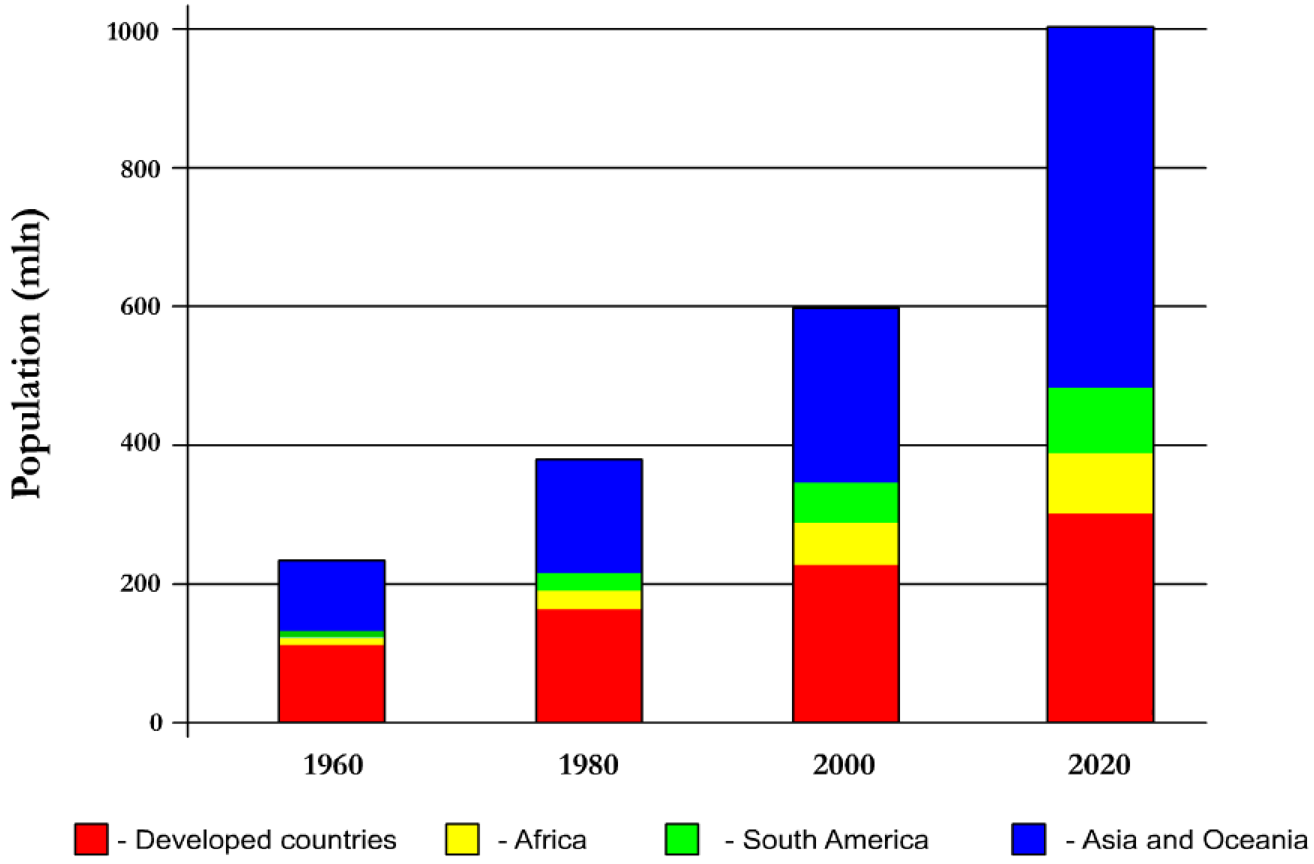
Interrelation of life style, disease, work, ecology, genetics and biological ageing



Population aged 60 and over in the main world regions 1960 – 2020

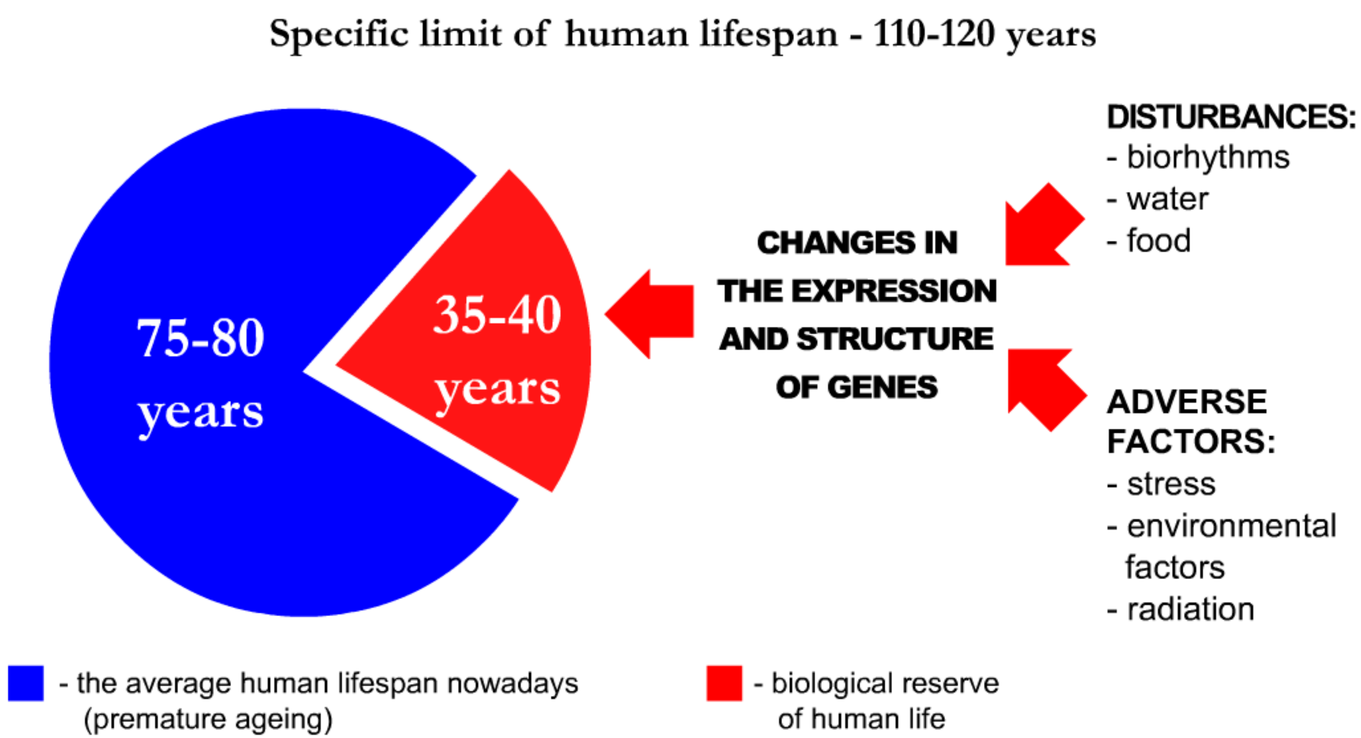


Source: UN unit of population

Documented centenarians

Country	Age	Name	Date of birth	Date of death
France	122	Jeanne Calment	23.02.1875	04.08.1997
Japan	120	Sigechio Izumi	29.06.1865	21.02.1986
Russia	117	Semennikova Varvara	10.05.1890	09.03.2008
USA	114	Martha Graham	12.1844	25.06.1959
Great Britain	114	Martha Eliza Williams	02.06.1873	02.06.1987
Canada	113	Pierre Joubert	15.07.1701	16.11.1814
Spain	112	Joseph Salas Mateo	14.06.1860	27.02.1973
France	112	Augustine Tessier	02.10.1869	09.03.1981

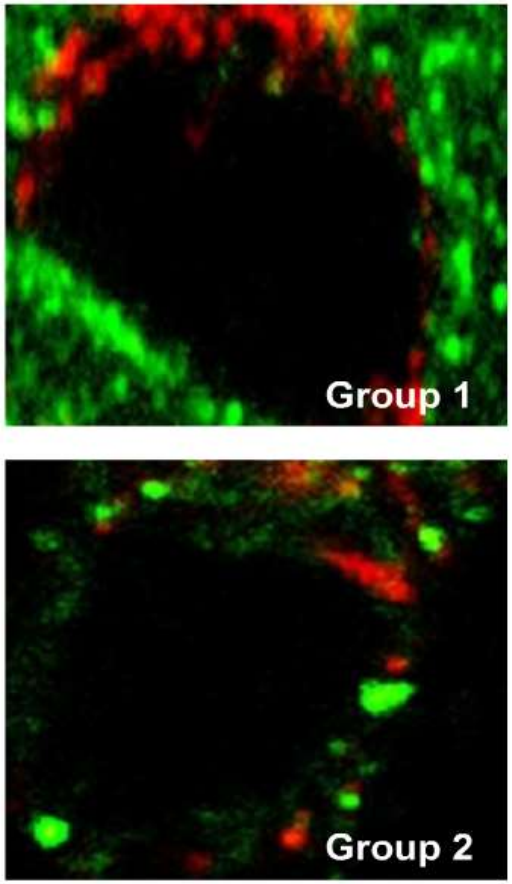
Potential increase in the average human lifespan up to specific limit (biological reserve)



Khavinson V. Peptidergic regulation of ageing (2009)

1. Peptide bioregulators: biological activity

The expression of transcription proteins (PAX1) in epithelial cells of human thymus



Groups (age)	Optic density (y.e.)	Area of expression (%)
1 (6 mon. - 4 years)	2.05±0.06	3.36±1.54
2 (65 - 79 years)	0.54±0.02 *	1.24±0.06 *
3 (80 - 95 years)	0.39±0.01 *	0.99±0.03 *

- $p<0.05$ as compared to group 1

Immunofluorescence laser confocal microscopy, x400
Red fluorescence – Rodamin G
Green fluorescence – FITC

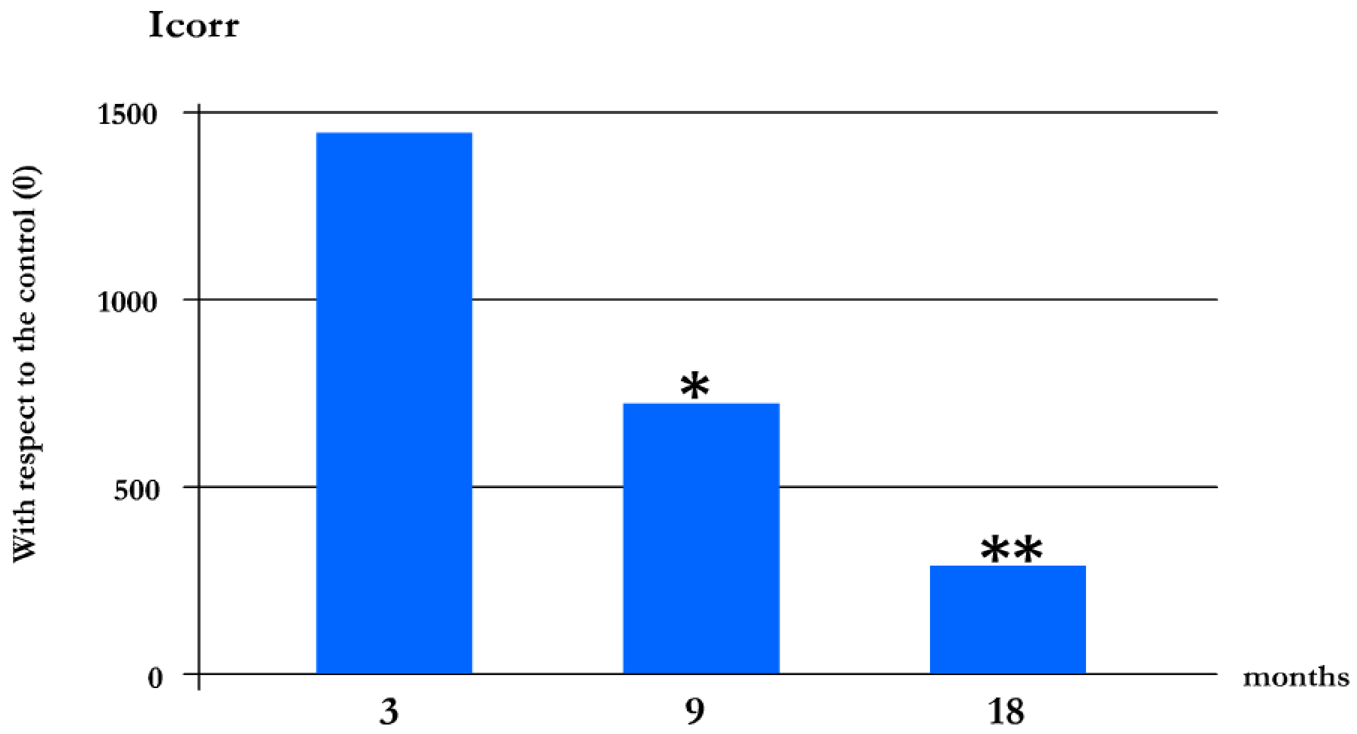
Khavinson V. Peptidergic regulation of ageing (2009)

Age-related changes in the expression of signal molecules in human thymus

Markers	Area of expression, %		
	60-74 years	**75-89 years**	**>90 years**
Ki67	0.58±0.07	0.19±0.03*	0.07±0.02*
P53	4.51±0.11	9.32±0.43*	4.41±0.20
AIF	0.07±0.02	1.35±0.02*	2.61±0.31*
MMP2	0.58±0.07	0.19±0.03*	0.07±0.02*
MMP9	4.51±0.11	9.32±0.43*	4.41±0.20
CGRP	0.07±0.02	1.35±0.02*	2.61±0.31*
CD4	2.70±0.54	1.58±0.18*	0.32±0.07*
CD5	2.48±0.31	1.66±0.31	1.05±0.12*
CD8	3.88±0.52	3.91±0.49	1.84±0.32*
CD20	0.69±0.12	0.56±0.11	0.65±0.13

* - $p<0.05$ as compared to corresponding indices in the group of patients aged 60-74

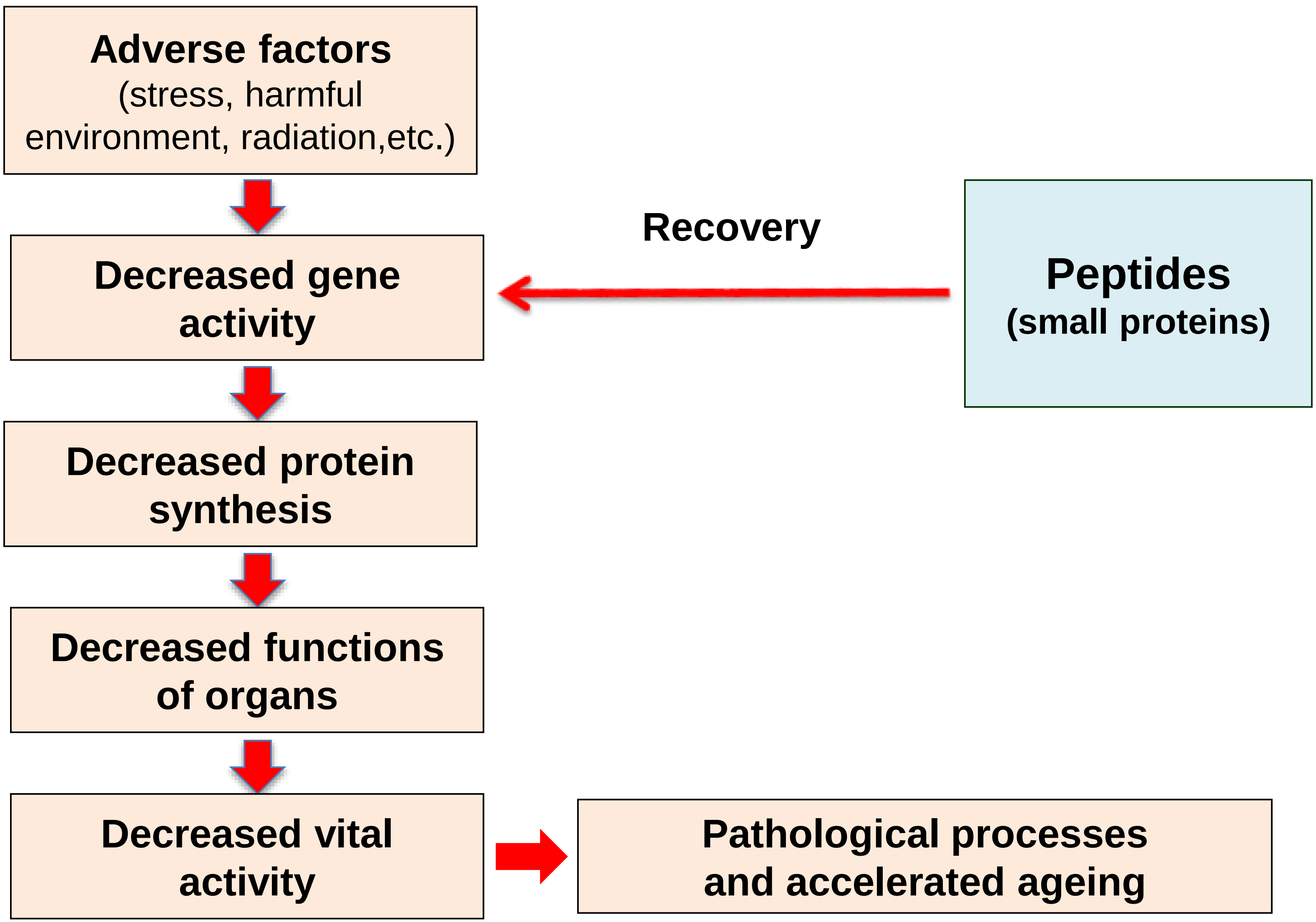
Protein synthesis in hepatocytes of rats of different age



* - p<0.05 as compared to the 3-month old rats; ** - p<0.05 as compared to the 9-month old rats

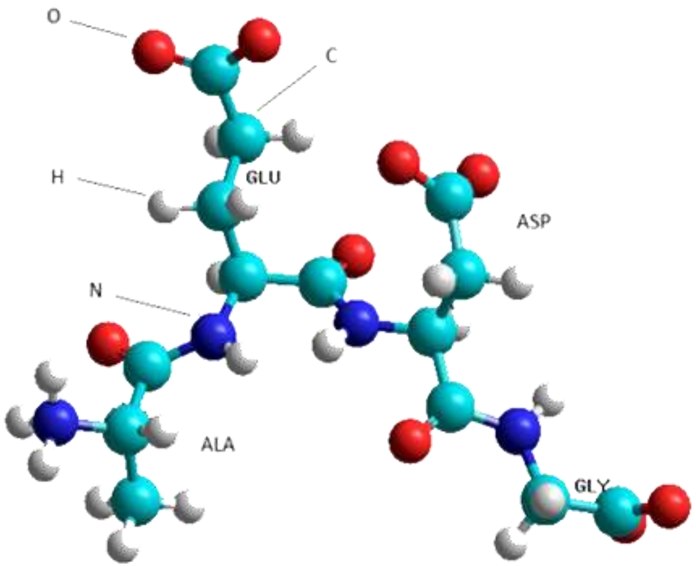
Khavinson V. Peptides and ageing (2002)

Peptide bioregulation of vital activity



Characteristics of the peptides

1. *Natural origin*
2. *Tissue-specific action*
3. *Safe to use*
4. *Microdoses*
5. *Availability of the product*



St. Petersburg Institute of Bioregulation and Gerontology

1. *Peptide preparations (over 30)*
2. *Peptide biologically active food supplements (over 40)*
3. *Peptide cosmetic products (over 10)*

Cytomedins® - peptide geroprotectors (pharmaceuticals)

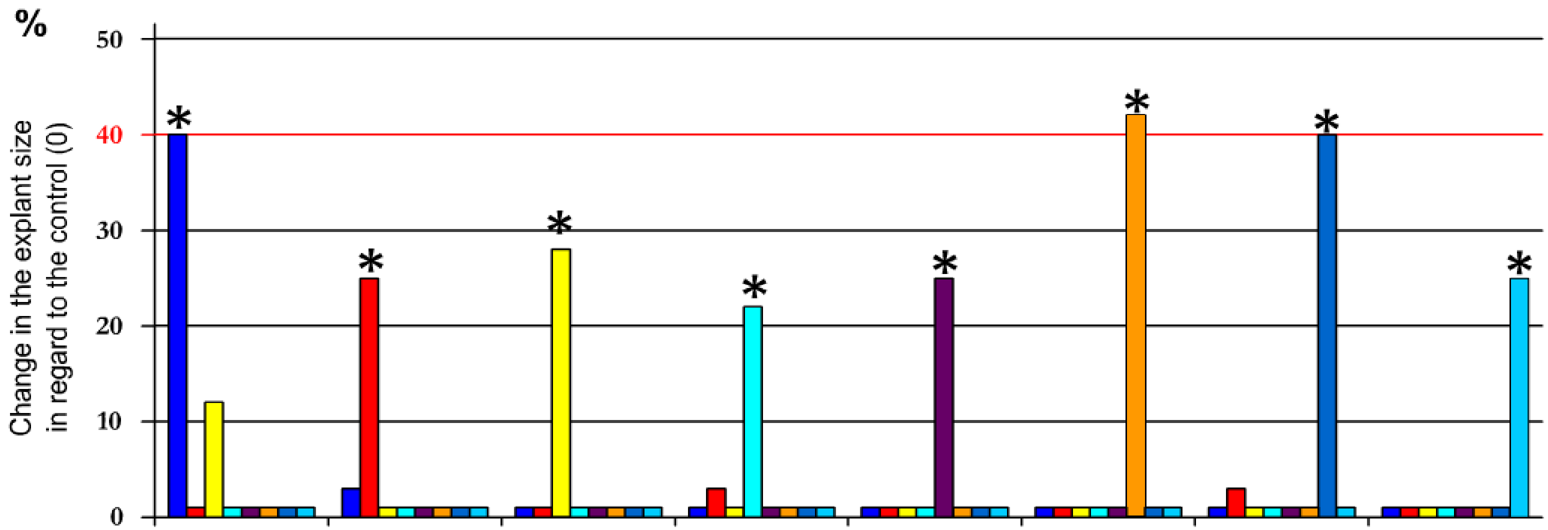
Preparation (State Pharmacopoeia of the Russian Federation)	**Source of the peptides**	**Patents**
Thymalin® (1982)	Thymus	*Morozov V., Khavinson V.* *US Patent № 5,070,076 (1991)*
Epithalamin® (1990)	Pineal gland	*Khavinson V. et al.* *RU Patent № 944191 (1993)*
Prostatilen® (1992)	Prostate gland	*Khavinson V. et al.* *RU Patent № 1417244 (1993)*
Cortexin® (1999)	Brain	*Khavinson V. et al.* *RU Patent № 1298979 (1993)*
Retinalamin® (1999)	Retina	*Khavinson V. et al.* *RU Patent № 1436305 (1993)*

Over 15 million patients were treated with these pharmaceuticals both for prevention and treatment of different pathological states (1982 – 2014).

Cytogens® - synthetic peptides (pharmaceutical and food supplements)

Preparations	Structure	Correction of functions	Patents
Thymogen®	Glu-Trp	Immune system	*Morozov V., Khavinson V.* *US Patent № 5,538,951 (1996)*
Vilon®	Lys-Glu	Regeneration processes	*Khavinson V. et al.* *US Patent № 6,642,201 (1996)*
Vesugen®	Lys-Glu-Asp	Vessels	*Khavinson V. et al.* *US Patent № 7,851,449 (2010)*
Livagen®	Lys-Glu-Asp-Ala	Liver	*Khavinson V.* *US Patent № 7,101,854 (2006)*
Epitalon®	Ala-Glu-Asp-Gly	Endocrine system	*Khavinson V.* *US Patent № 6,727,227 (2004)*
Bronchogen®	Ala-Glu-Asp-Leu	Bronchopulmonary system	*Khavinson V. et al.* *US Patent № 7,625,870 (2009)*
Pancragen®	Lys-Glu-Asp-Trp	Pancreas	*Khavinson V. et al.* *US Patent № 7,491,703 (2009)*
Cardiogen®	Ala-Glu-Asp-Arg	Cardiovascular system	*Khavinson V. et al.* *US Patent № 7,662,789 (2010)*

Peptide tissue (gene)-specific regulation

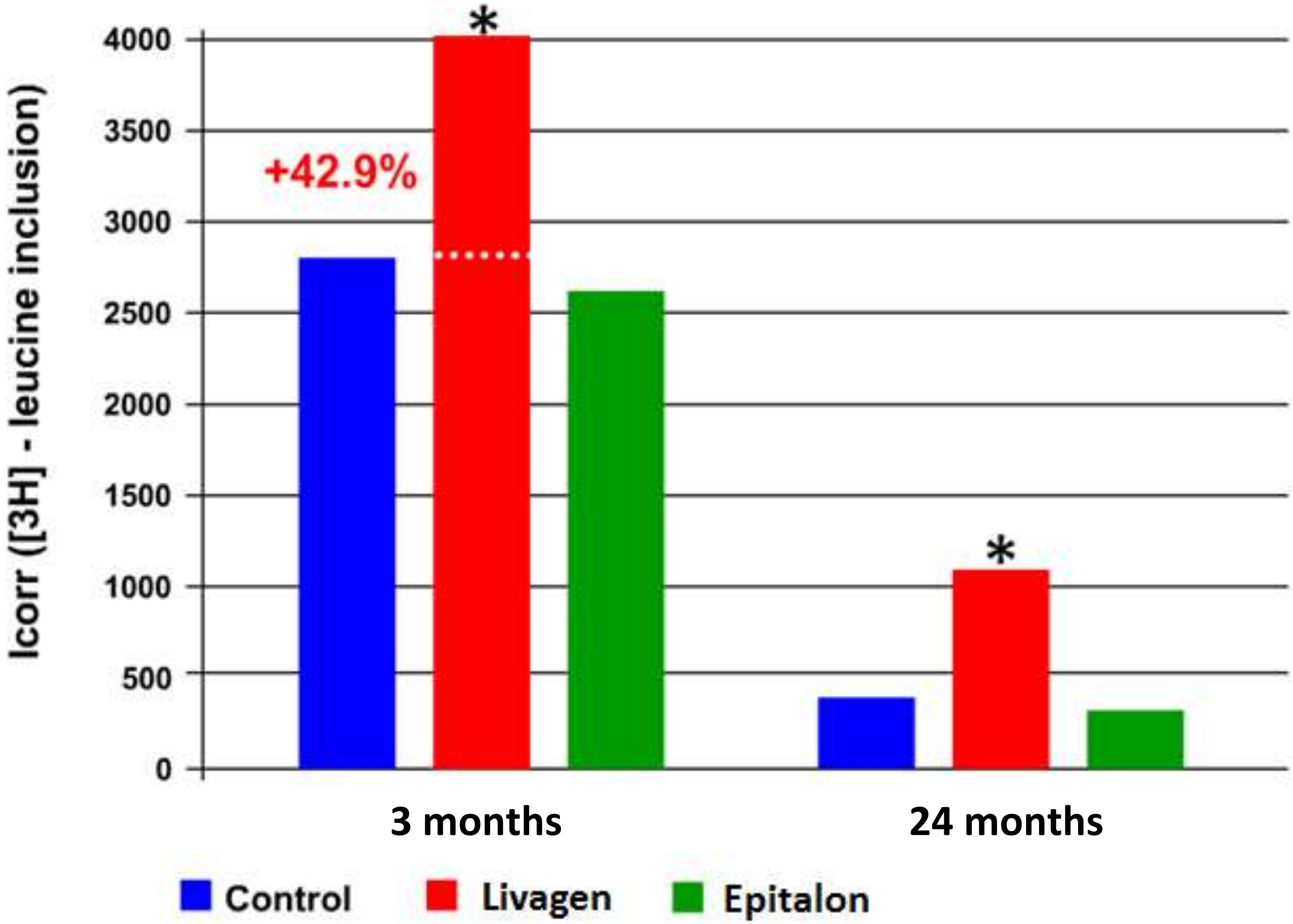


Tissue / Peptide	Cerebral cortex	Subcortical structures	Liver	Urinary bladder	Thymus	Pancreas	Heart	Bronchi
Brain	+	-	-	-	-	-	-	-
Pineal	-	+	-	-	-	-	-	-
Liver	-	-	+	-	-	-	-	-
Prostate	-	-	-	+	-	-	-	-
Thymus	-	-	-	-	+	-	-	-
Pancreas	-	-	-	-	-	+	-	-
Heart	-	-	-	-	-	-	+	-
Bronchi	-	-	-	-	-	-	-	+

* - $p<0.05$ as compared to the control

Khavinson V. Bull. Exp. Biol. Med. (2002)

Livagen increases protein synthesis in rat hepatocytes



* - $p<0.05$ as compared to the control

Khavinson V. Neuroendocrinology Lett. (2002)

Peptide immune modulators
(Saint Petersburg Institute of Bioregulation and Gerontology)

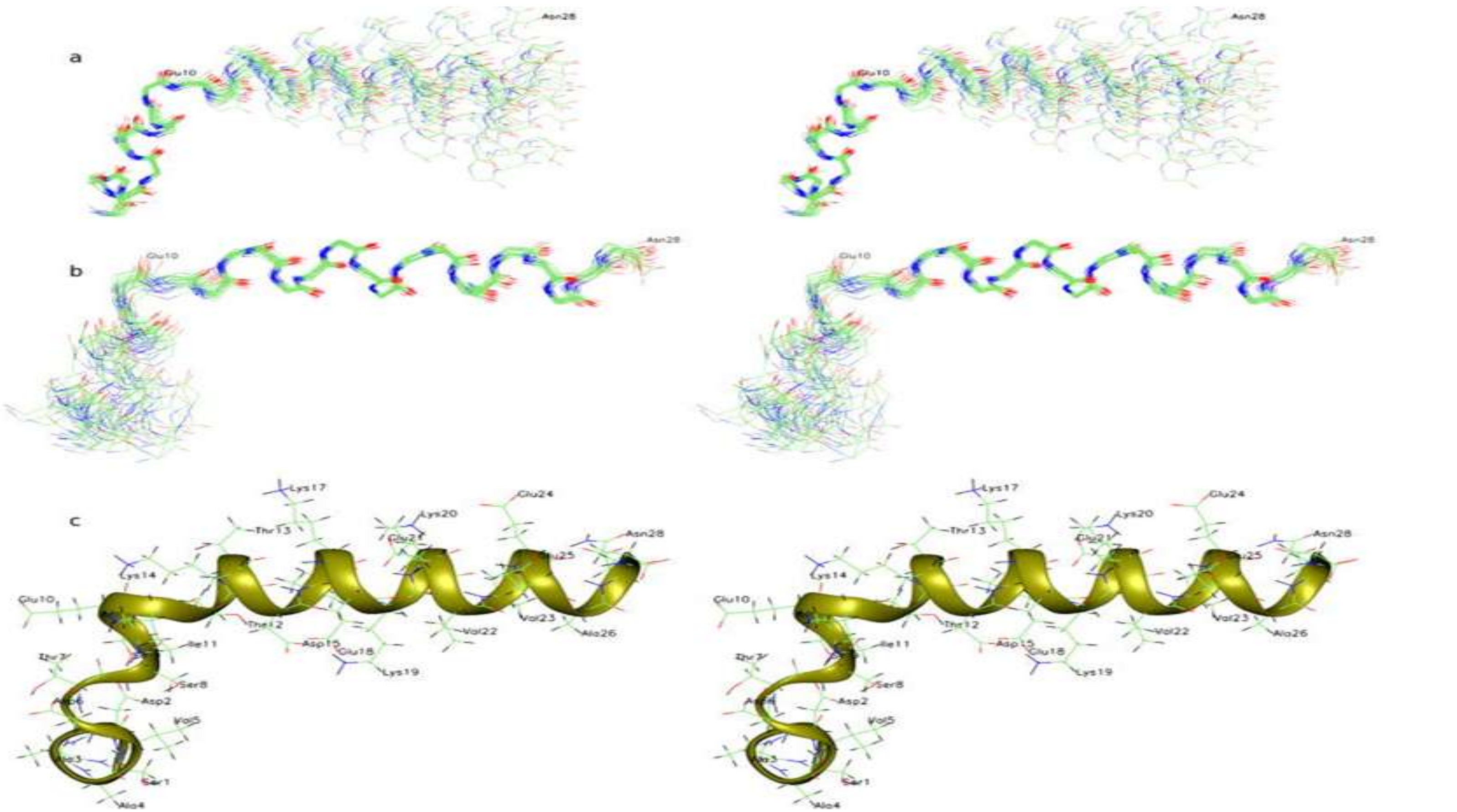
Preparation	Structure	Patents
Thymalin®	**Polypeptides from thymus**	*Morozov V., Khavinson V. US Patent № 5,070,076 (1991)*
Thymogen®	**Glu-Trp**	*Morozov V., Khavinson V. US Patent № 5,538,951 (1996)*
Vilon®	**Lys-Glu**	*Khavinson V. et al. US Patent № 6,642,201 (1996)*
Crystagen®	**Glu-Asp-Pro**	*Khavinson V. et al. US Patent № 8,057,810 (2011)*

Similarity in structures of the peptide immune modulators

Preparation	Structure	Publications
Vilon	**Lys-Glu**	*(Morozov V., Khavinson V.,1997)*
Splenopentin	Arg-**Lys-Glu**-Val-Tyr	*(Audhya G. et al., 1984)*
Thymosin alpha-1	(ACE)-Ser-Asp-Ala-Ala-Val-Asp-Thr-Ser-Ser-Glu- Ile-Thr-Thr-Lys-Asp-Leu-**Lys-Glu**-Lys-**Lys-Glu**-Val-Val-Glu-Glu-Ala-Glu-Asn	

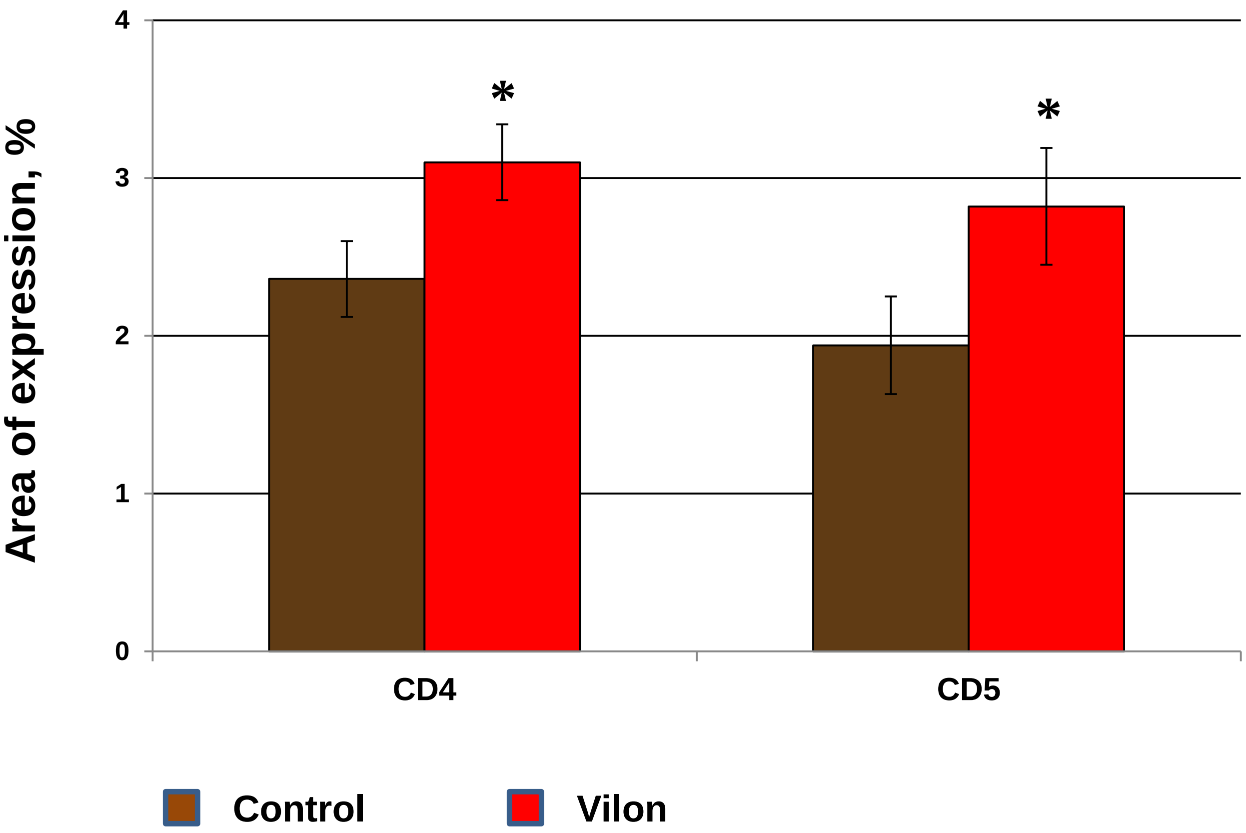
Thymosin alpha-1

(ACE)-Ser-Asp-Ala-Ala-Val-Asp-Thr-Ser-Ser-Glu-Ile-Thr-Thr-Lys-Asp-Leu-Lys-Glu-Lys-Lys-Glu-Val-Val-Glu-Glu-Ala-Glu-Asn



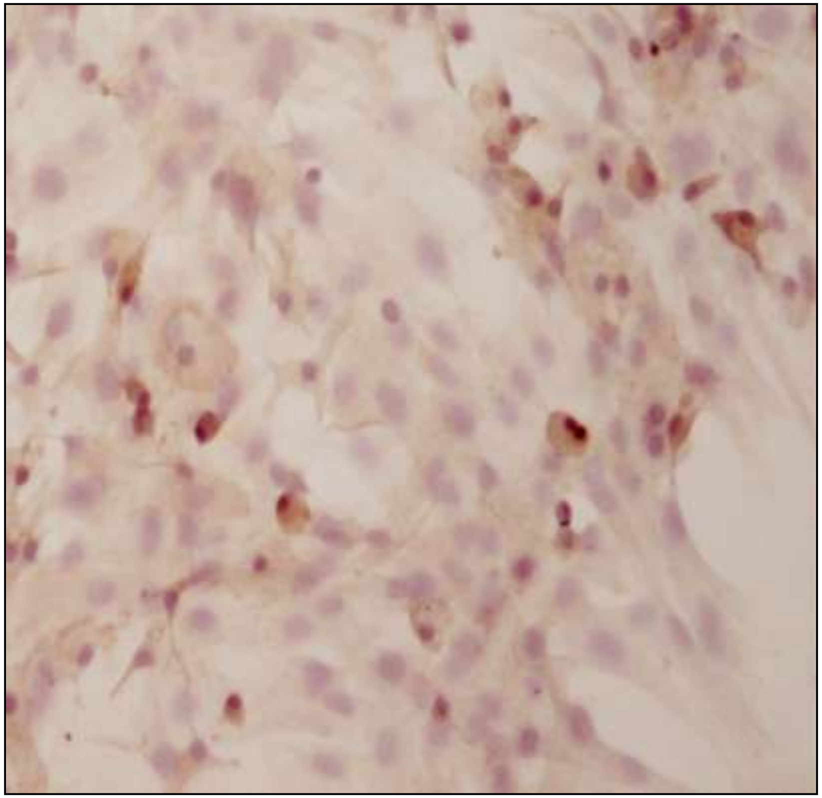
Elizondo-Riojas M.-A. et al., Biochem. Biophys. Res. Commun. (2011)

The influence of Vilon (KE) on the expression of signal molecules in dissociated human thymus cultures



*- p<0.05 as compared to the control

The influence of Vilon on CD5 expression in dissociated human thymus cultures



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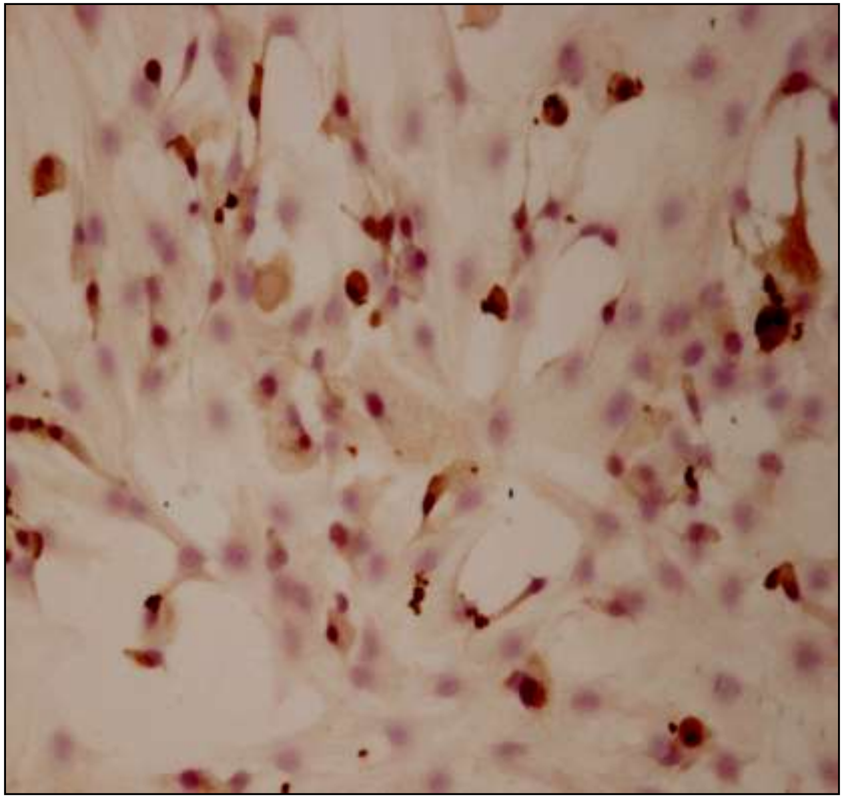
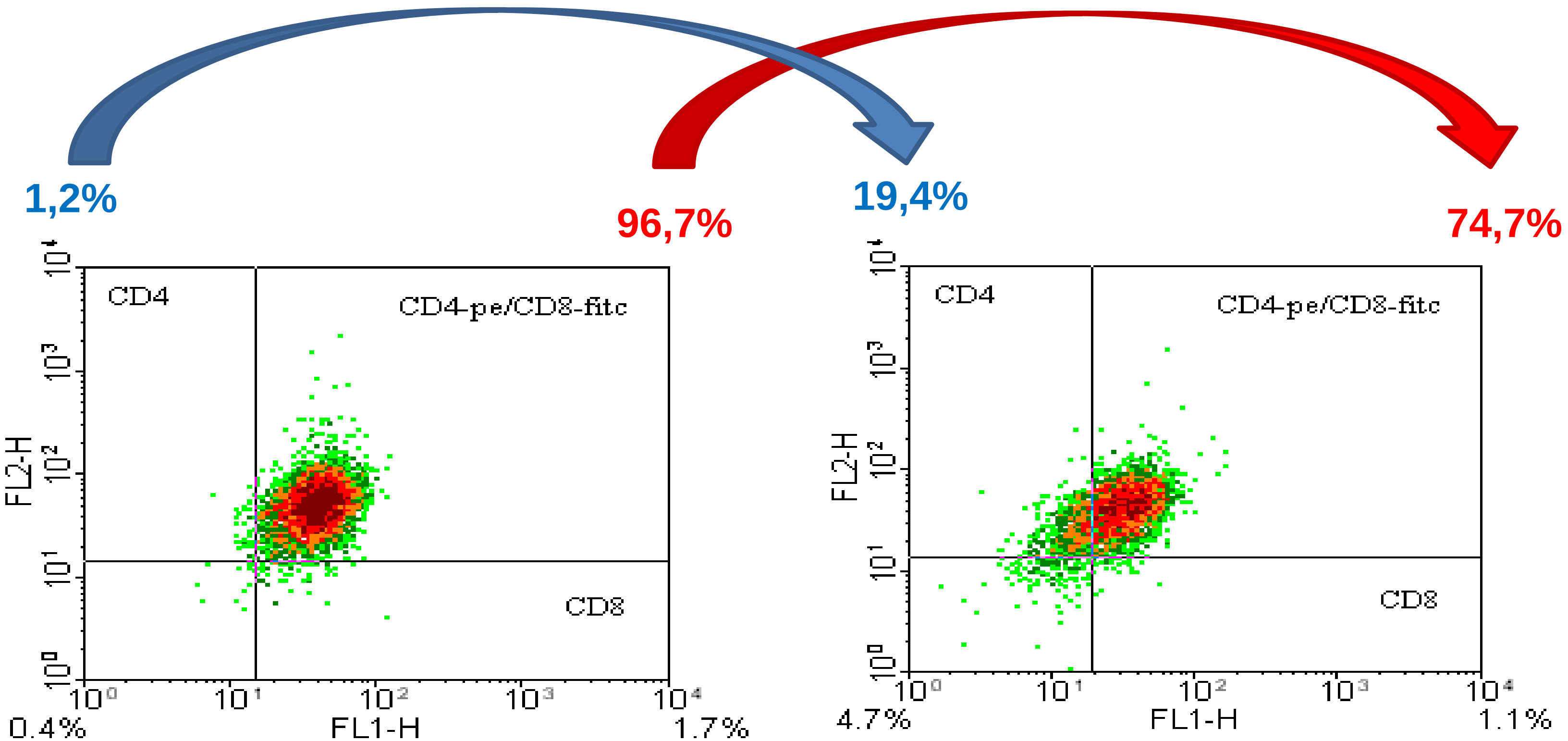


Vilon

Immunohistochemistry with hematoxylin and eosin stain, x200

The influence of Vilon on differentiation of CD4$^+$CD8$^+$ thymocytes into CD4$^+$ lymphocytes



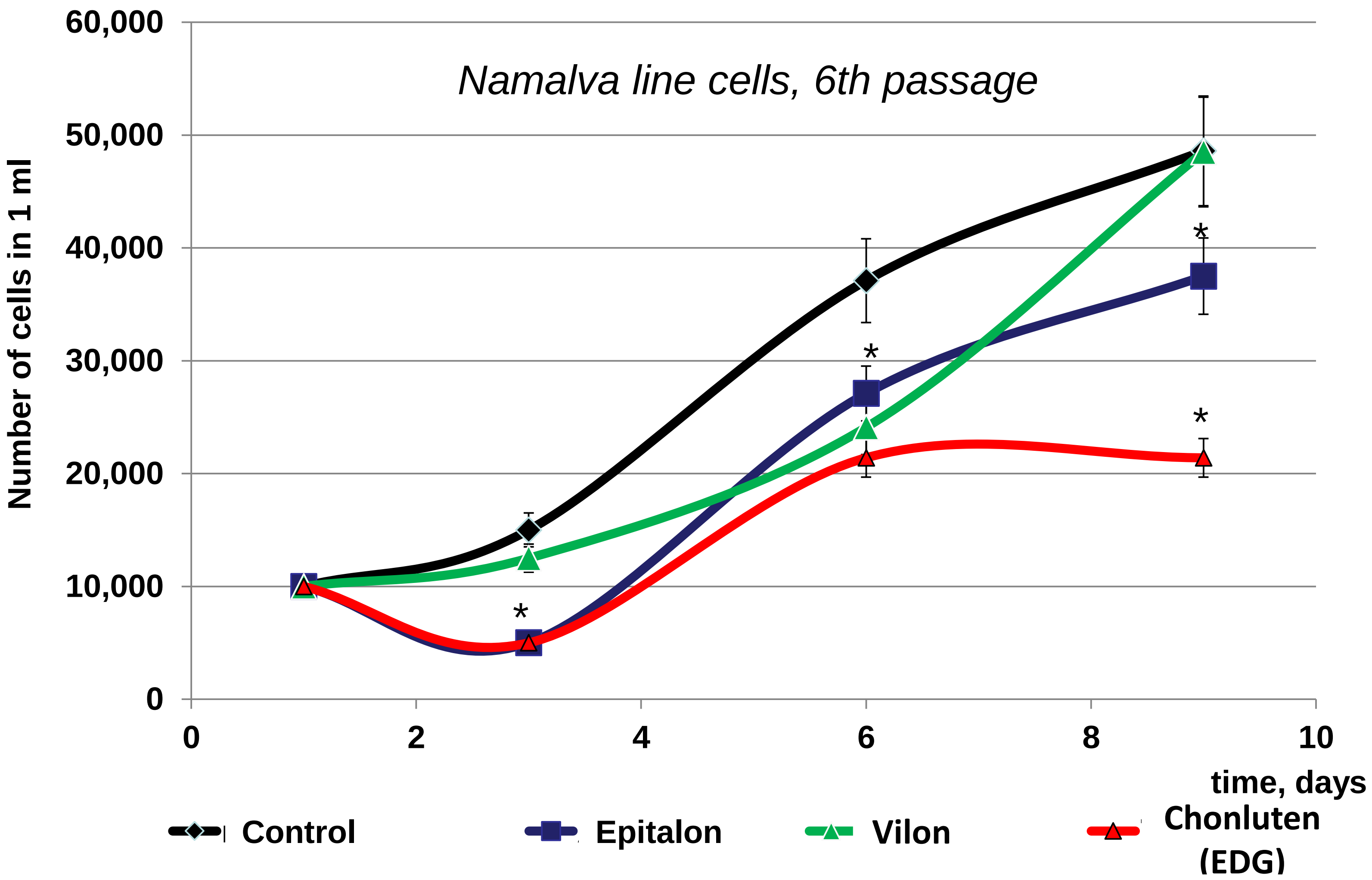
Control **Vilon**

Flowcytometry

Vilon binding sites in gene promoter regions

Gene	Sites of binding (n)	Alteration in expression	Gene	Sites of binding (n)	Alteration in expression
Genes of cell division			**Regulatory genes**		
Mybl1	15	-2.03	Tcf20	11	3.04
MCM10	13	2.12	Banp	21	2.71
Cul5	10	3.35	Zfp61	12	2.49
HS_NOL1	11	2.34	Id2	11	2.01
Ccnl2	18	2.08	Spink4	13	-2.06
H3f3b	12	2.03	Tpp2	15	2.06
Cell signal transmission genes			Rps6kb1	14	2.94
Selplg	12	-2.79	**Genes related to metabolism**		
Thbs3	11	2.31	SLC25A13	11	2.95
Stk11	15	2.38	ABCB1	17	2.78
Cell structure and mobility genes			SLC7A6	11	2.42
Fmn2	11	3.28	**Unclassified genes**		
Dst	24	2.69	Tcf25	13	4.91
Wnt4	12	2.81	Ift122	11	2.67
Cell protection genes			MECR	15	-2.29
Rfwd2	14	3.35	KCTD10	13	2.28
Hsp90ab1	14	2.49	MTMR12	15	2.13
H2-Aa	10	3.05	BICD2	20	2.07
PRRC2A	12	2.23	BICD2	20	2.07

Peptides suppress the cellular growth curve of human B-cell lymphoblastic leukemia

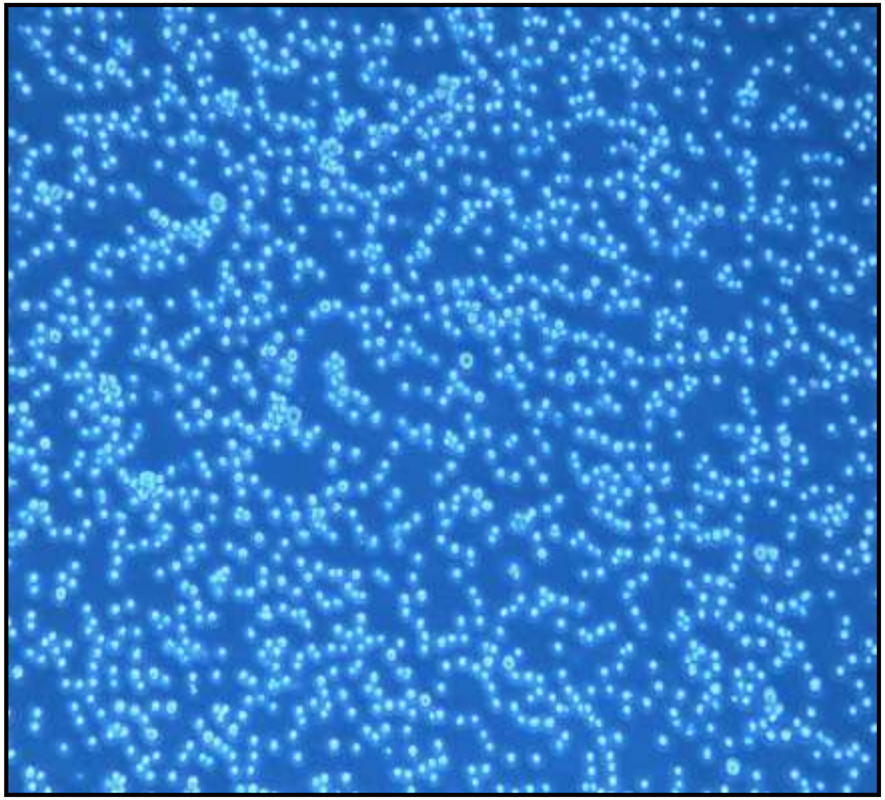


* - $p<0.05$ as compared to the control

Peptide concentration – 20 ng/ml

The peptide decreases the number of B-cells of human lymphoblastic leukemia

(Namalva line, 6th passage)



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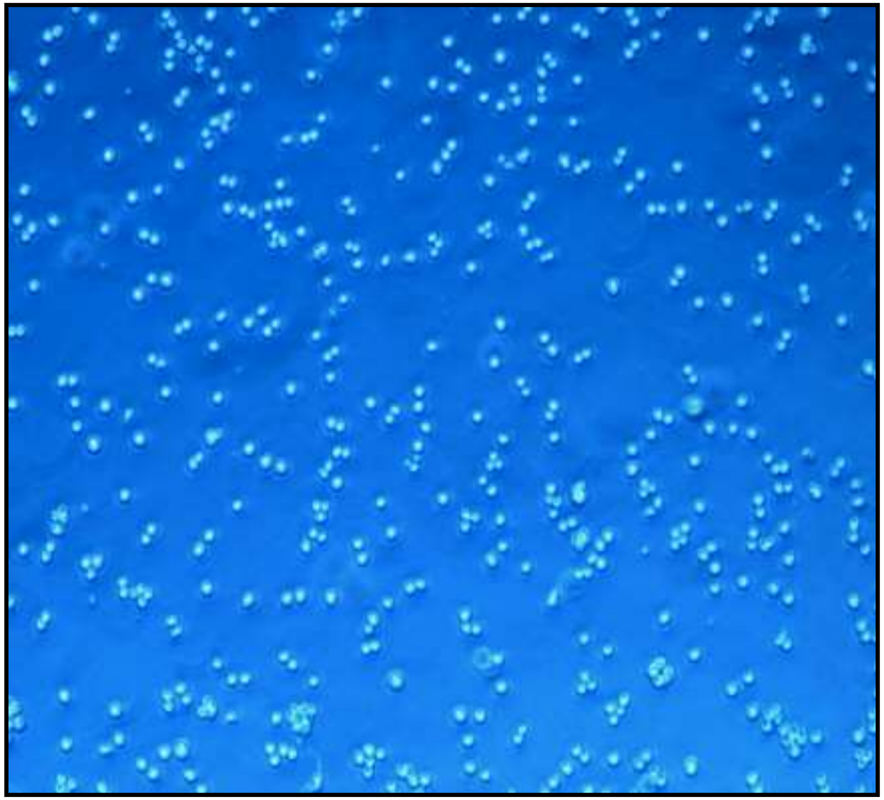
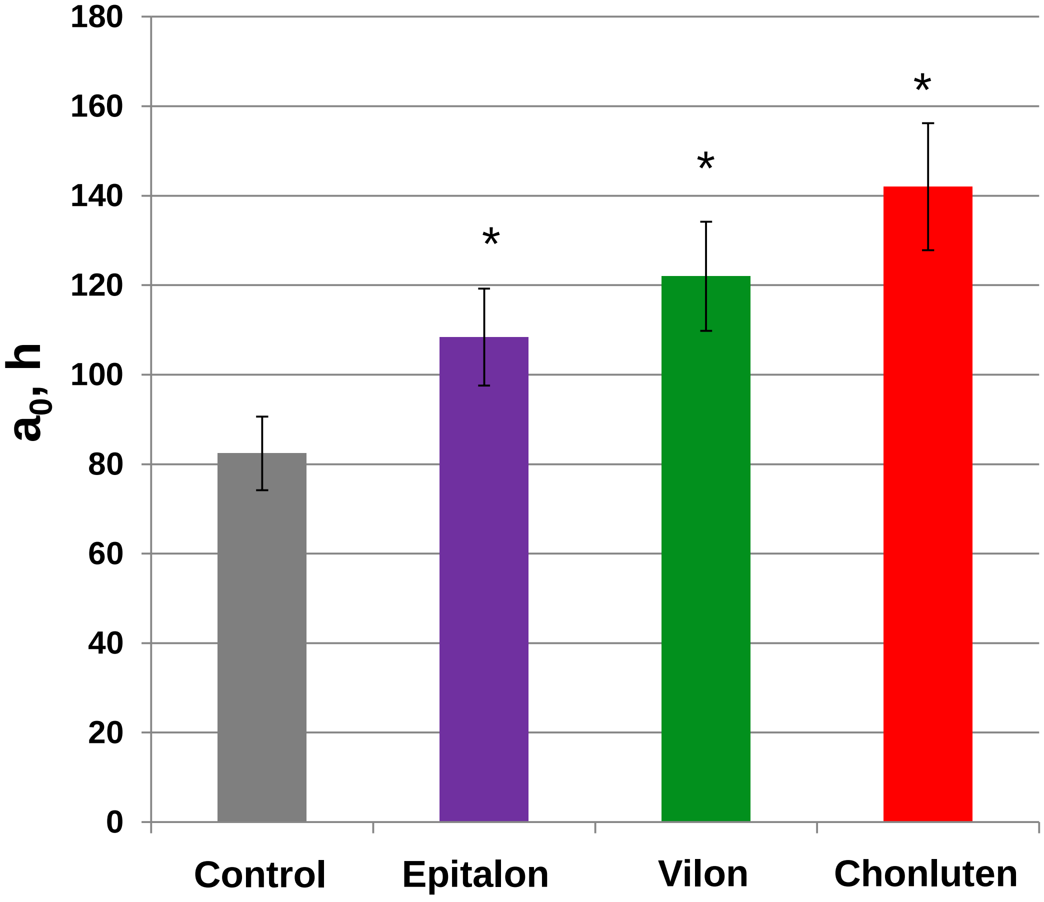


Chonluten, 20 ng/ ml

Life light microscopy, x 100, 3rd day of the experiment

Peptides increase average doubling time of cell population of human B-cell lymphoblastic leukemia

(Namalva line, 6th passage)



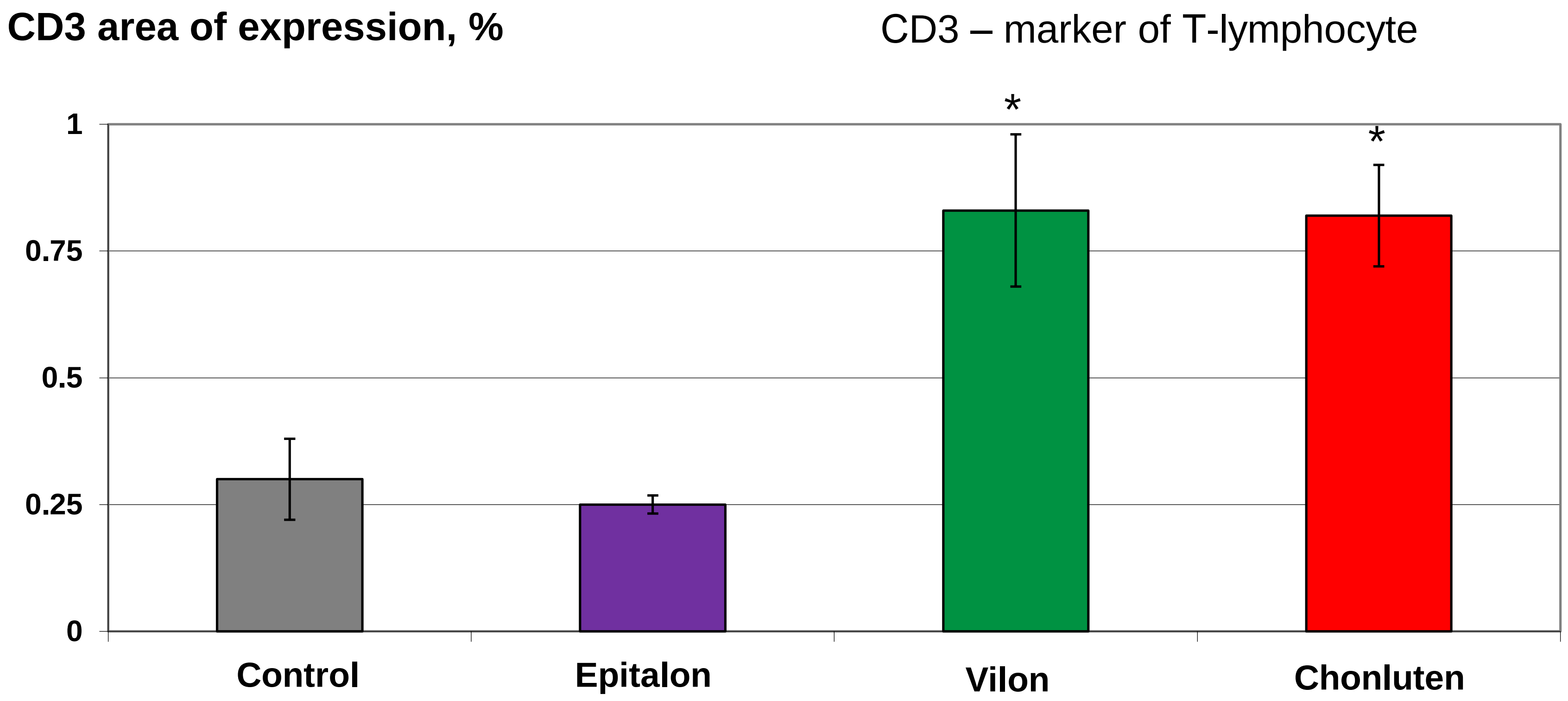
$$a_0 = \frac{t \cdot \ln 2}{\ln \frac{M_t}{M_0}}$$

a_0 – average doubling time of cell population, h
t – time of logarithmic phase of culture growth, h
Mt - number of cells at the moment of time t
M_0 - number of cells at the initial time

* - $p<0.05$ as compared to the control

Peptides increase CD3 expression in cell population culture of human B-cell lymphoblastic leukemia

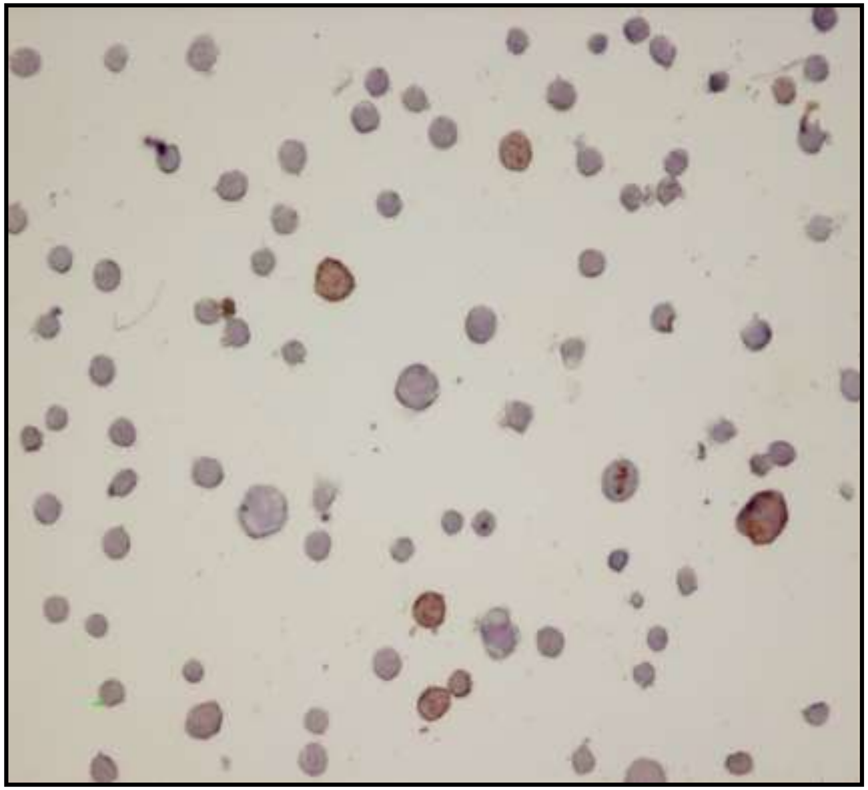
(Namalva line, 6th passage)



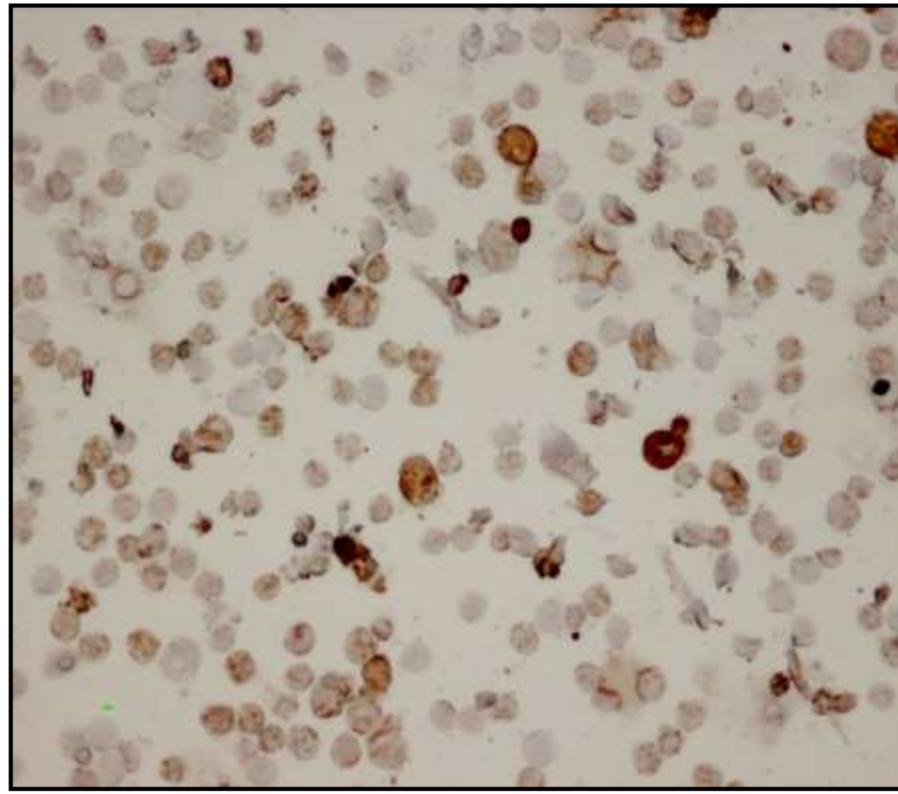
* - $p<0.05$ as compared to the control

Peptides increase CD3 expression in cell population culture of human B-cell lymphoblastic leukemia

(Namalva line, 6th passage)



Control



Vilon, 20 ng/ ml

Light microscopy, immunocitochemistry with haematoxylin coloration, x 200

Peptides decrease induced carcinogenesis in animals

Peptides	Animal species	Carcinogenic action	Tumour site	Tumour incidence, % Control	Peptide
Epithalamin (pineal peptides)	rats	DMBA	mammary gland	81	26*
		X-ray irradiation	mammary gland	16	3*
Thymalin (thymus peptides)	rats	DMBA	mammary gland	69	18*
		X-ray irradiation	mammary gland	21	3*
	C3H mice	X-ray irradiation	leukaemia	46	14*
Thymogen (Glu-Trp)	rats	isotops ^{90}Sr and ^{137}Cs	any malignant tumours	16	8*
Vilon (Lys-Glu)	C3H mice	DMH	kidney	60	14*
Epitalon (Ala-Glu-Asp-Gly)	C3H/He mice	MMTV	mammary gland	9	5*
	female rats	Constant lighting	mammary gland	41	27*
	male rats	Constant lighting	leukaemia	12	0*

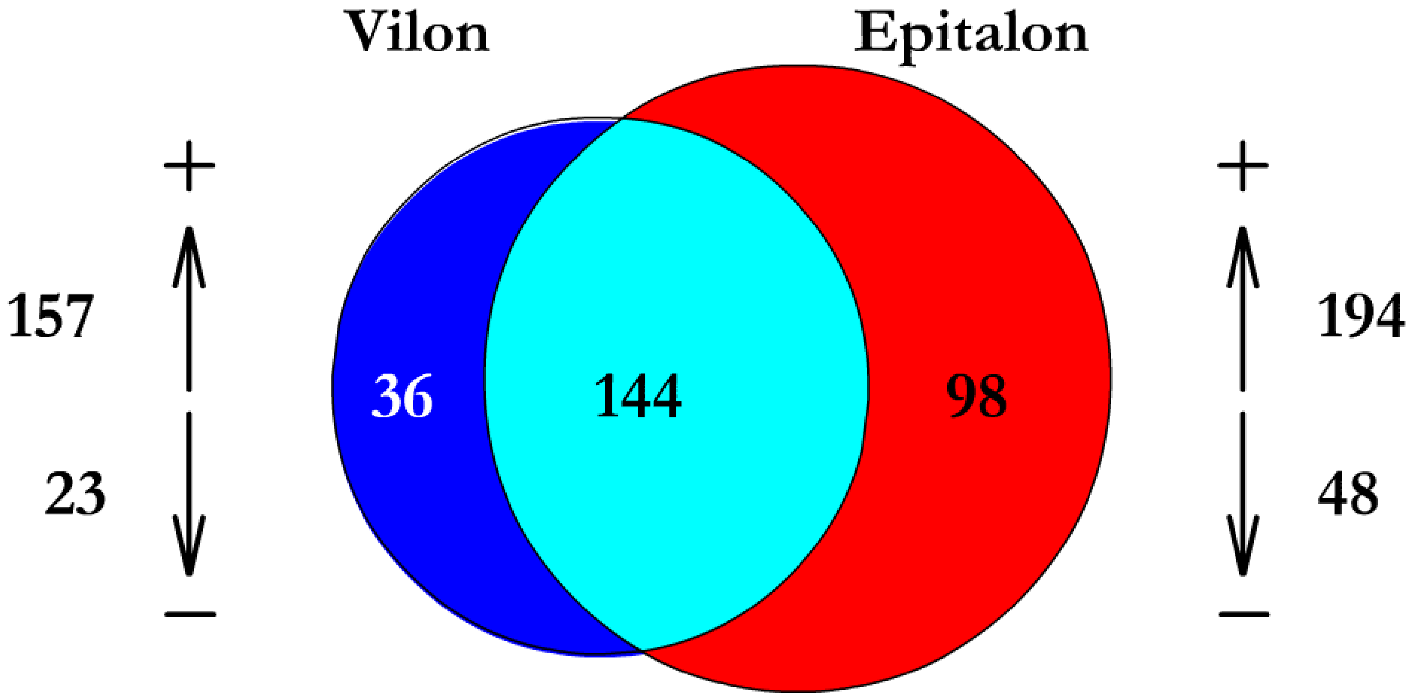
* - $p<0.05$ as compared to the control

Anisimov V., Khavinson V. Biogerontology (2010)

Peptides regulate gene expression in the mice heart

15247 genes tested,

maximum increase - 6.61 fold; maximum decrease - 2.71 fold



Euler-Venn Diagrams

Khavinson V. Neuroendocrinology Lett. (2002)

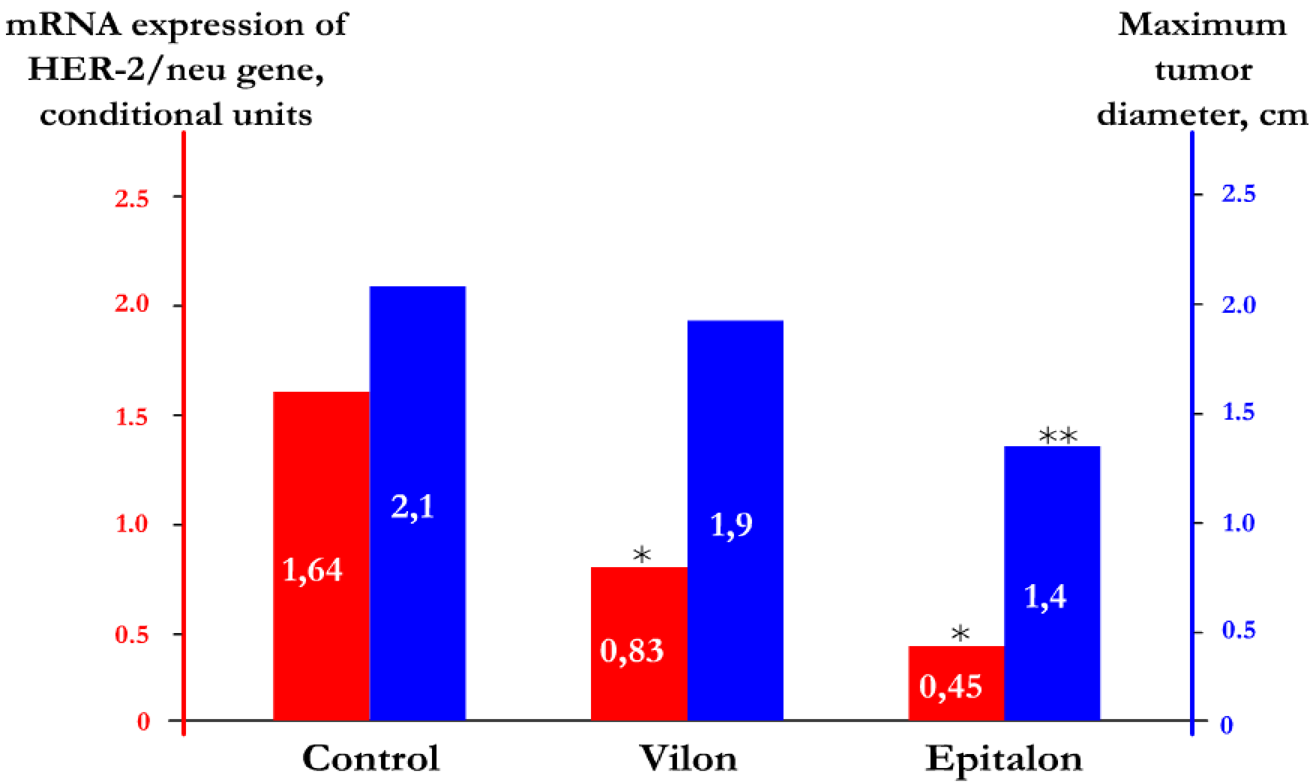
The influence of peptides on chromatin in human lymphocytes

Peptides	Associating acrocentric chromosomes (per cell)	Deheterochromatinization of optional heterochromatin (per cell)	Total heterochromatin	Structural heterochromatin of the 1st chromosome
Control (20-40 years old)	1.33±0.06	7.7±0.4	Stabile state	Stabile state
Control (75-88 years old)	1.17±0.05 *	5.9±0.2 *	Hetero-chromatinization	Hetero-chromatinization
Vilon	2.39±0.11 **	9.9±0.6 **	Dehetero-chromatinization	Hetero-chromatinization
Epitalon	2.32±0.12 **	8.4±0.5 **	Dehetero-chromatinization	Dehetero-chromatinization

* - $p<0.05$ as compared to the control (20-40 years old); ** - $p<0.05$ as compared to the control (75-88 years old)

Khavinson V., Malinin V. Gerontological aspects of genome peptide regulation (2005)

Peptides suppress HER-2/neu oncogene expression in transgenic mice

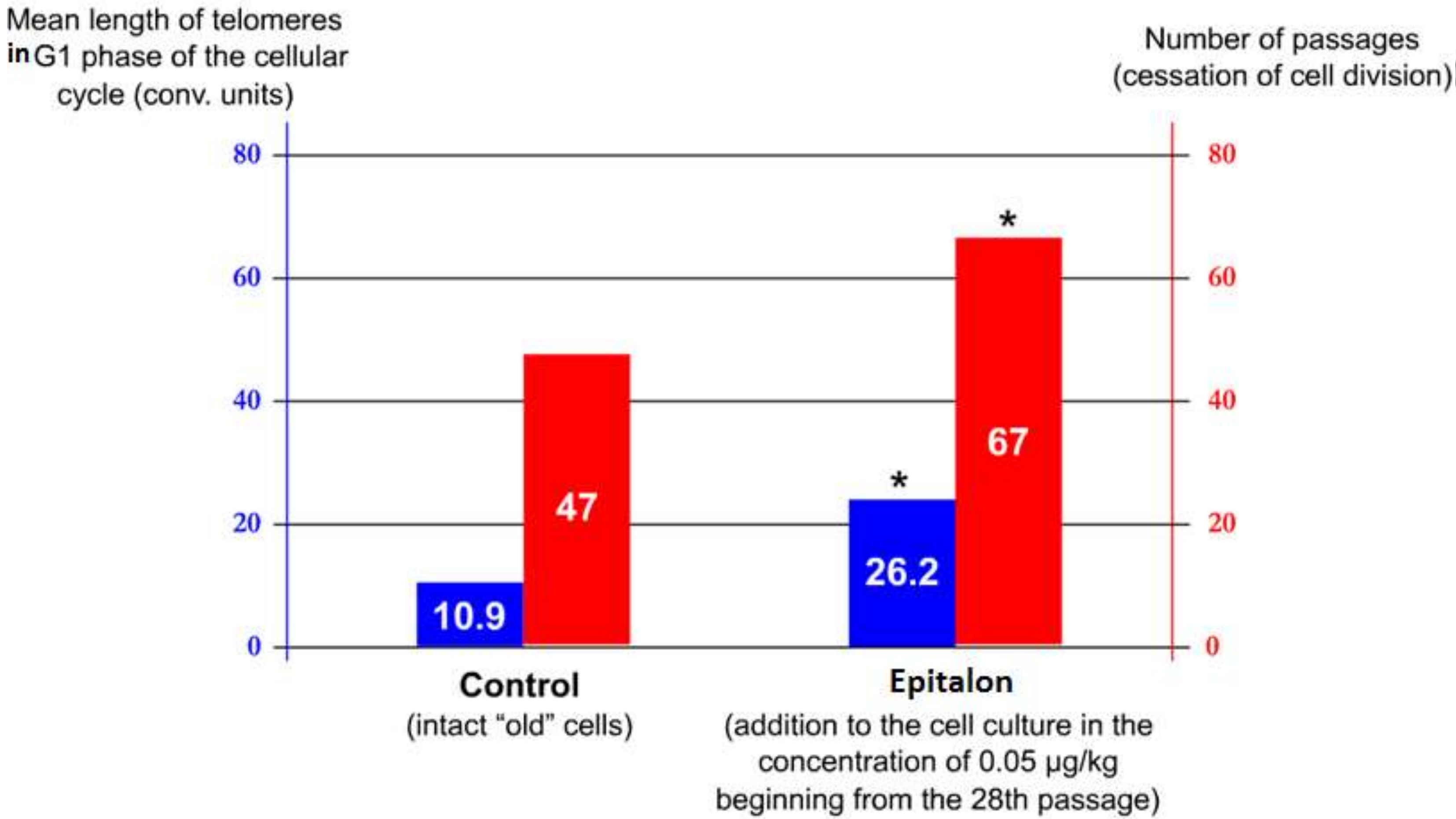


* - $p<0.001$ as compared to the control; ** - $p<0.05$ as compared to the control

Anisimov V. et al. International J. Cancer (2002)

Epitalon increases telomere length and the limit of fibroblasts division

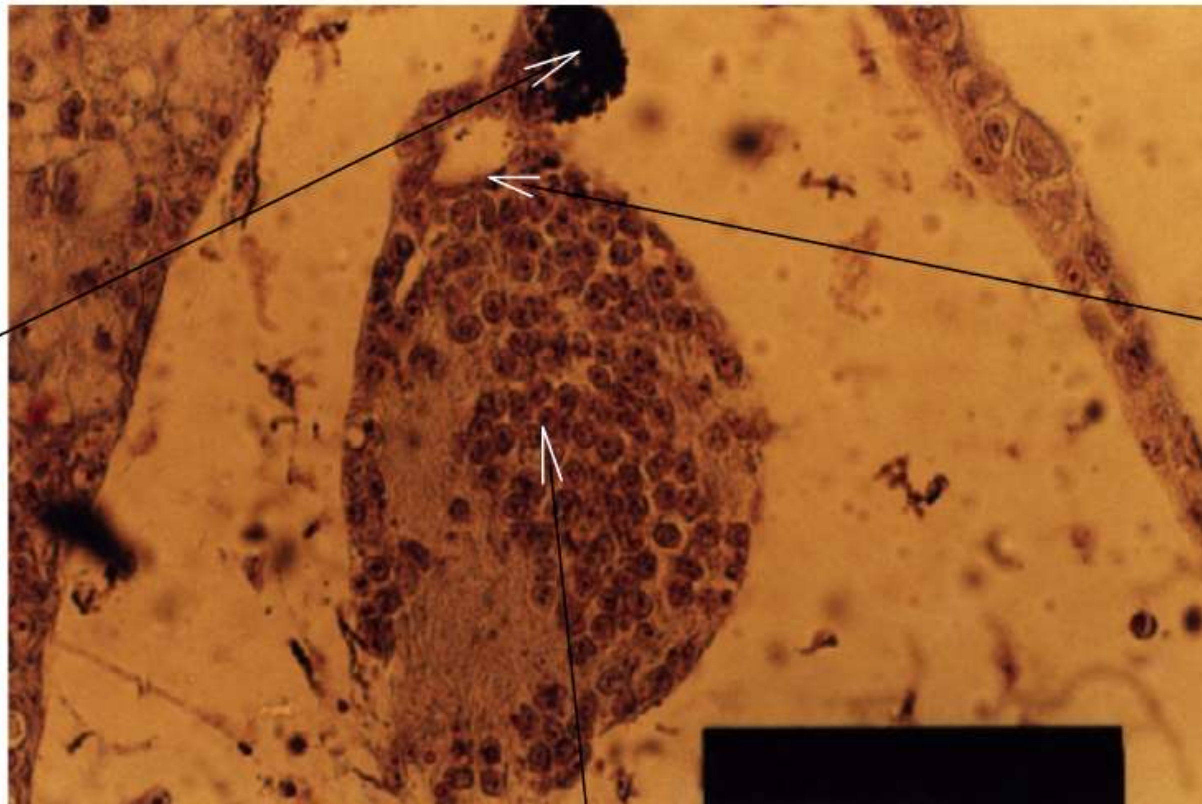
Increase in the number of cell divisions in the experiment by **42.5%** as compared to the control.



* - $p<0.05$ as compared to the control

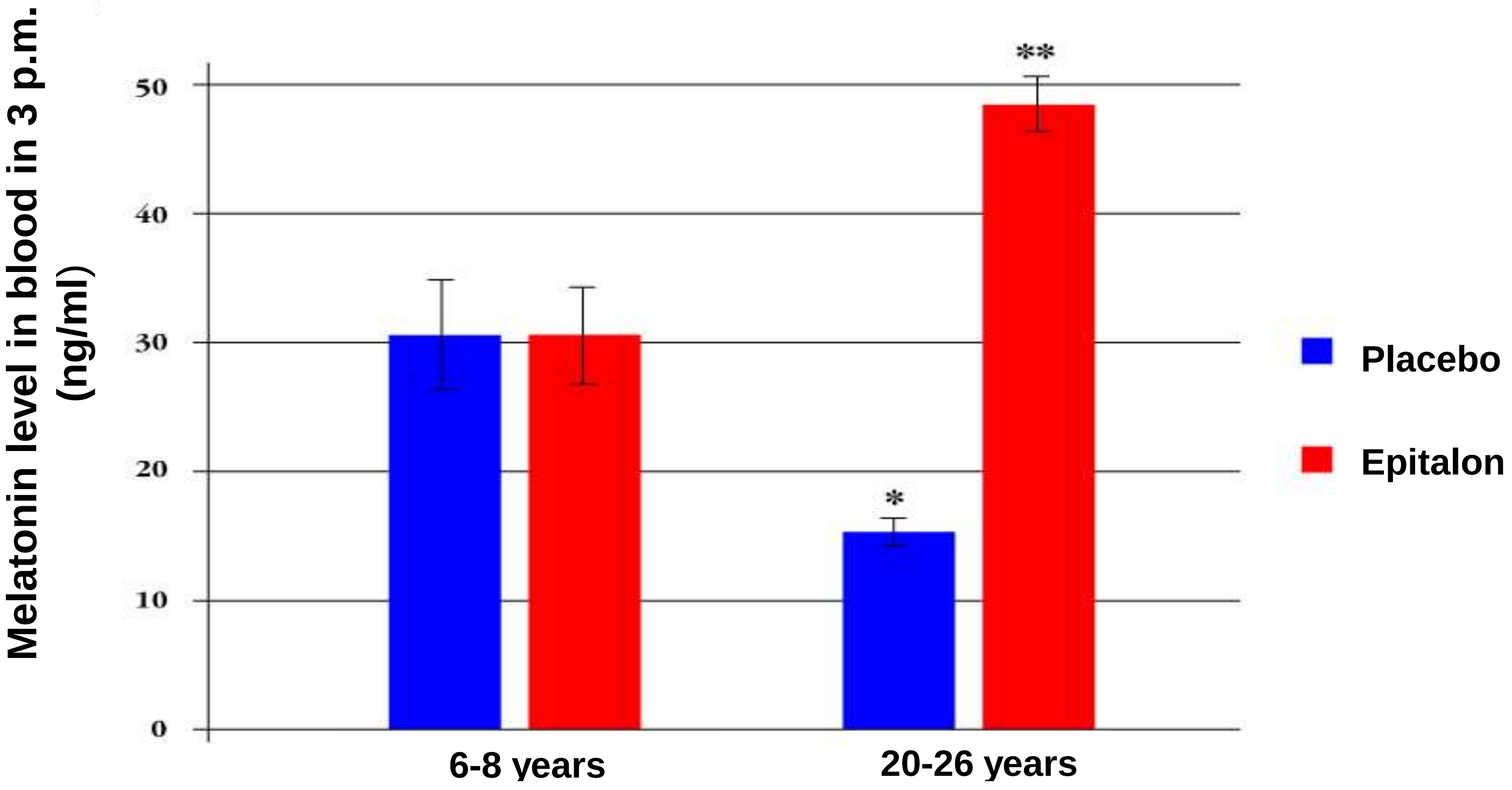
Anisimov V., Khavinson V. Biogerontology (2010)

Retinal peptides induce the differentiation of polypotent ectoderm of *Xenopus laevis*



Khavinson V. Peptidergic regulation of ageing (2009)

Epitalon increases melatonin synthesis in old monkeys

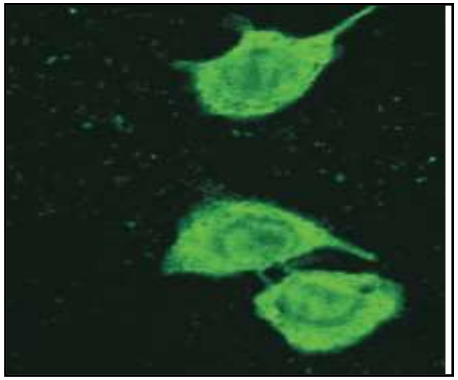


* - $p<0.05$ as compared to the young animals, placebo; ** - $p<0.05$ as compared to the old animals, placebo

Khavinson V. et al. Neuroendocrinology Lett. (2001)

Ezhekort (EDG) decreases apoptosis

1. mito-GPF expression in mice fibroblasts



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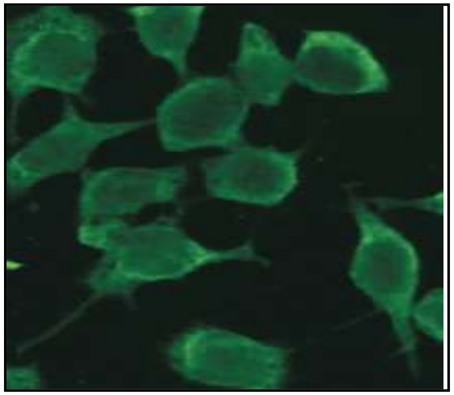
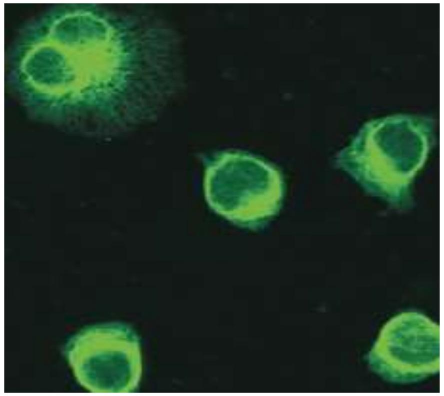


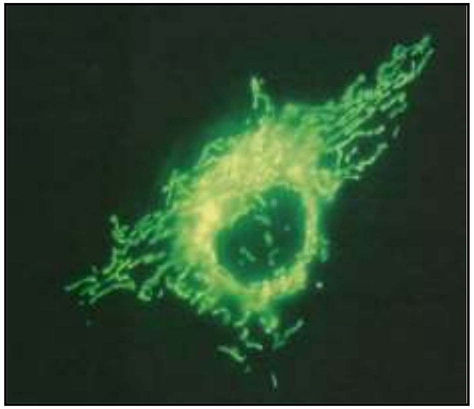
H. Pylori



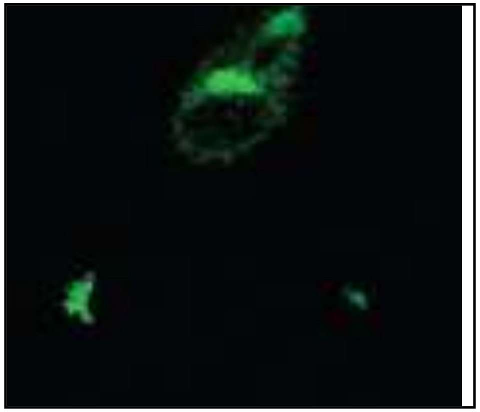
H. Pylori + Ezhekort

Confocal microscopy, x400

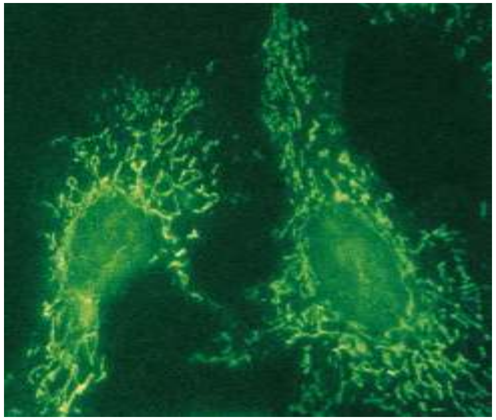
2. mito-GPF expression in human gastric epithelial cells



Control



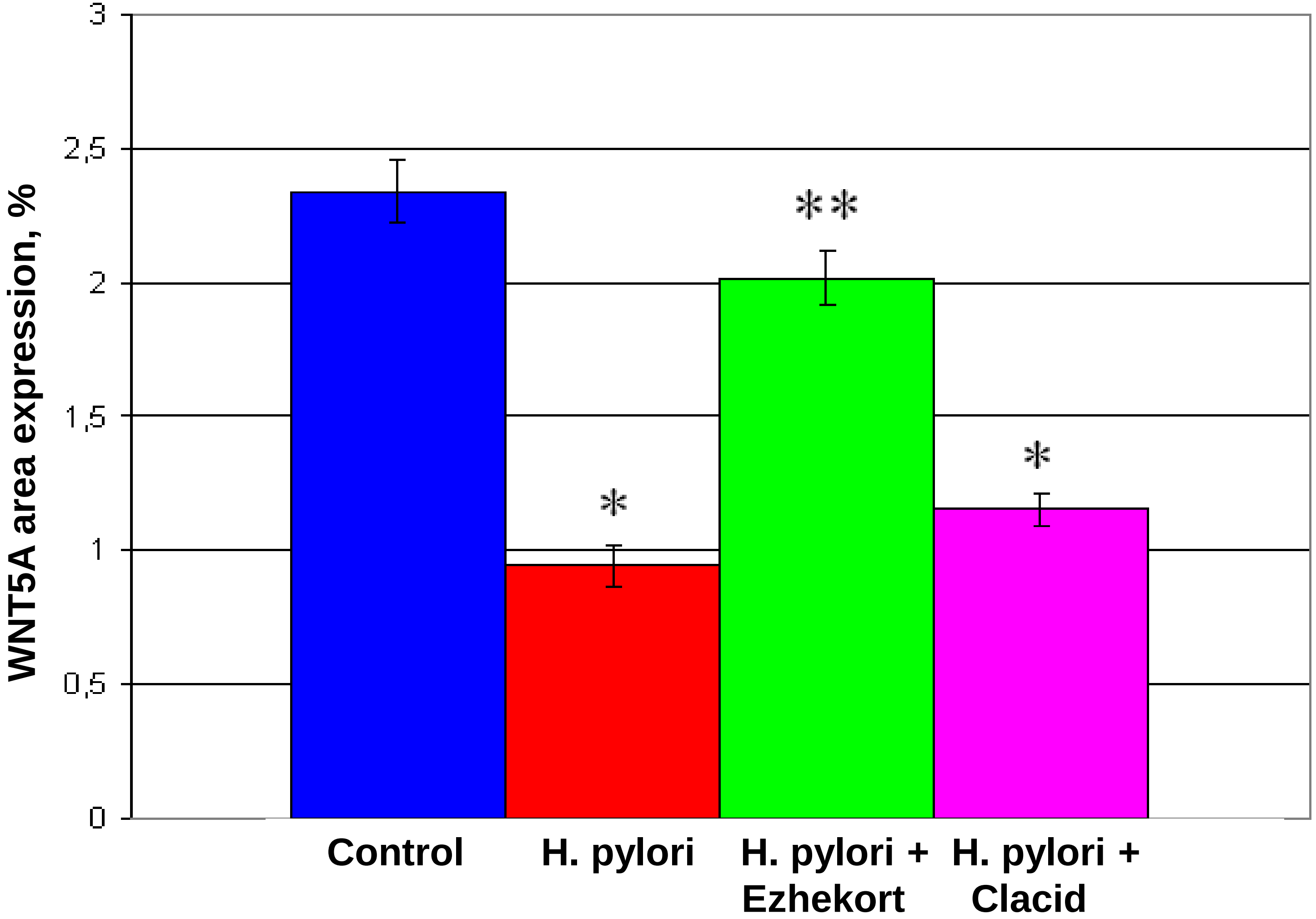
H. Pylori



H. Pylori + Ezhekort

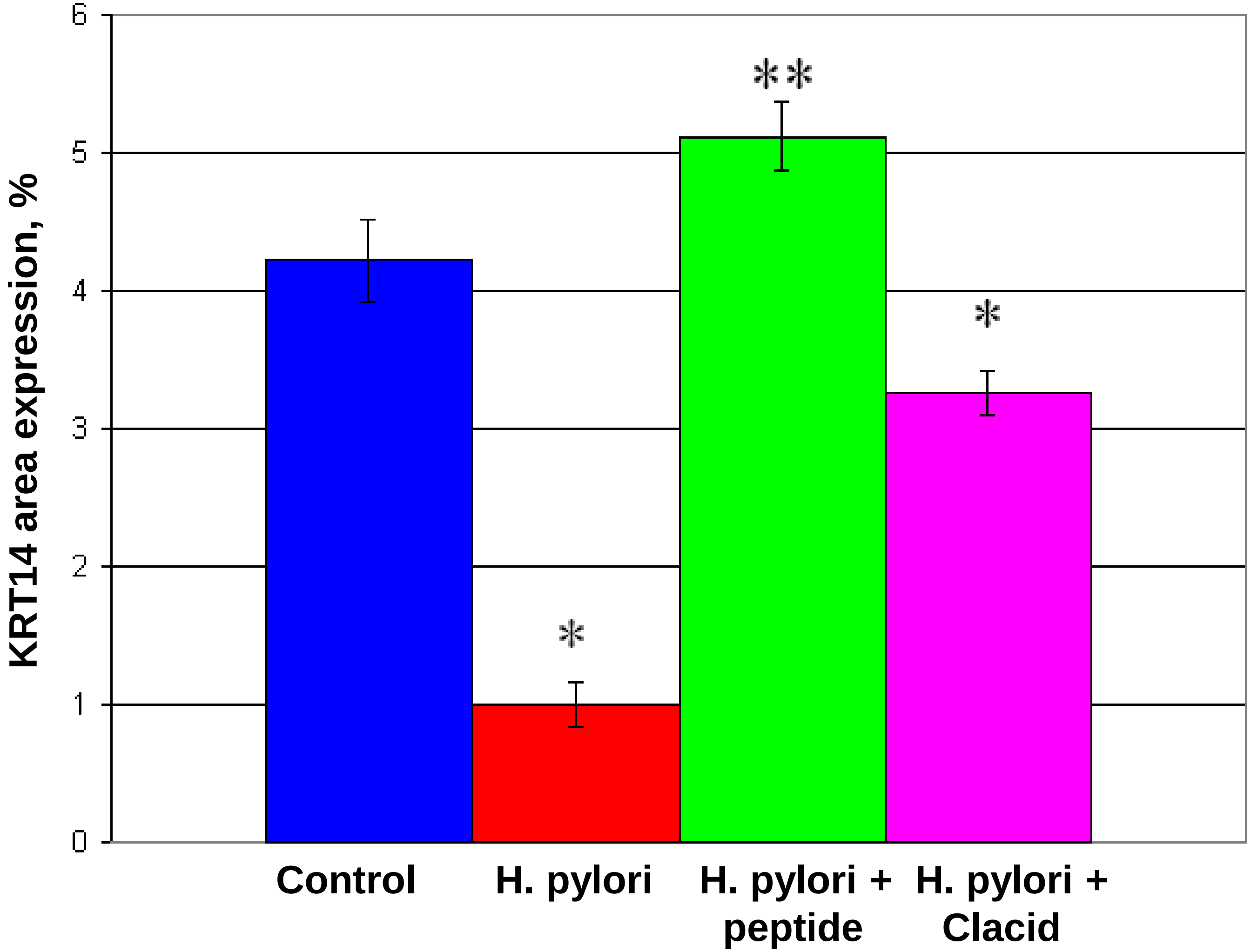
Confocal microscopy, x600

Ezhekort increases the expression of marker WNT5A in culture of human gastric epithelial cells



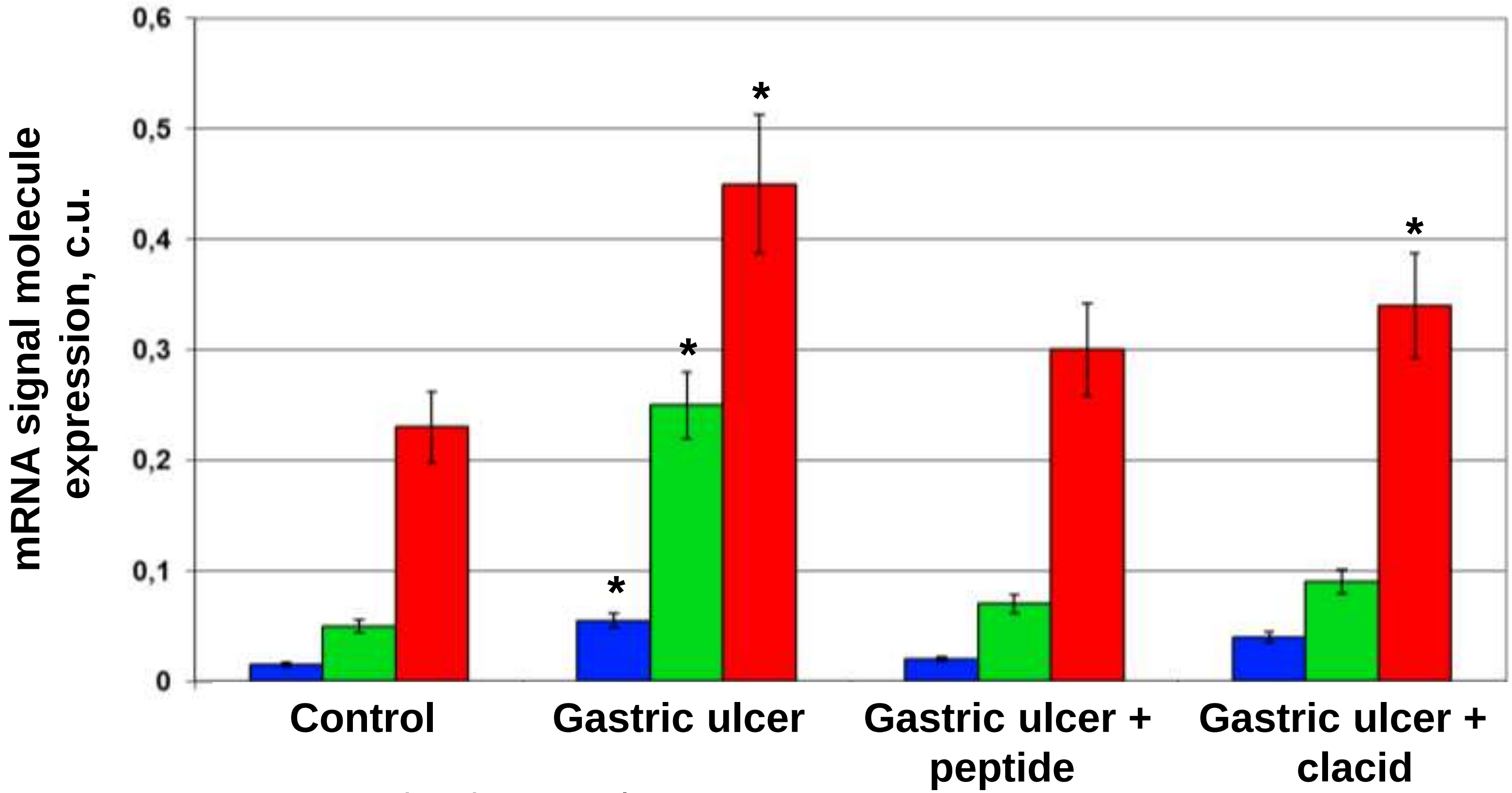
* - $p < 0.05$ as compared to the control; ** - $p < 0.05$ as compared to H. pylori

Ezhekort increases the expression of KRT14 protein (cytoskeleton marker) in culture of human gastric epithelial cells



* - $p < 0.05$ as compared to the control; ** - $p < 0.05$ as compared to H. pylori

Ezhekort decreases the expression of mRNA signal molecules in gastric mucous membrane



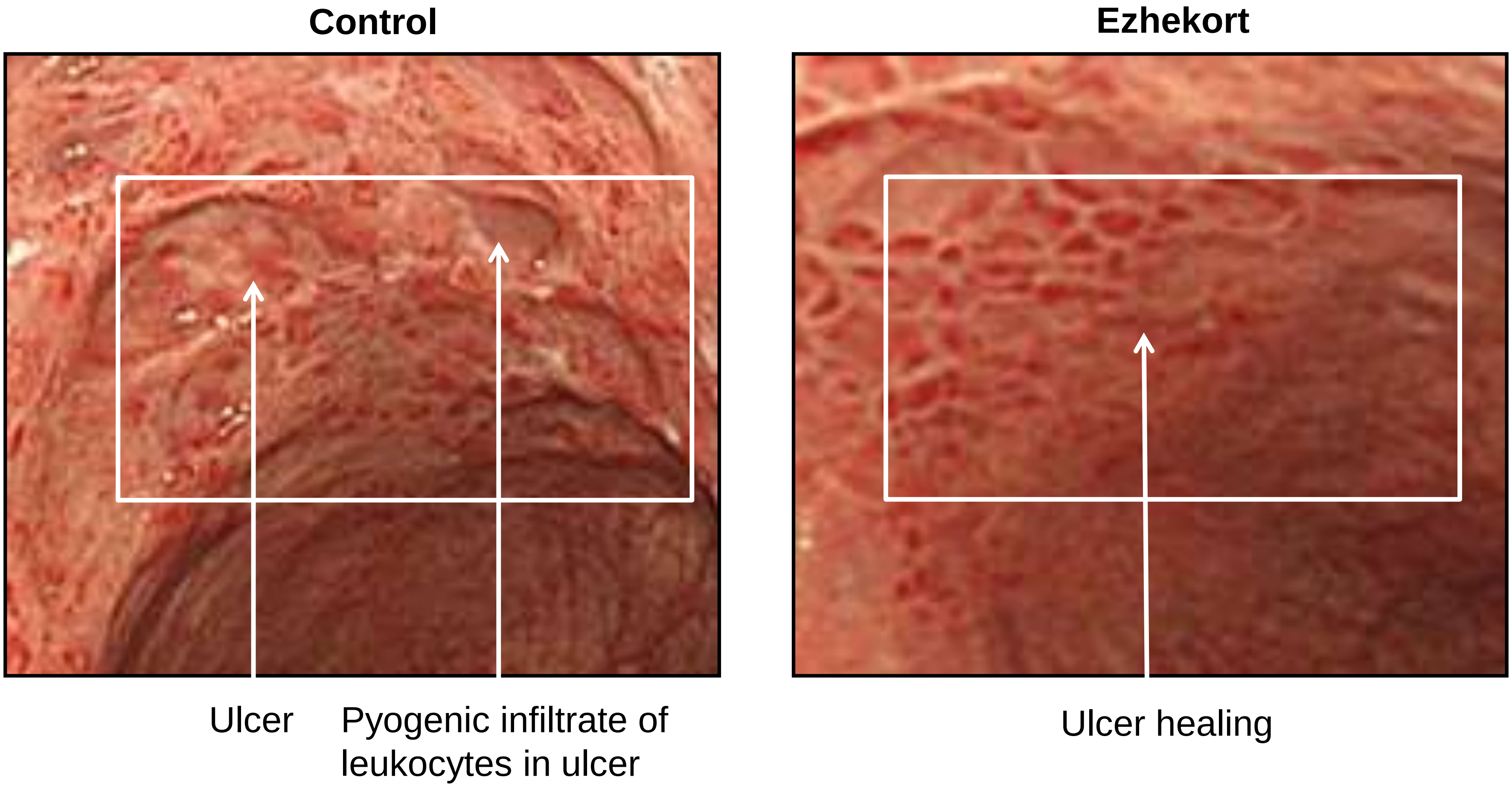
* - $p<0.05$ – as compared to the control

TNFα SOD Cox-2

TNFα - tumor necrosis factor, SOD – superoxide dismutase, Cox-2 – cyclooxygenase

Khavinson V., et al. Bull. Exp. Biol. Med. (2011)

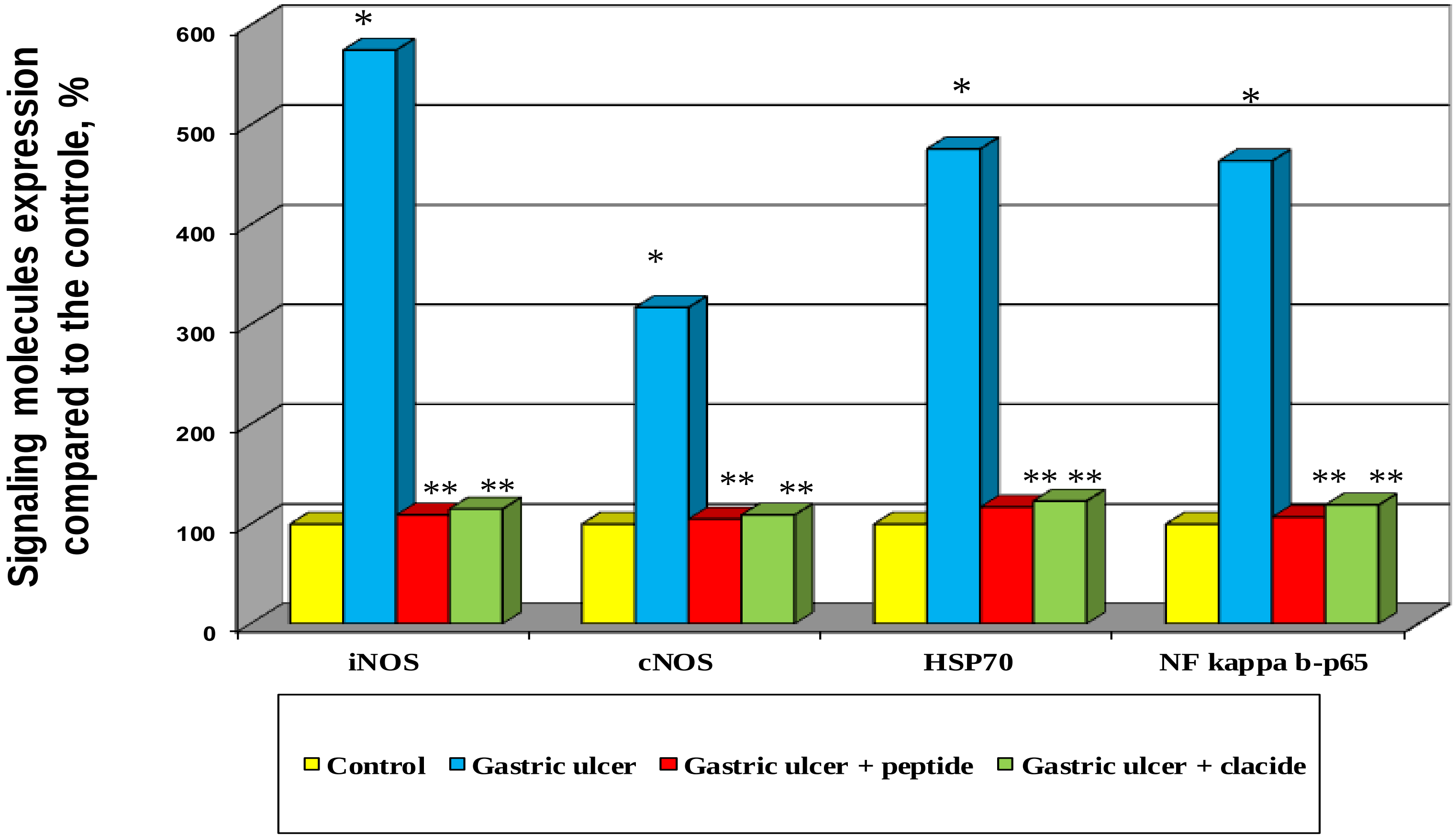
Ezhekort contributes to epithelialization of gastric ulcer



Pathomorphosis of induced gastric ulcers, x 20

Khavinson V., et al. Bull. Exp. Biol. Med. (2011)

Ezhekort decreases the mRNA gene expression in gastric mucous membrane



* - p < 0.05 as compared to the control; ** - p < 0.05 as compared to the gastric ulcer

iNOS – NO-synthase 2 (inducible), eNOS - endothelial NO-synthase,
HSP70 - heat shock protein, NF kappa b-p65 - transcription factor

The influence of Pancragen on glucose content in the blood of rats with alloxan-induced diabetes *(treatment)*

Animal group	Glucose concentration in blood (mg %)						
	Initial level	In 15 days after alloxan injection	8th day of Pancragen course	After the completion of Pancragen course (days)			
				9	28	44	58
Control (NaCl)	84.0±5.7	345.0±15.4	360.0±12.3	351.4±11.2	375.7±11.2	347.2±12.8	332.1±13.7
n	10	10	9	7	5	5	3
Pancragen	81.1±3.8	333.6±12.4	254.5±16.2*	183.9 ±10.5*	221.5±11.2*	210.8±9.3*	198.9±11.5*
n	11	11	11	9	8	8	7

* - p < 0.001 as compared to the control

Pancragen decreases glucose content in the blood of rats with alloxan-induced diabetes

(prevention and treatment)

Animal group	Glucose concentration in blood (mg %)					
	Initial level	7th day of Pancragen course (1st course)	After alloxan injection (days)			
			14	21	28	40
				2nd Pancragen course (from 18th till 28th day)		
Control (NaCl)	82.7 ± 0.9	96.4 ± 1.0	333.7 ± 55.8*	345.6 ± 57.8*	156.4 ± 26.4*	405.0±89.8*
n	7	7	6	5	5	4
Pancragen	76.8 ± 1.1	94.0 ± 0.8	261.5 ± 39.5**	159.0 ± 32.6**	77.3 ± 1.3**	107.7±6.4**
n	8	8	8	6	6	6

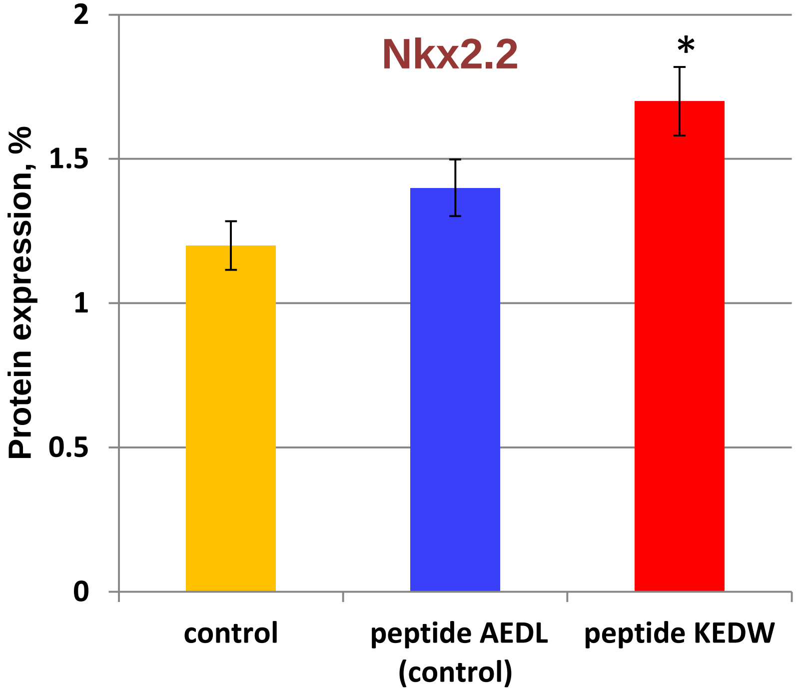
* - $p < 0.001$ as compared to the initial level; ** - $p < 0.02$ as compared to the control

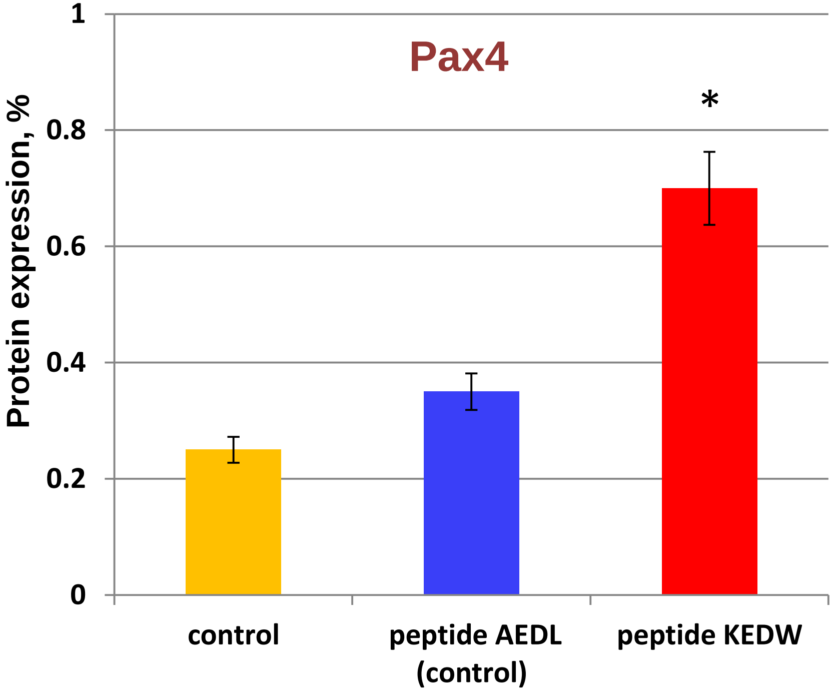
Pancragen increases insulin content in blood of rats with alloxan-induced diabetes

Animal group	Insulin content in the blood (μMU/ml)				
	Initial level	8th day of the treatment of alloxan diabetes	After alloxan injection (days)		
			9	18	44
Control (NaCl)	24.3 ± 2.1	0.8 ± 0.25	0	0	0
n	8	8	6	5	5
Pancragen	23.8 ± 2.8	3.1 ± 1.1*	3.2 ± 0.5**	4.3 ± 0.5**	3.9 ± 1.1**
n	10	10	7	7	7

* - $p < 0.05$ as compared to the initial level; ** $p < 0.001$ as compared to the control

Pancragen (KEDW) increases the protein expression in senescent pancreatic cells



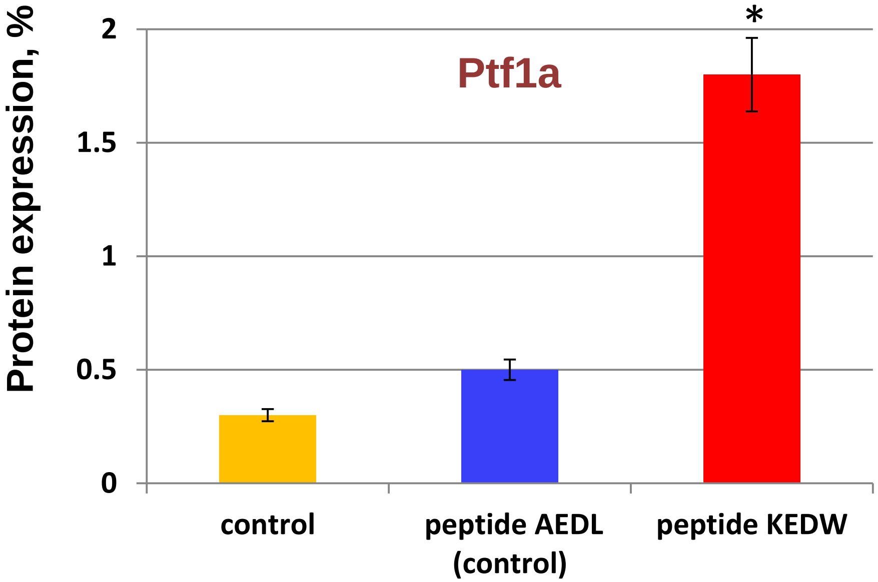


* - $p<0.05$ as compared to the control

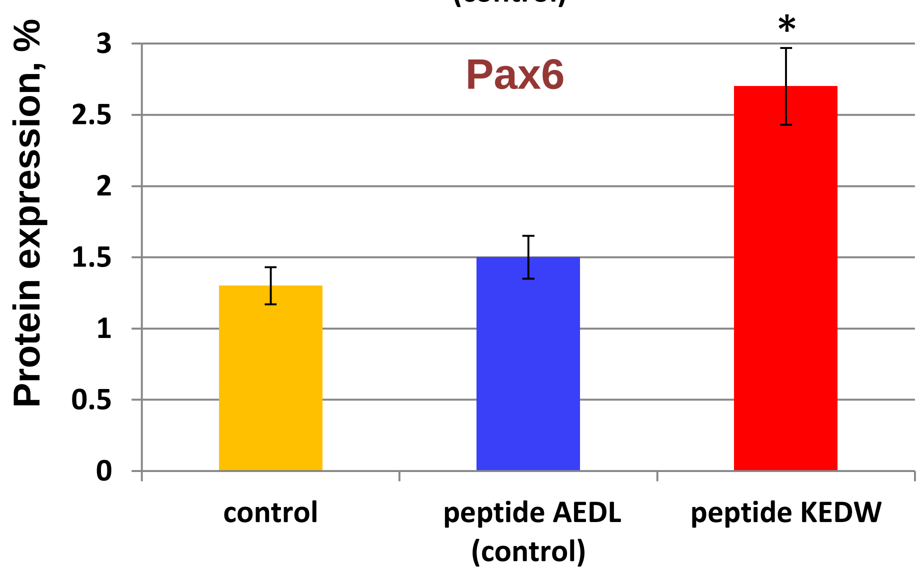
Peptide KEDW stimulates the expression of β-cell differentiation factors (Nkx2.2, Pax4) in human pancreatic cell cultures

Khavinson V. Advances in Gerontology (2013)

Pancragen increases the protein expression in senescent pancreatic cells



Peptide KEDW stimulates the expression of acinar differentiation factor Ptf1a in human pancreatic cell cultures

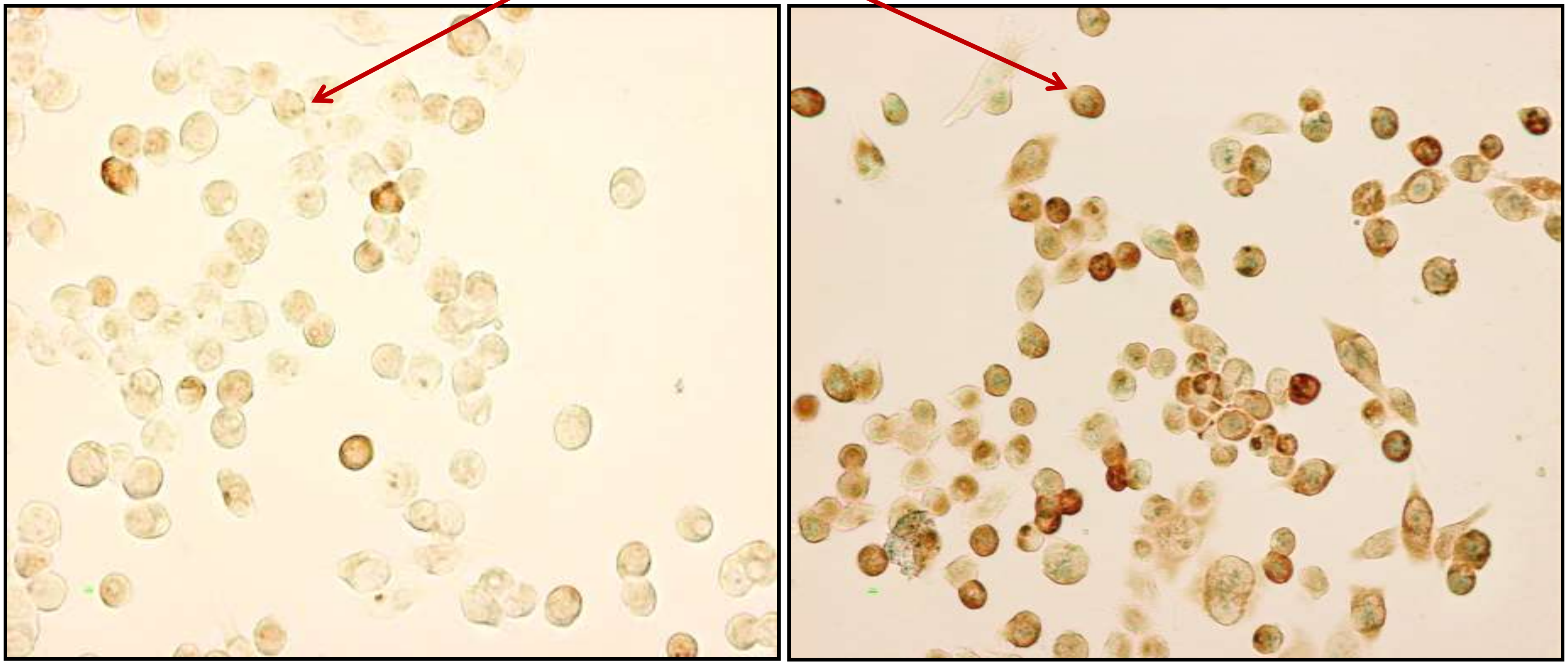


Peptide KEDW stimulates the expression of α-cell differentiation factor Pax6 in human pancreatic cell cultures

* - $p<0.05$ as compared to the control

Khavinson V. Advances in Gerontology (2013)

Pancragen increases the expression of differentiation factor Pax6 in senescent pancreatic cells



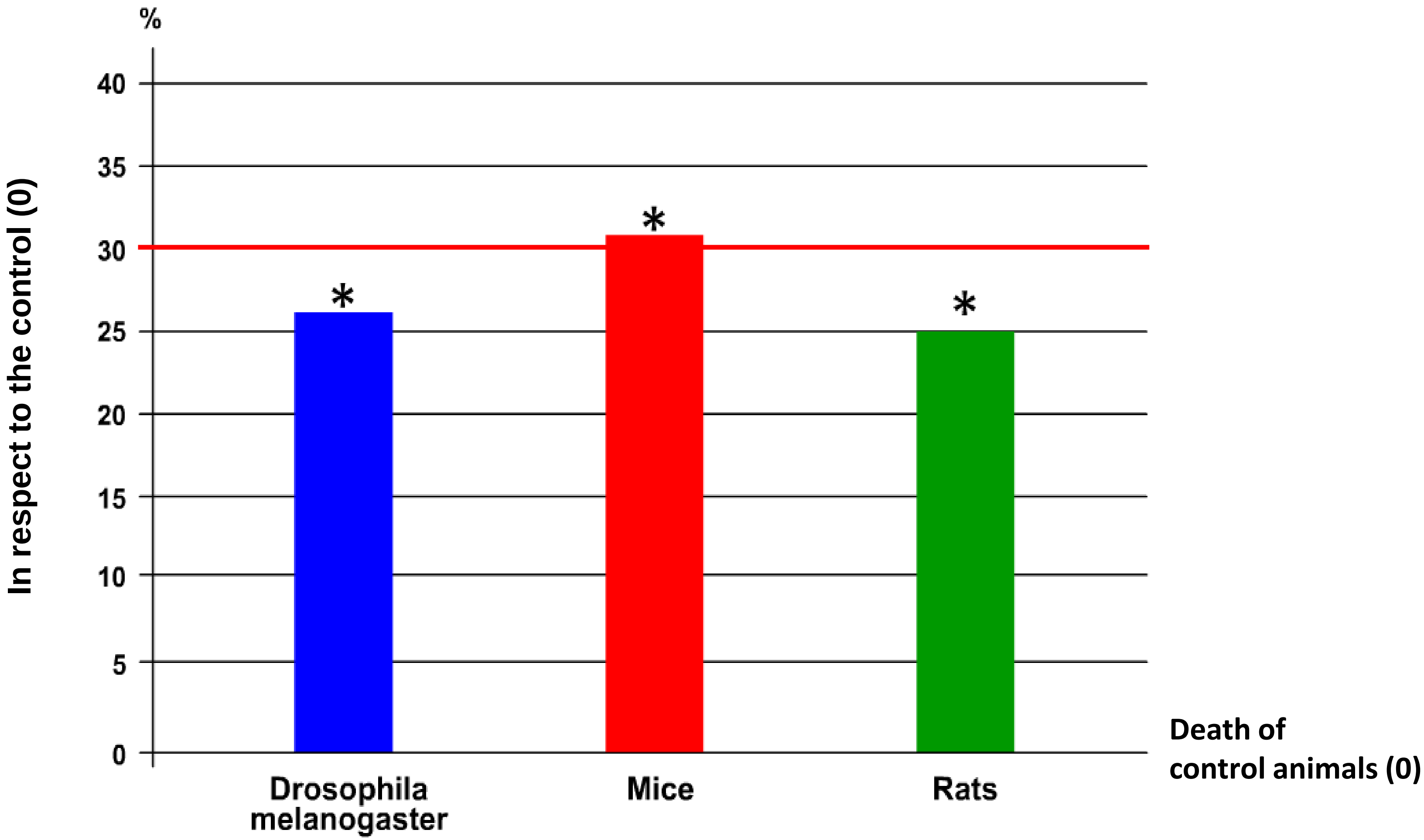
Control **Peptide KEDW**

Immunocytochemistry, x200,
aged (14th passage) cell culture of human pancreas “MIA PaCa-2”

Khavinson V. Advances in Gerontology (2013)

Peptides increase average lifespan

(the results of 25 experiments)

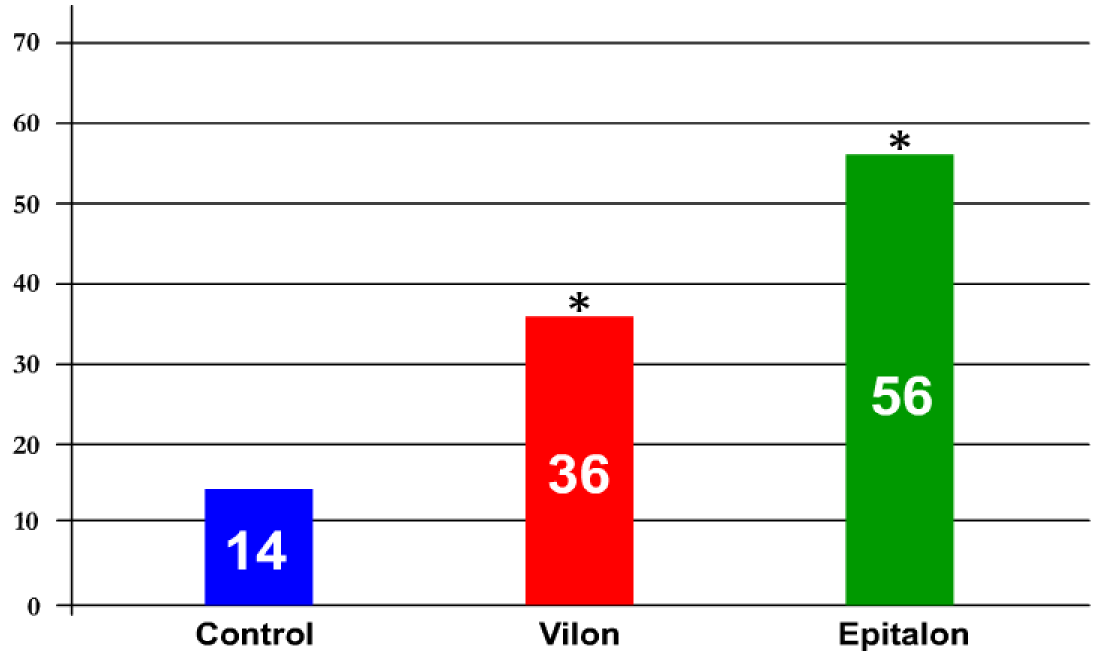


* - $p<0.05$ as compared to the control

Khavinson V. Peptidergic regulation of ageing (2009)

The influence of peptide bioregulators on mice lifespan

NUMBER OF MICE, WHICH REACHED 2 YEARS OF AGE



LIFESPAN

Index	Control	Vilon	Epitalon
Mean, days	685 ± 9	694 ± 13	721 ± 11 *
Maximum, days	740	792	1053
increase (%)	-	7,0	42,3

* - $p<0.05$ as compared to the control

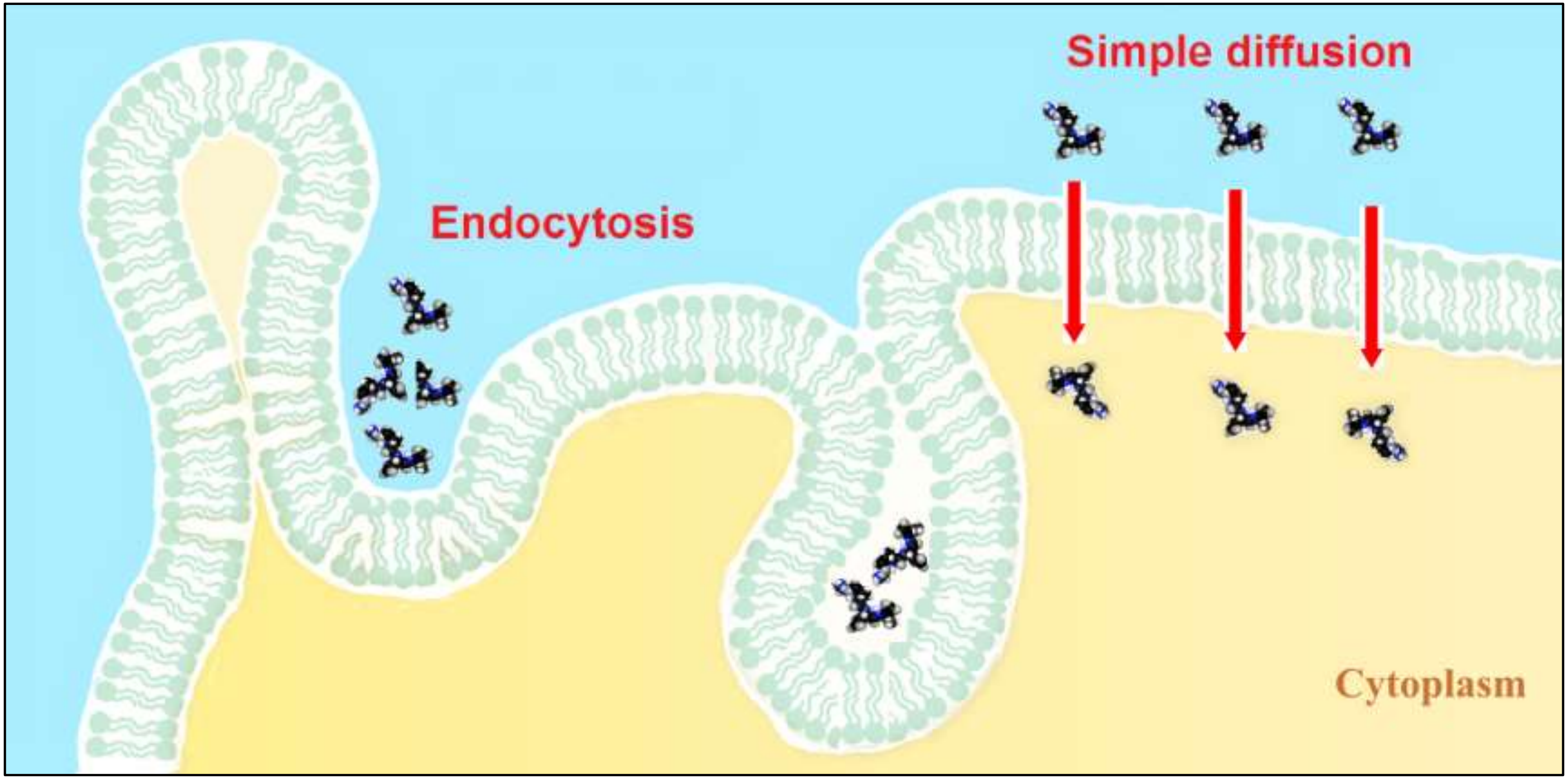
Khavinson V. Peptides and ageing (2002)

Peptides increase organism vital resource

Biological Activity	Publications
Increase in the protein synthesis in rat hepatocytes by **42.9%**	*Khavinson V. Peptides and ageing. NEL (2002)*
Increase in the growth of explants in organotypic cultures of cells of the animal by **22-42%**	*Khavinson V. Peptides and ageing. NEL (2002)*
Increase of the amount of optional heterochromatin in lymphocytes of elderly people by **42.4%**	*Khavinson V. et al. NEL (2003)*
Increase in the number of divisions of human fibroblasts by **42.5%** and a **2.4-fold** increase in the average length of telomeres	*Khavinson V. et al. Bul. Exp. Biol. Med. (2004)*
Increase in the lifespan of animals by **20-40%** and maximal lifespan by **42.3%**	*Anisimov V., Khavinson V. Biogerontology (2010), Anisimov V. et al. Mech. Ageing Dev. (2001)*
A **3.1-fold** decrease in the frequency of tumors induced by a carcinogenic agent in animals	*Anisimov V., Khavinson V. Biogerontology (2010)*

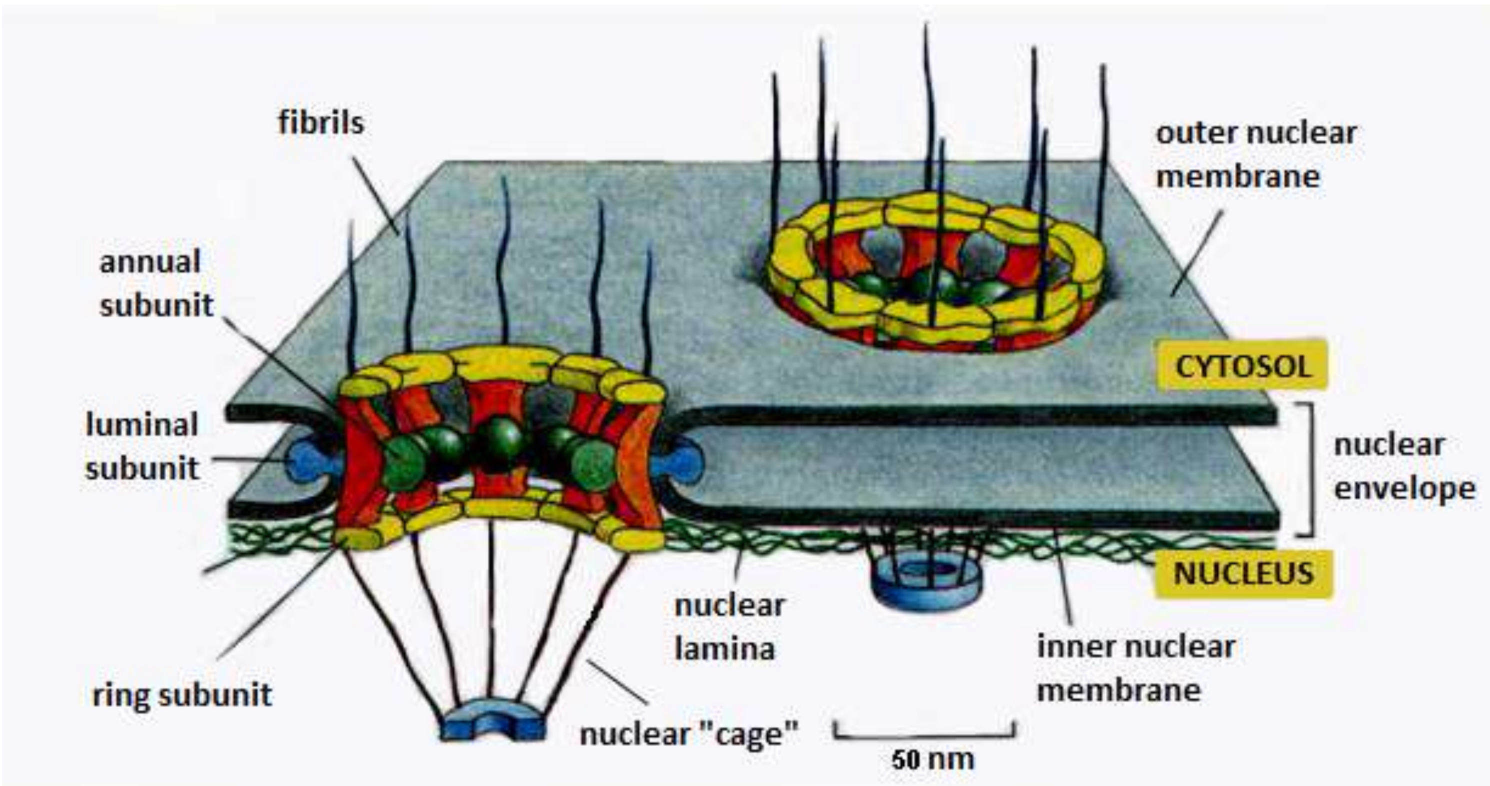
2. Peptide bioregulators: mechanism of action

Penetration of small peptides (CPPs) into cell



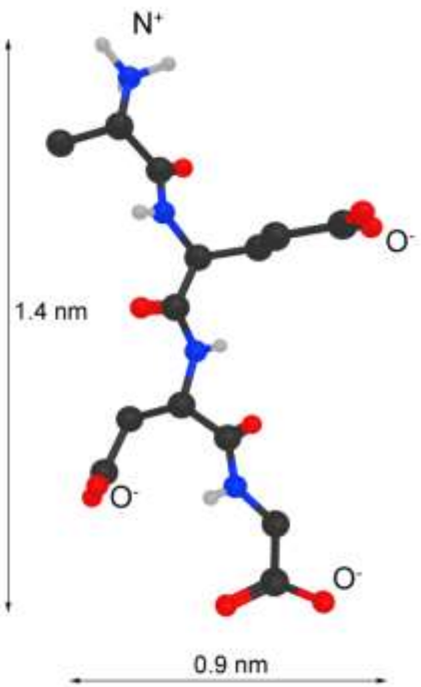
Trabulo S. et al. Pharmaceuticals in modification (2010)

Schematic representation of nucleopore

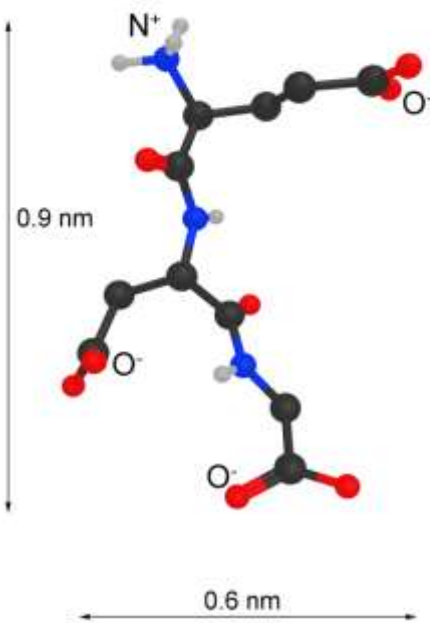


Alberts B. et al. Molecular Biology of the Cell (1994)

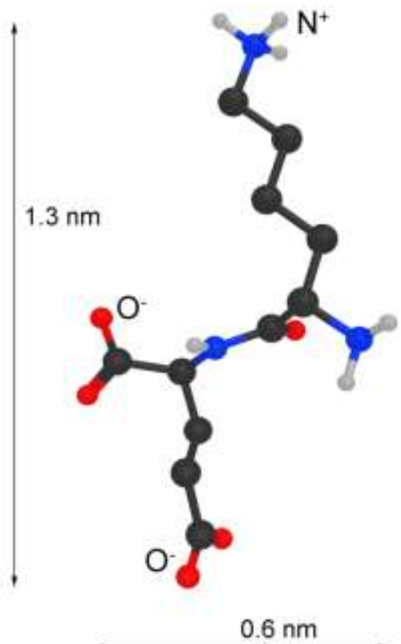
The structures of peptides (3D models)



Epitalon



Chonluten

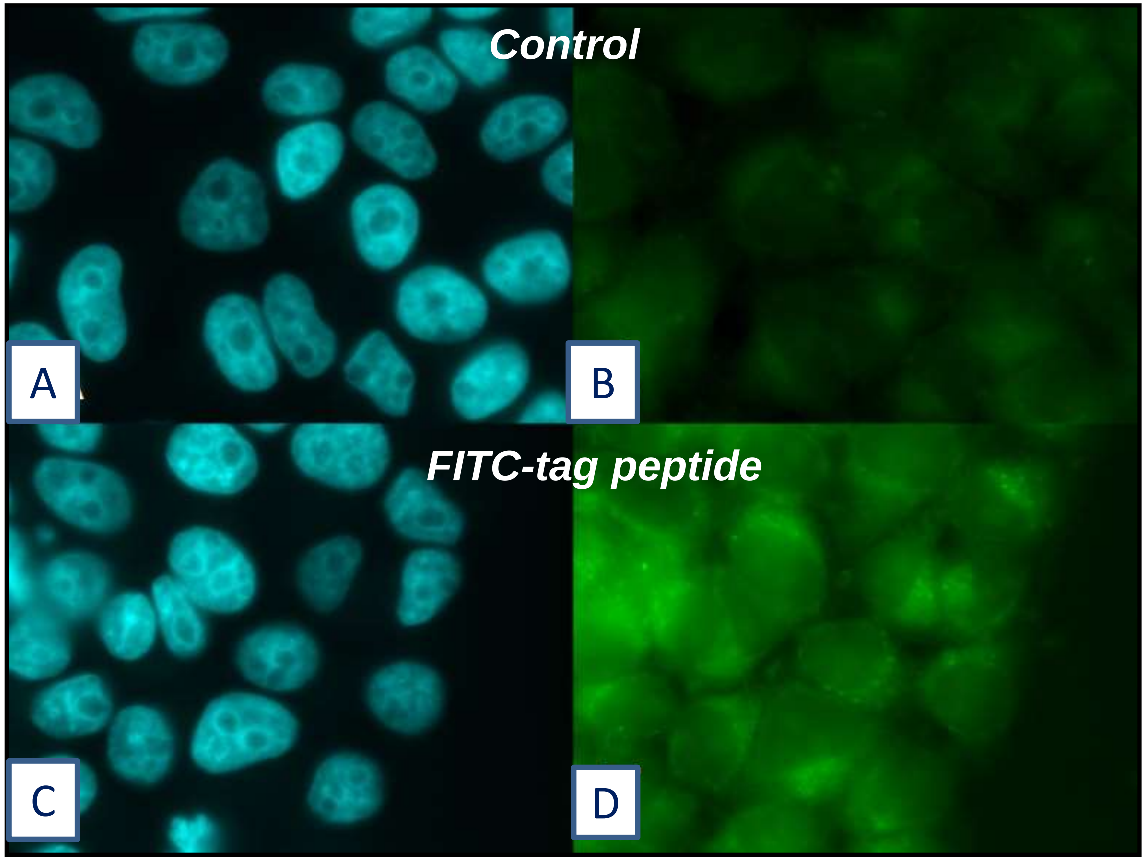


Vilon

Conformations of the peptides ***Ala-Glu-Asp-Gly*** *(Epitalon)*, ***Glu-Asp-Gly*** *(Chonluten)*, ***Lys-Glu*** *(Vilon)* with optimal minimization energy.
Red colour indicates oxygen molecules, **blue** – nitrogen molecules, **black** – carbon molecules, **light grey** - polar hydrogen molecules. Nonpolar hydrogen molecules are not displayed

Penetration of FITC-labeled peptide into HeLa cells

A, C – staining of DAPI nuclei (DNA identification)



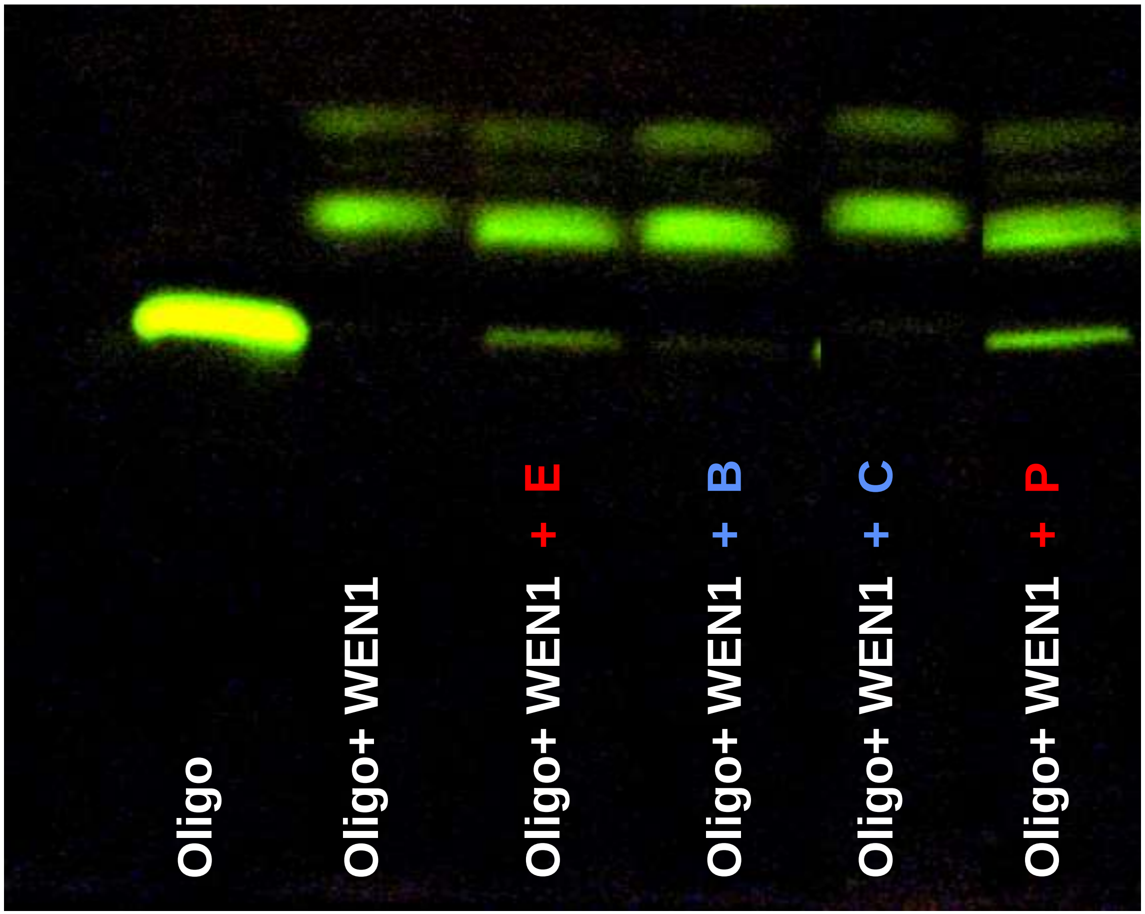
B, D – fluorescence

6-hour cell incubation with
FITC-labeled peptide (1,2 x 10^{-6} M)

Fedoreeva L. et al. Biochemistry (2010)

The influence of peptides on hydrolysis of fluorescence-labelled deoxyribooligonucleotide with endonuclease WEN1

Oligonucleotide - (5’) FAM – CGC CGC CAG GCG CCG CCG CG (3’)
(FAM – carboxyfluorescein)



E - Epitalon
(Ala-Glu-Asp-Gly)

B - Bronchogen
(Ala-Glu-Asp-Leu)

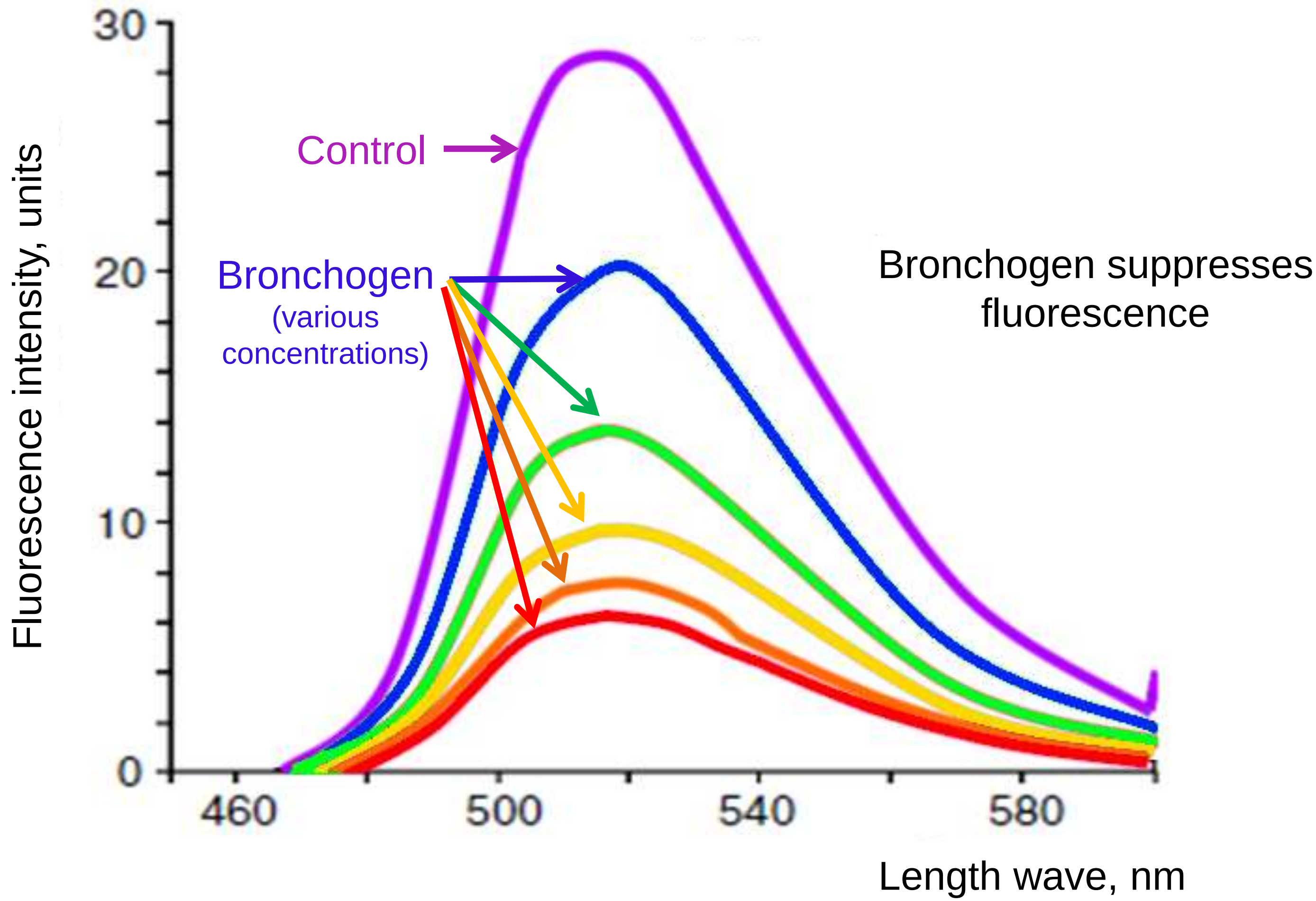
C - Cardiogen
(Lys-Glu-Asp-Arg)

P - Pancragen
(Lys-Glu-Asp-Trp)

Khavinson V. et al. Bull. Exp. Biol. Med. (2011)

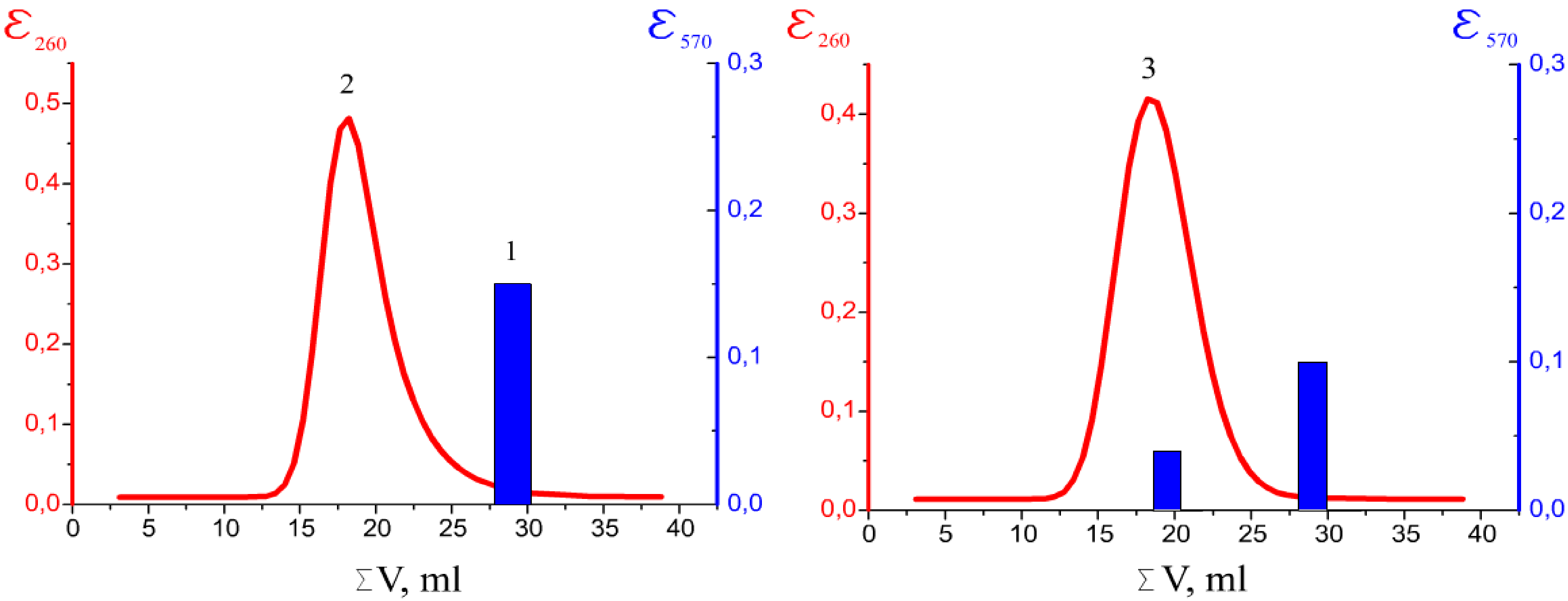
The influence of Bronchogen (ADEL) on deoxyribooligonucleotide fluorescence with CNG and CG sites which could be metilated

(5') (FAM)-*cg-<u>ccg</u>-c<u>ca-g</u>gc-gcc-gcc-gcg* *(3')*



Fedoreeva L. et al. Biochemistry (2011)

HPLS of peptide and DNA on sefandex G-25 in physiological solution at room temperature

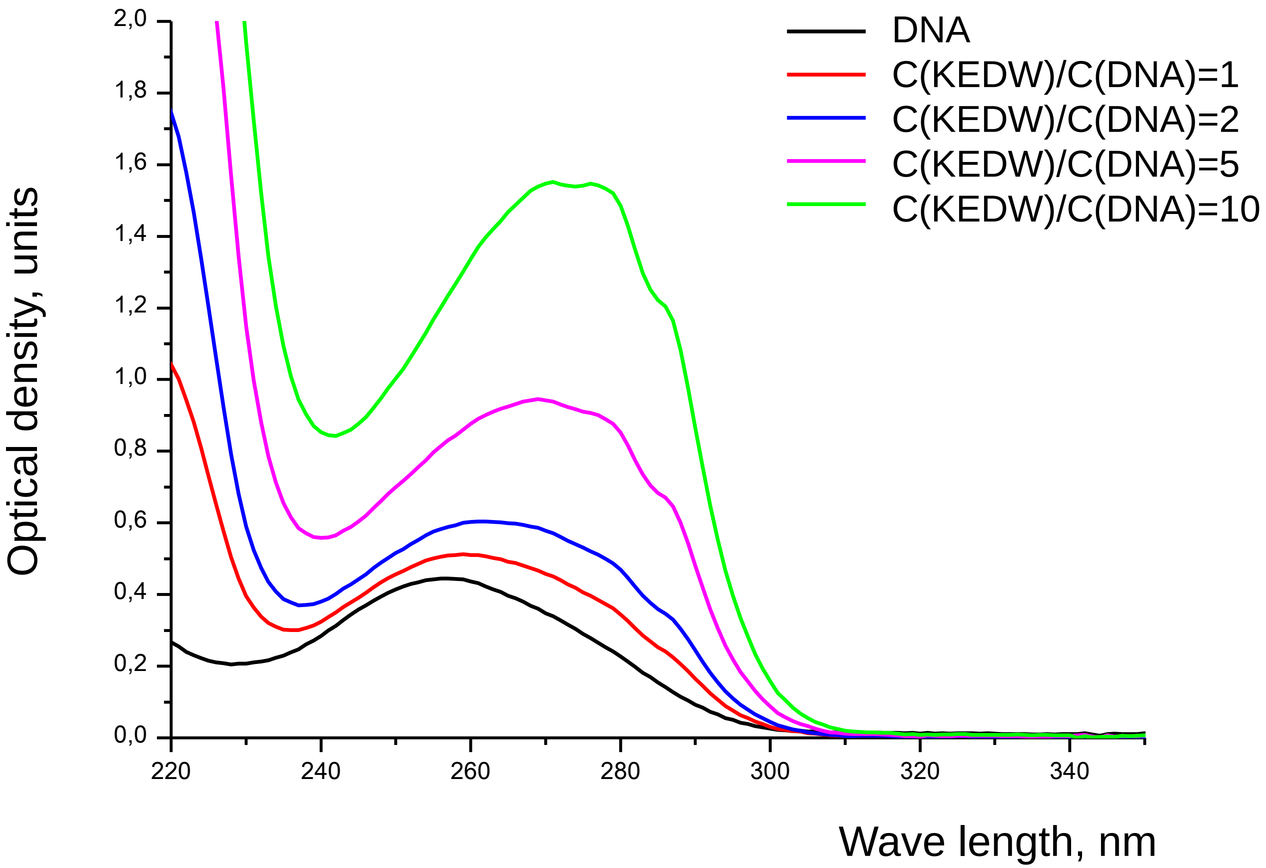


1 - individual Ala-Glu-Asp-Gly peptide
2 - free DNA double helix [poly (dA-dT) : poly (dA-dT)]

3 - mixture of peptide and DNA

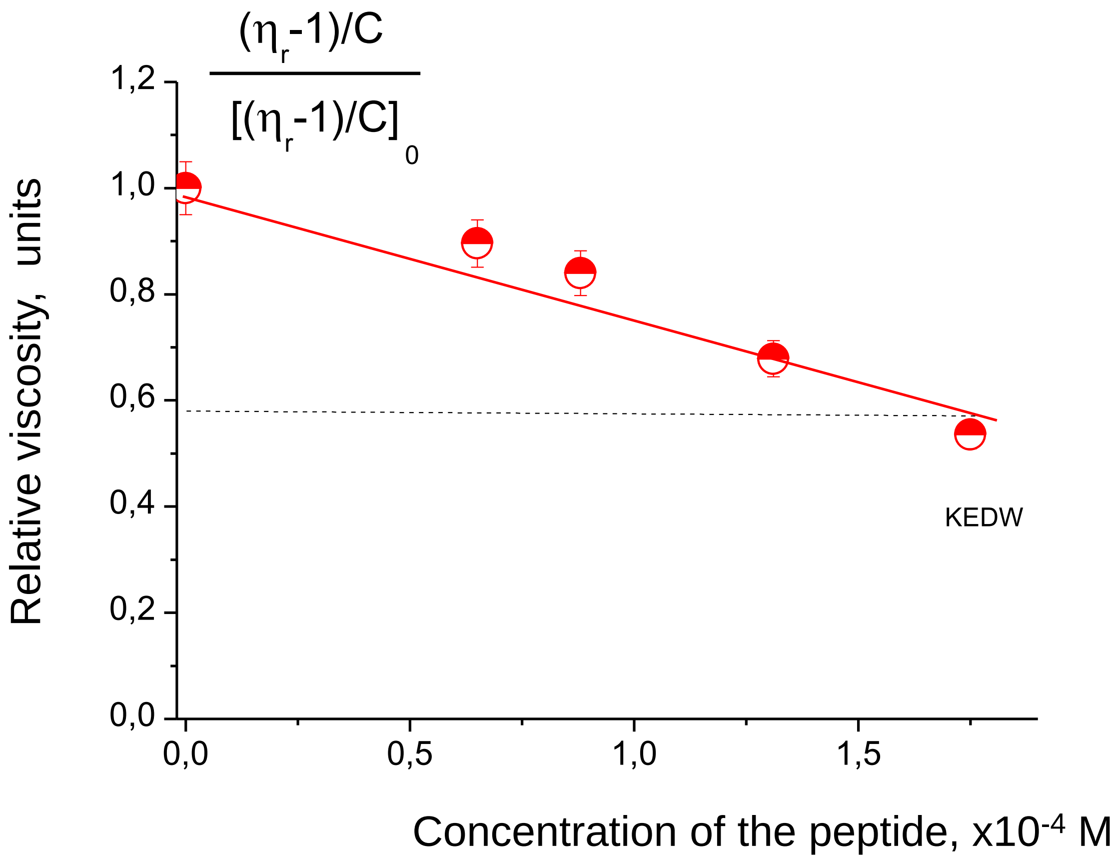
Khavinson V. et al. Bull. Exp. Biol. Med. (2006)

DNA - Pancragen (KEDW) interaction (spectrophotometric method)



The peptide influences the secondary structure of the macromolecule.
Changes in DNA spectral properties are observed in KEDW presence.

DNA - KEDW interaction (viscosimetry method)

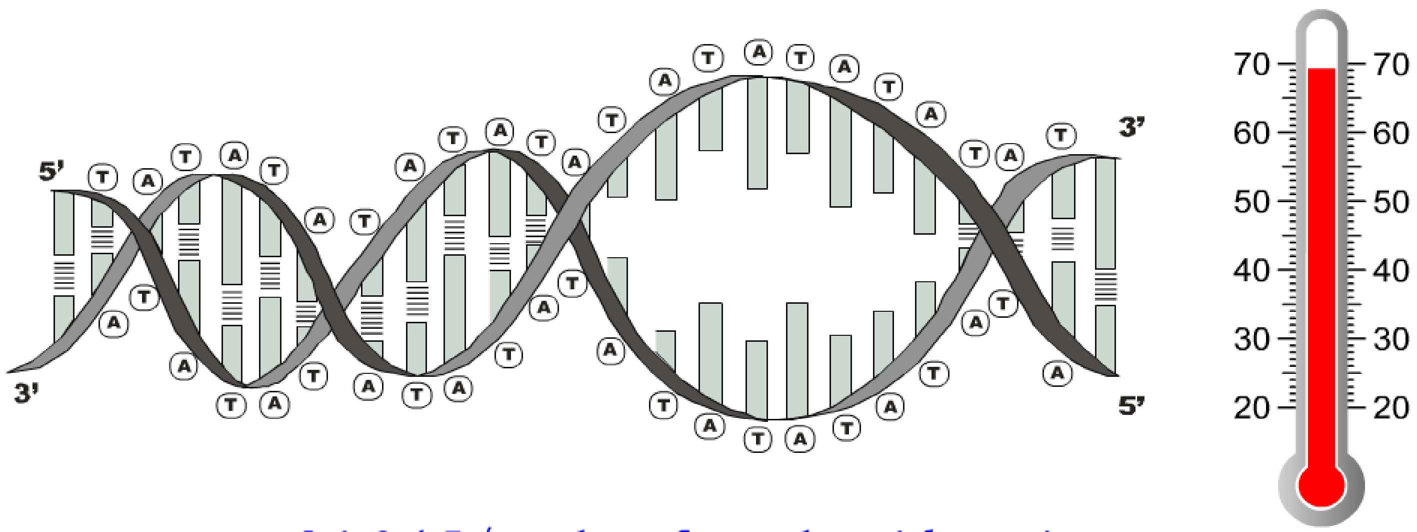


KEDW binding with DNA leads to the decreased viscosity.
Thus the peptide influences the tertiary DNA structure.

Local separation of strands [poly (dA-dT):poly(dA-dT)] as a result of DNA double helix melting

Free DNA

69.5°C



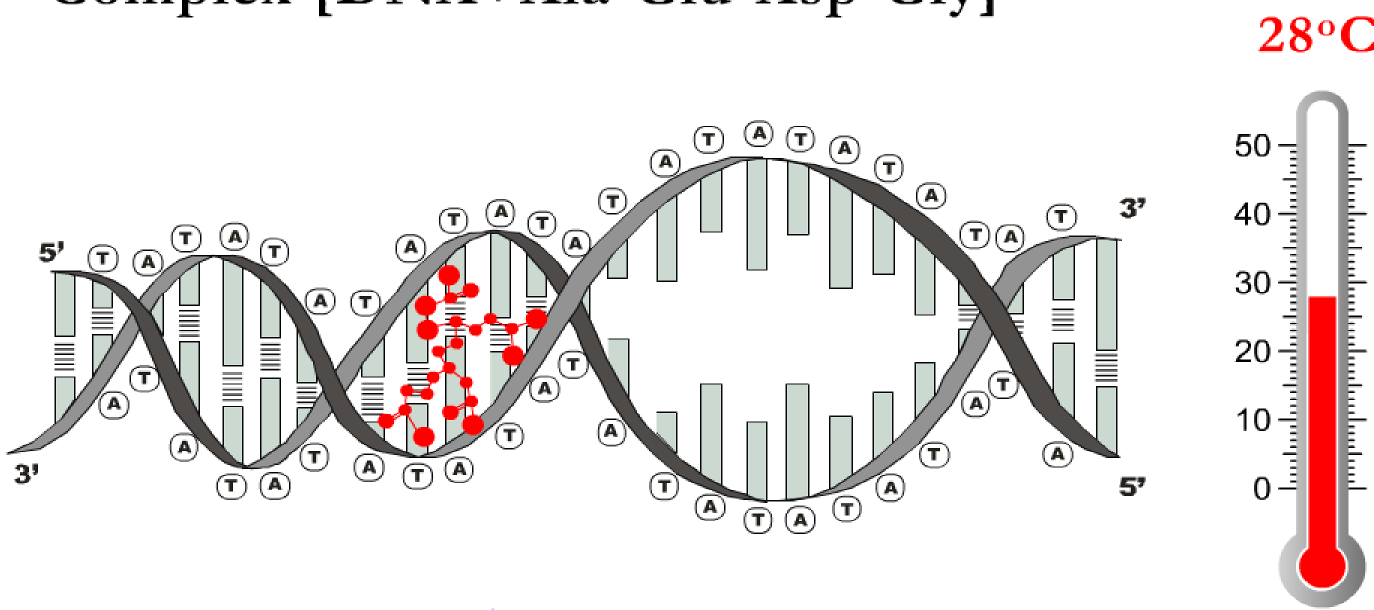
-36,0 kJ/mole of nucleotide pairs

Changes in the free energy (ΔG) necessary for the double helix strands separation

Khavinson V. et al. Bull. Exp. Biol. Med. (2008)

Local separation of strands [poly (dA-dT):poly(dA-dT)] as a result of DNA double helix melting

Complex [DNA+Ala-Glu-Asp-Gly]



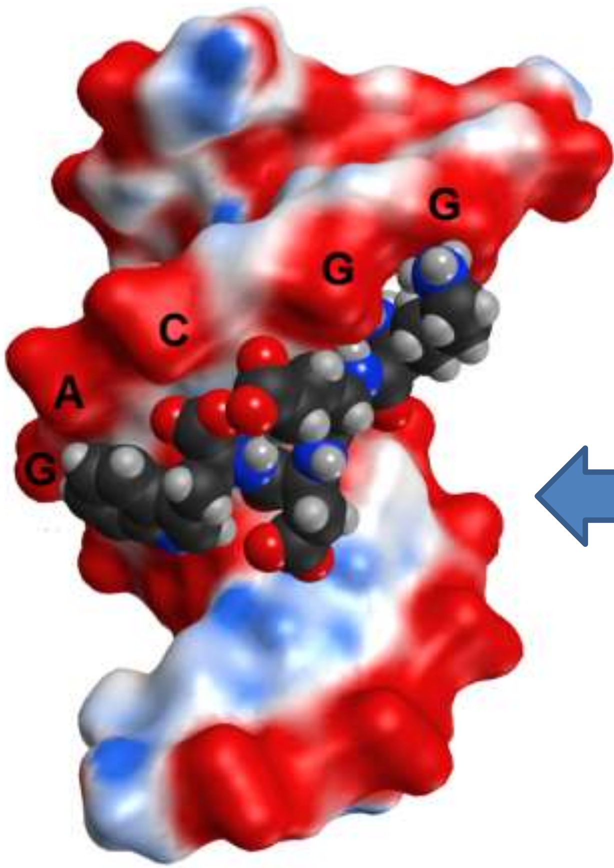
-30,9 kJ/mole of nucleotide pairs

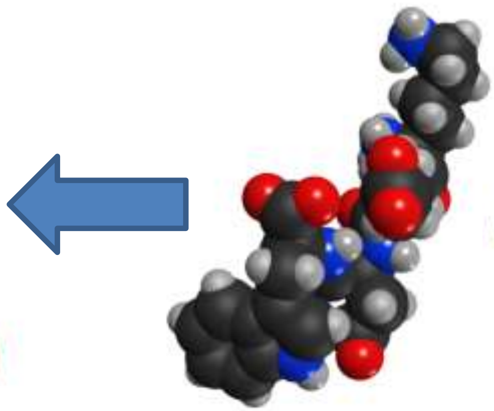
Changes in the free energy (ΔG) necessary for the double helix strands separation

Khavinson V. et al. Bull. Exp. Biol. Med. (2008)

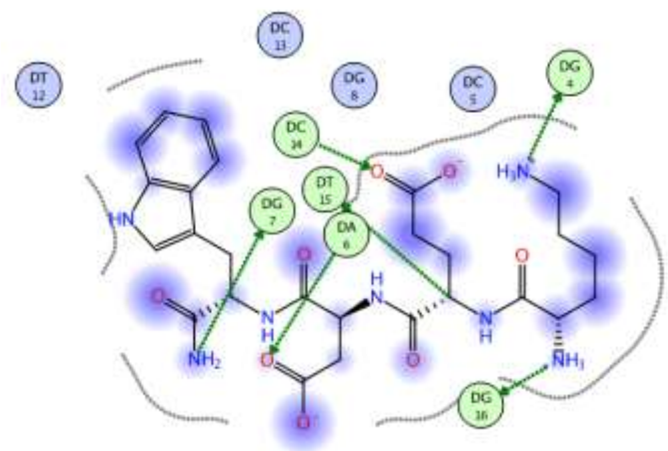
Model of DNA-peptide complex

(red)	positively charged part of DNA
(blue)	negatively charged part of DNA
(white)	neutrally charged part of DNA





Pancragen



Major groove
5'-GGCAGG-3'
3'-CCGTCC-5'

aaggcaggaa: aaggcaggaa / untitled

Ligand		Receptor			Interaction	Distance	E (kcal/mol)
N	1	O6	DG	16	H-donor	2.93	-9.6
NZ	21	N7	DG	4	H-donor	3.04	-4.8
CA	27	O4	DT	15	H-donor	3.34	-0.9
OXT	58	O6	DG	7	H-donor	3.21	-2.5
OE1	38	N4	DC	14	H-acceptor	3.01	-4.4
OD1	50	N6	DA	6	H-acceptor	2.94	-5.1
NZ	21	OP2	DG	4	ionic	3.73	-1.1

Pancragen sites of binding in gene promoter regions

Gene	Gene regulatory region, range -499 to 100 bp	GenBank №
Pdx1	ATCAAATGCTTCTGACCTAGAGAGCTGGGTCTGCAAACTTTTTTTTTATCGTATTCCGCAACAGTTAAATAAAAAATTAAAAACTCA ACATGTCTCCTTGTAAACTACATCAATTAACAAACACACTATGTCCATTATCAAATATAATAGAAAAAATATAGGAAAATAGAAAAT AGAAAAATATAGGAAAATAGAAACTTTTAAGCCACGGTGAAAATGTTTCTATAAATGAGTGGTTCTAATGTTTTCGTGAGCGCCCAT TTTGGGGAGCACCGCCAGCTGCCCGTTCAGGAGTGT**GCAG**CAAACTCAGCTGAGAGAGAAAATTGGAACAAAA**GCAG**GTGCTCGCGG GTACCTGGGCCTAGCCTCTTAGTGCGGCCAGCCAGGCCAATCACGGCCCCCGGCTGAACCACGTGGGGCCCCGCGGAGCCTATGGTG CGGCGGCCGGCCCGCCGGTCCGCGCTGGCTGTGGGTTCCCTCTGAGATCAGTGCGGAGCTGTCAAAGCGA**GCAG**GGGTGGCGCCGGG AGTGGGAACGCCACACAGTGCCAAATCCCCGGCTCCAGCTCCCGACTCCCGGCTCCCGGCTCCCGGCTCCCGGTGCCC	NM_000209.3
Pax6	GGCCCGAAGCCGCCGAGAGAGCTCGGGACAGC**GCAG**GACCAG**GCAG**CCGCTCGCTCTCCTGTCACCTTAACT**GCAG**GCTCCGAGGGG CGCCTTTGGAGTGTACTGAGGTGTGTCCTAATCGTGCGGCATTCAACAAATGGACTTCTGGTGTGTGGTCAGAAGAGAAAAGCCATT TACTTACTTTCCTCCCCGGTTTTCTGGCAACAGCTGAAGGGGAGCTGCCTCCGTGGACTGA**GCAG**ACCCAGGAGAGGGAGTCGTGGT GCGGAGACACACGCACCACACACAGATGACCGGTGGCACACACGACACACGCTGACATACCGACATCGCCAGTGGGACACACACACA CACACACACACACACACACACACACACAGAGAGAGAGAGAGAATCCCTCCCAGCATTGGTCATCCGCCCCCCCACCCAGGCTTCCAC TCCCCCTCCCCTCTTATCTCCCCTGGCTTCCCCTCCTCTCGGGCGCTGCGAAAA**GCAG**CCGCACTTAGTCAACAAATGGCACGTGGG AGAAGTTGGTGAGTGTTTGGTGAGGACTCTTCAGGGCTTTTCACAAGAACCCTCTGTACACAAAGTAAGTGGCGTGTT	NM_000280
Pax4	GCCAGCTCTCAAAGAAA**GCAG**CTTGCGTTGACAGCCTGGGG**GCAG**CAAGGAT**GCAG**TCTCCCAGGAGAGGATGCACTCGGTGGTGGG AAGCCAGGCTGGAGGGGCCTGAGTGACCCTCTCCACAGGCGG**GCAG**G**GCAG**TGGGAGAGGTGGTGTGTGGATACCTCTGTCTCACGC CCAGGGATCA**GCAG**CATGAACCAGCTTGGGGGGCTCTTTGTGAATGGCCGGCCCCTGCCTCTGGATACCCG**GCAG**CAGATTGTGCGG CTA**GCAG**TCAGTGGAATGCGGCCCTGTGACATCTCACGGATCCTTAAGGTAATGGGCCAGCACCTTTACCCAGTGATGGGGACAGGA A**GCAG**GGAGAAAGGGCTCCTCTGAAGGCAAGAGCCTGGGGCTGTT**GCAG**GCTCTGAGGGCTTCTGGGACTTGGGTCACTTCCTGGGA GATCCTCTCGGAGGTTGAAAAGGGGAGCCTCAGGCCCTCAAAGGTGAGGCTGGACTCCCGACTTCATGGCCTGGTCCAGTAAGTCTT GGCTTTGTCTTATAGCCTCCTCCTGTCCCAGGGACACTCTCCTTCCTTCTGCCCATCATGCCTCACCTGTCCCTGCTT	NM_006193
Nkx2.2	TCCCCTCCTCCTCCCCACCCCCACCTTTTTTAAGATGCAATTTGTTAAAACGGCCCTTTCAAGTGTGTGGACTCGCGAGCGACGCGG TGGCCCTTTGTATGTAAATACTGGGTTTAAAAAAAAAAAGGCTCGCCCCGTCTTTGCAATTAATTGACACGTTACACCTCTCATCTT GCTCTAGAGGGCCGTTGGCTGGGAGCGCGGAGCTCCCCAAAACCCACAATTTCACATCTGCAAATACTGTCTTCATCCACTTGACTC CCAAGACCCGCCCACACGTGGCCAACCTTTGCGTTTTTAATGTCTCTTCCCCCTTTTTTCCACCCTTCTCCCGCTCCCTCTCTCGCT CCCCCTCCCTCCCTCTTTCTTTCCCTCCCTCCCTCTCTTTCTCCCCCTCTCCCCGCCTCCCCAGGTTCGTGAGTGGAGCCCAGCCTT ATATGGACTGATCGCTCGGGCAATGGCCCATTTTTTCCTCGCCACCAGCCGCCACCGCGCGCCGAGCGGCCGCCGGAGCCCGAGCTG ACGCCGCCTTGGCACCCCTCCTGGAGTTAGAAACTAAGGCCGGGGCCCGCGGCGCTCGGCGC**GCAG**GCCGCCCGGCTT	NM_002509.3
Foxa2	CGAAGCTCCGTGTCTGCCATCTCGCCTGTCTTCTGCCACCATCGCCCCCAATTTTGGACAGGTGGGCTGGATGCCCACTAGTTCCTA TGCATTCTCTGTGTCTGAGGGGGTGGGTACAGGGCTGGATCCCCAAGGTCCAGCCAGGTTTTCAGAACCAAGAAAGAGCCTCCACAT CCAAACACCTGCAATATCCCCCCACTCCAAATCTGGGCTCACAGGCTAACCCAGAACAGAAGACAATTTTTGAACCCAAGAGCTGCT GGGGAAATAAAAGTATACGATTGCTGGAGTTTCTAATTTCTATTAA**GCAG**TCCCTCTGGAAGACAGAGAGGACAGAGACGCTCTTGA AGTCAACTCCATATGCCCCATCATTGATTCCTGGATTCTTCTCTCCTCACCCCTCCCTCCCCACCTCCTGCCCTGTTTGTTTTAGTT ACGAAATGCTGTGGGCACCTCGGTTGTGACTGAAAAGTAACCTTGAAACACGCCGGCCTGAATATCAGAGACAAATCTCAGCCTCCC AACCGTCGGCCGCTGCTAGAGGGGCTGCTTGCGCCAGGCGCCGGCCGCCCCACTGCGGGTCCCTGGCGGCCGGTGTCT	NM_153675.2

Red colour indicates Pancragen sites of binding

Mechanism of peptide regulation of the living matter development

Complementary interaction between small peptides and DNA

↓

Chromatin decondensation

↓

Changes in gene conformation and expression

↓

Synthesis of tissue-specific proteins

↓

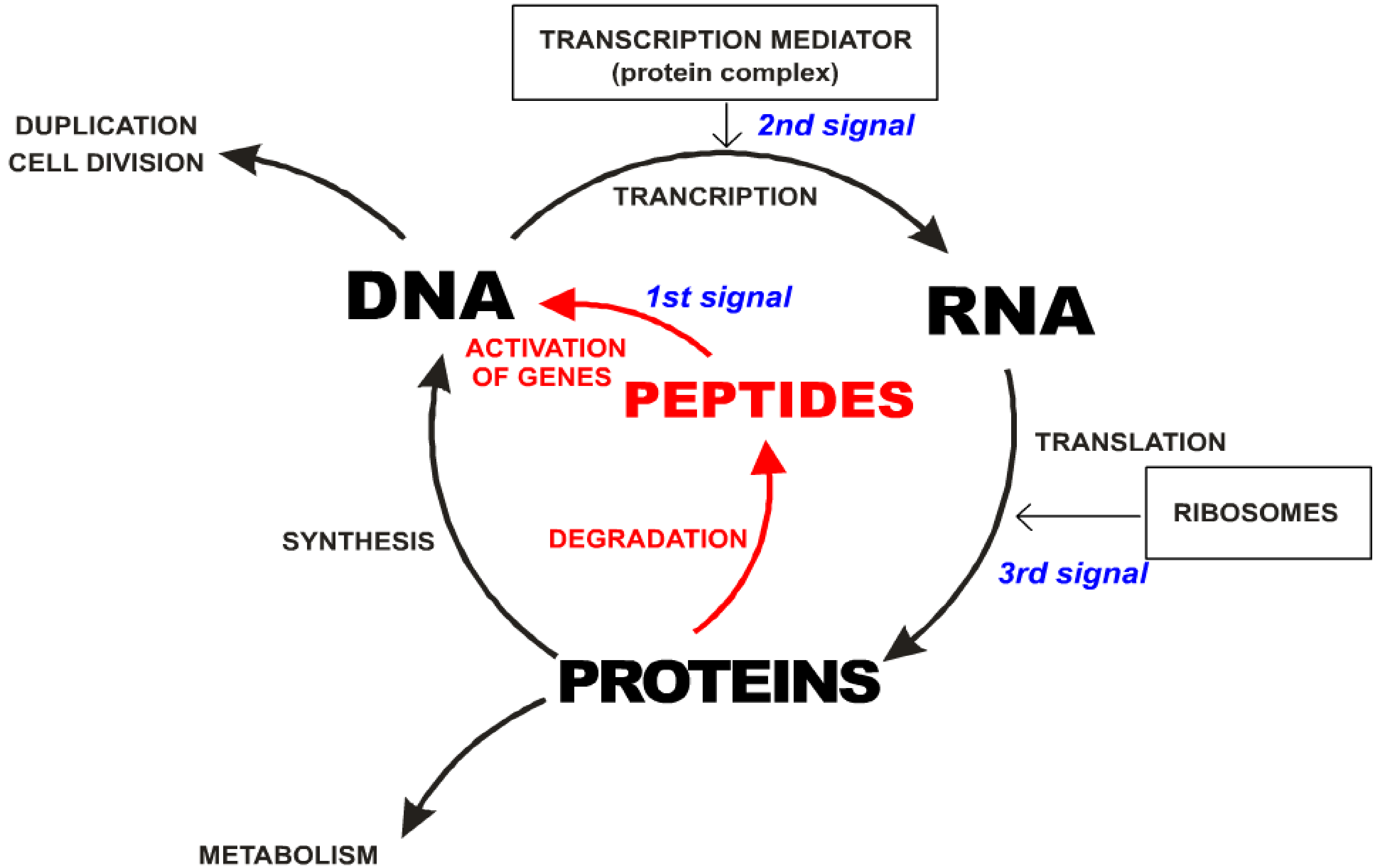
Cellular proliferation and differentiation

↓

Regulation of biochemical and physiological processes

Khavinson V. Peptidergic regulation of ageing (2009)

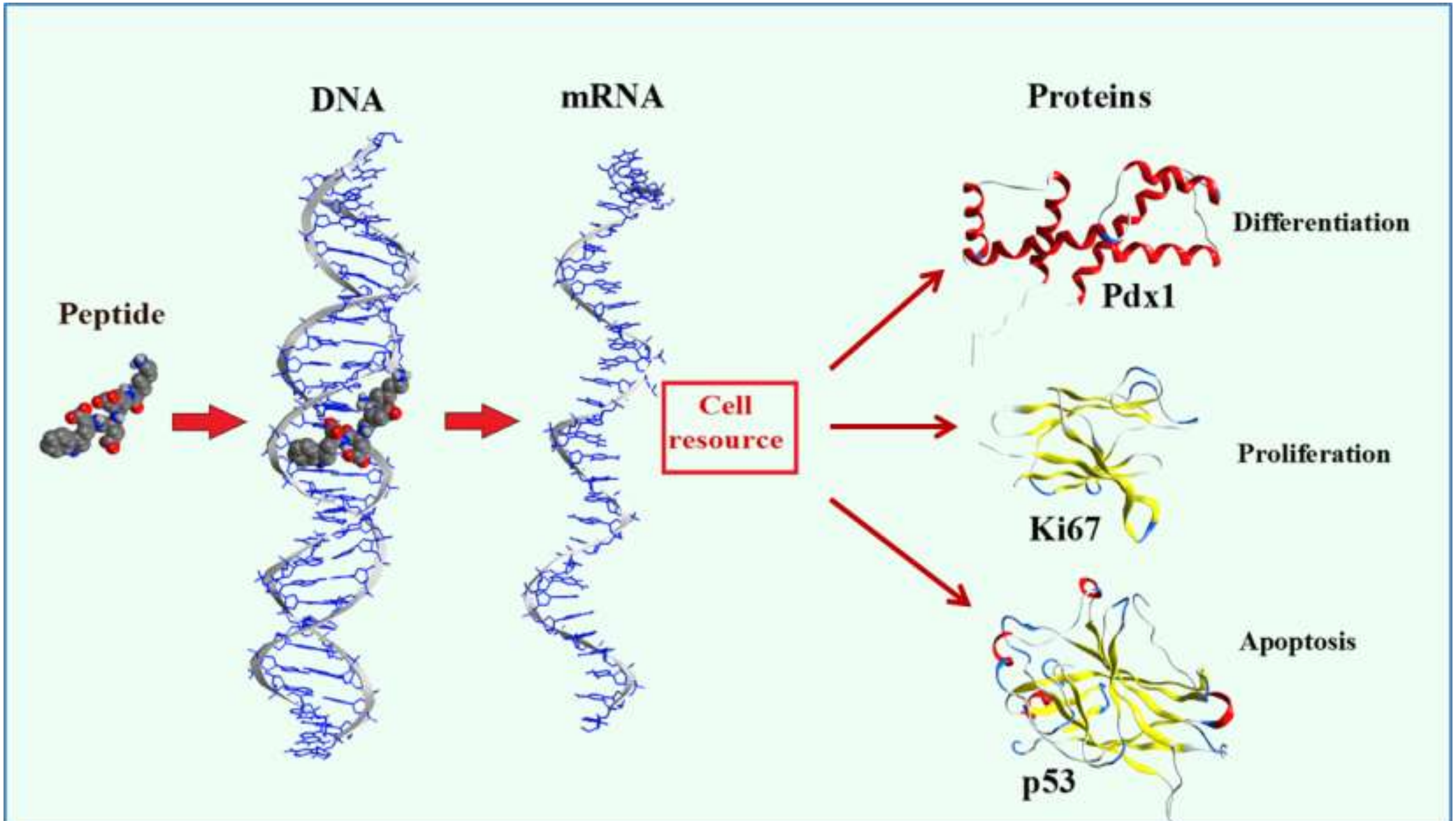
The role of peptides in the cycle of DNA, RNA and protein biosynthesis



Khavinson V. Peptidergic regulation of ageing (2009)

Peptide regulation of protein synthesis

(proposed mechanism)



Peptide activates selective gene transcription during its binding with DNA. This can lead to mRNA formation and the synthesis of apoptotic, proliferative and differentiation proteins. This increases cell resource and slows down cellular senescence.

Khavinson V. et al. Biology Bulletin Reviews (2013)

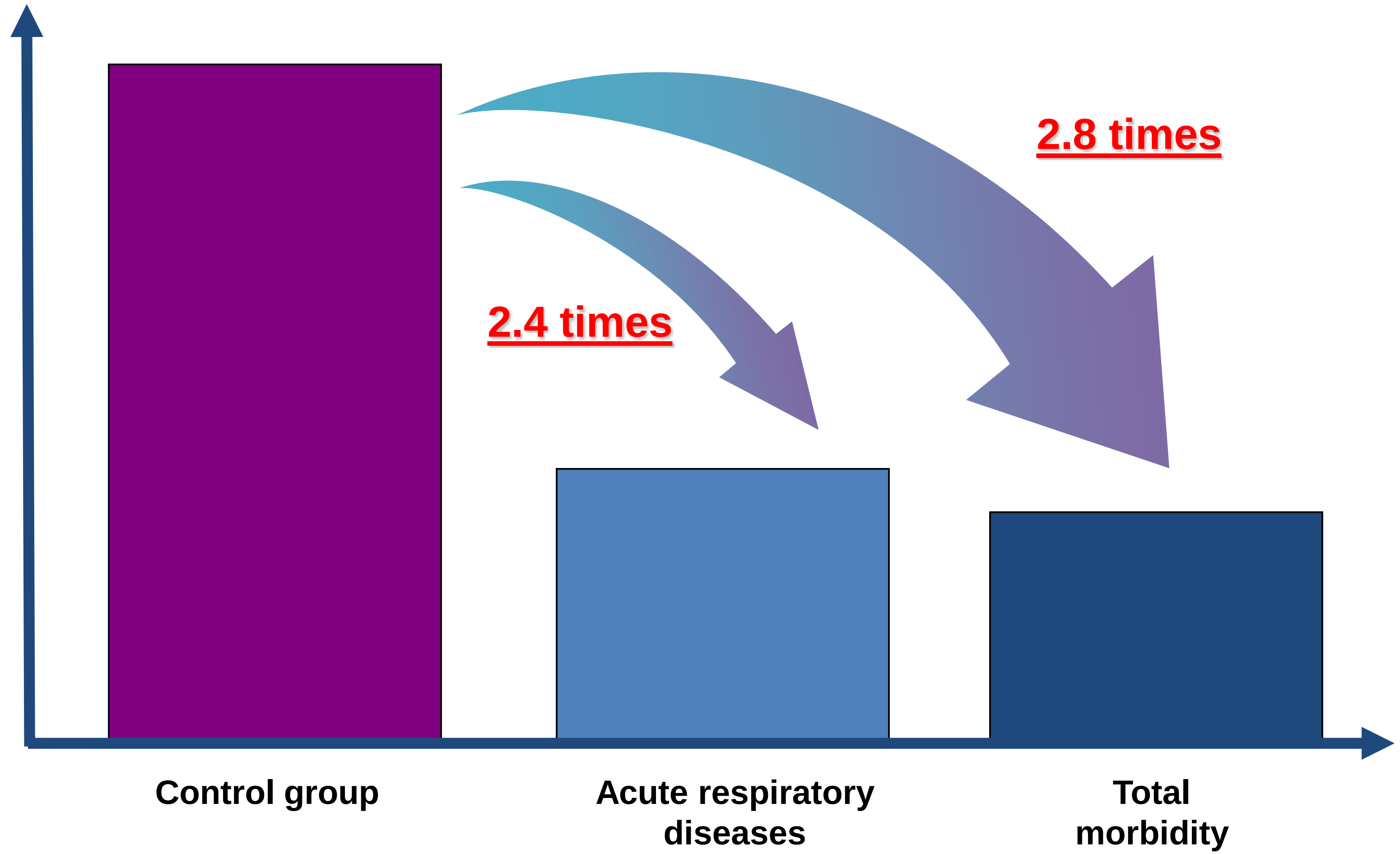
3. Peptide bioregulators: clinical studies

The influence of peptide bioregulators on morbidity of the employees (40-60 years) “Avtovaz” (Tolyatti) when exposed to harmful factors.

<u>Main group</u> (**450** employees) received injections during 30 days (10 injections sequentially), to improve the functions of brain - **Cortexin**, and to immune system - **Thymalin**, normalize endocrine system – **Epithalamin**.

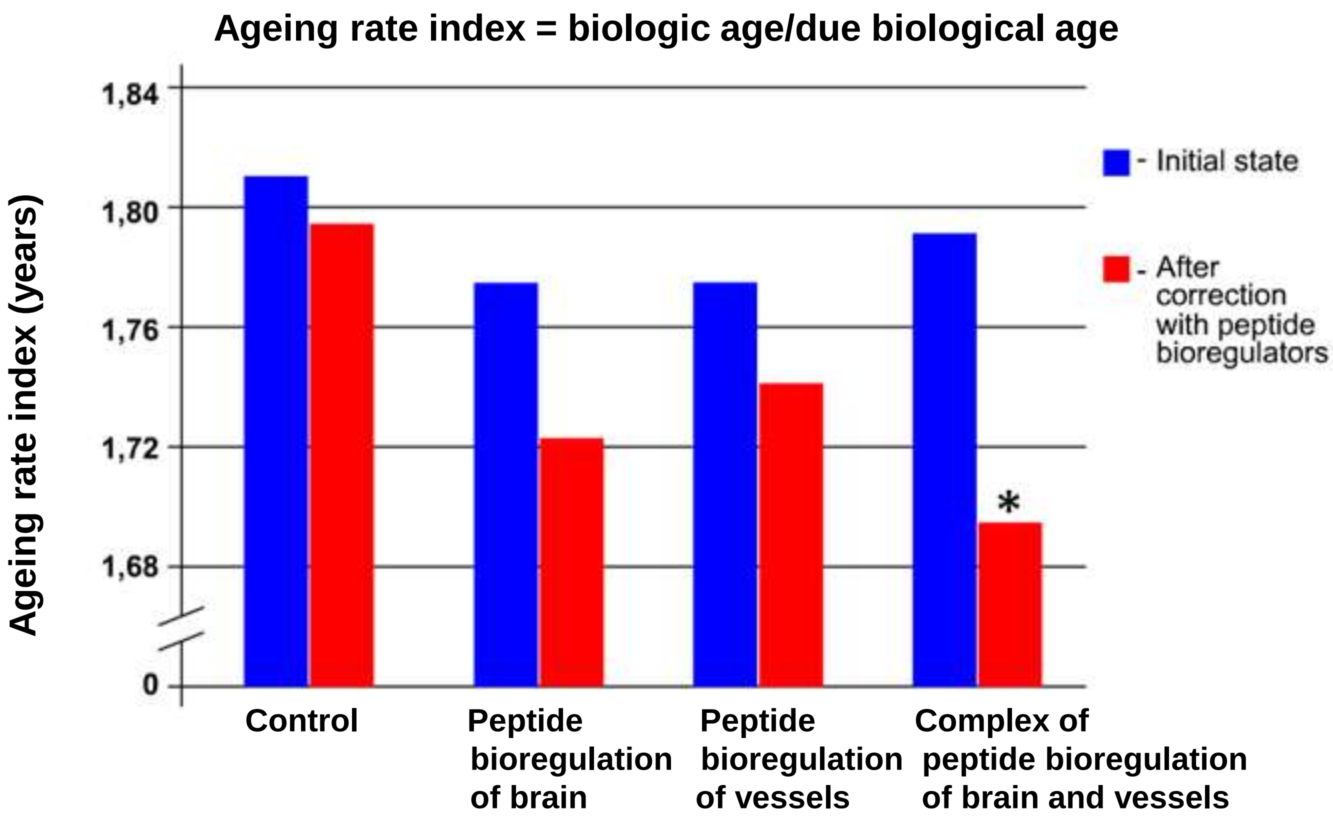
<u>Control group</u> (**400** employees) received injections during 30 days (10 injections sequentially) of vitamins B 12, C, D.

Changes in morbidity levels



The observation period - 1 year

The influence of peptide bioregulators on ageing rate of the employees under the influence of adverse factors



* - $p<0.05$ as compared to the control

Administration of the bioregulators - 300 people, Control (multivitamins) - 200 people

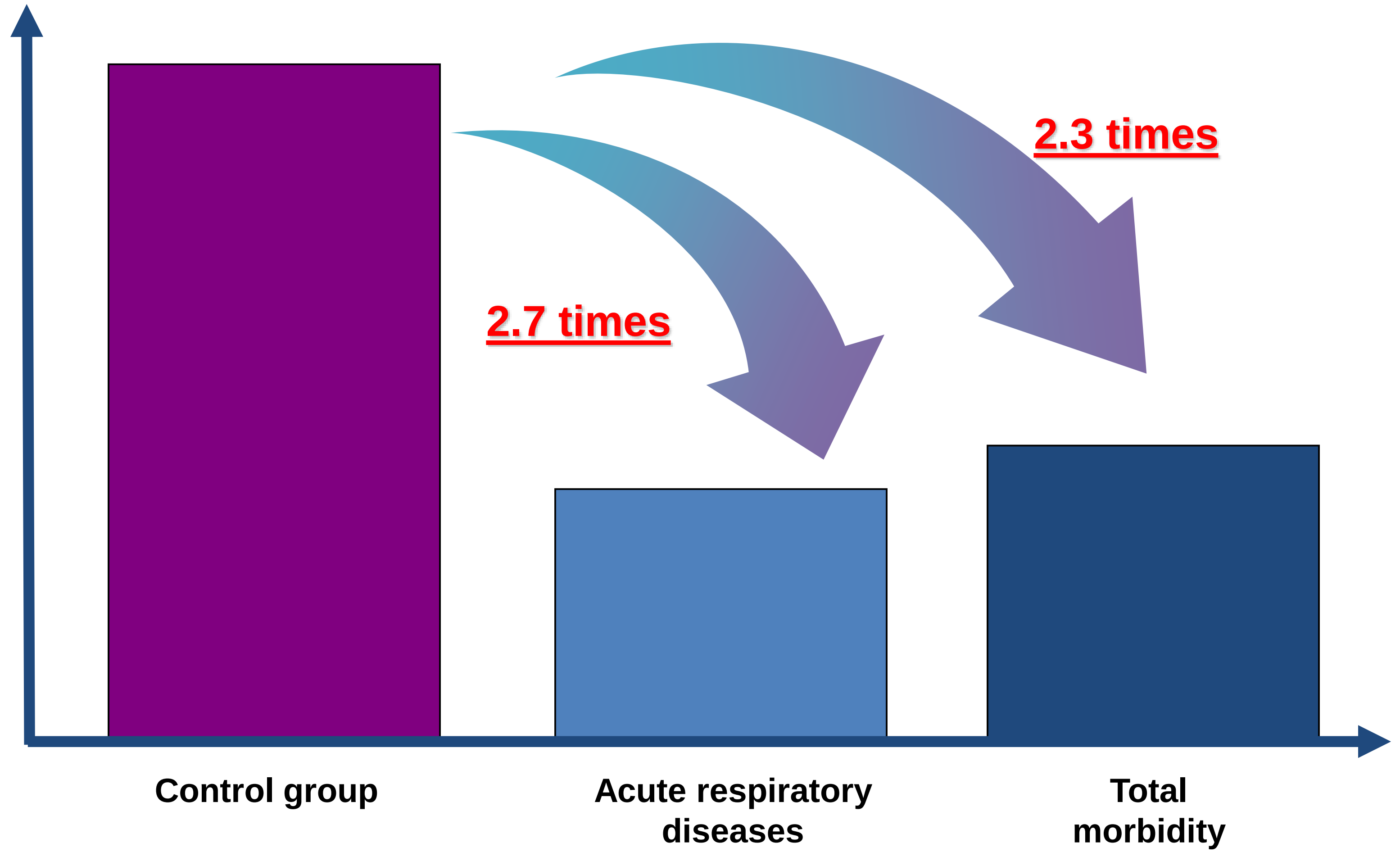
The observation period - 1 year

The influence of peptide bioregulators on morbidity of employees of «Gazprom» under the influence of adverse factors

<u>Main group</u> (**11 192** employees) received a complex of 6 peptide bioregulators to improve the functional state of immune system, brain, blood vessels, bronchi, liver, cartilage tissue (in capsules for oral administration).

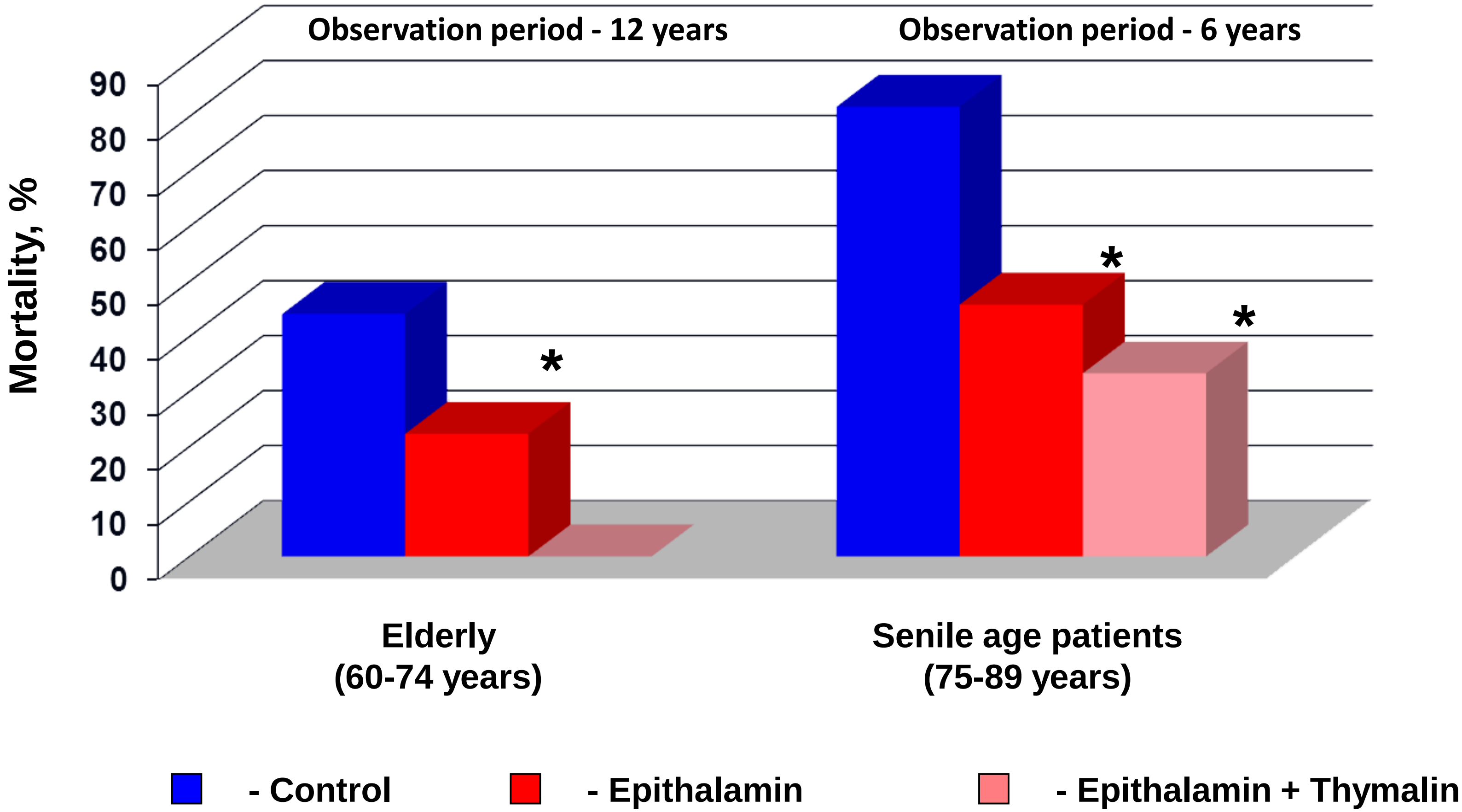
<u>Control group</u> (**3 000** employees) received multivitamins for 30 days (for oral administration).

Changes in morbidity levels



The observation period - 1 year

The influence of peptide bioregulators on mortality in elderly and senile age patients



* - $p<0.05$ as compared to the control

Khavinson V., Morozov V. Neuroendocrinology Lett. (2003)

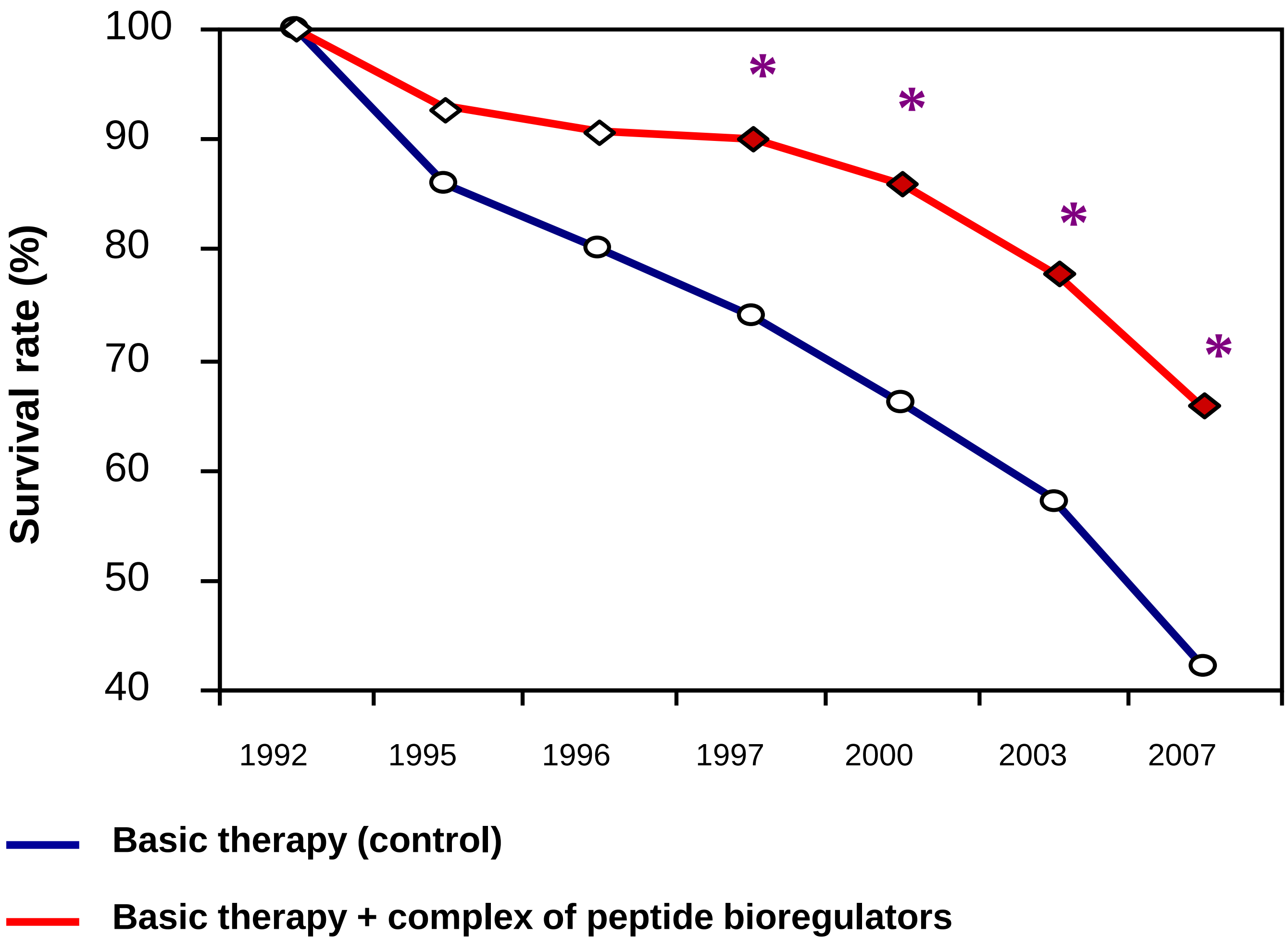
The influence of Epithalamin on survival of elderly patients

Indices	**Control group** (Basic treatment)	**Main group** (Basic treatment + **Epithalamin**)
Number of patients	40	39
Age of patients before the study	65.1 ± 1.1	64.5 ± 0.9
Survival rate at 15 years	16 (40%)	**26 (66.7%)***
The cause of death of the patients (myocardial infarction or stroke (%)	83.3	**46.2***

* - $p<0.05$ as compared to the control

Korkushko O. et.al. (2011)

The influence of Epithalamin on survival of elderly patients



* - $p<0.05$ as compared to the control

Korkushko O. et.al. (2011)

Epithalamin increases melatonin level in elderly people

		Melatonin level in blood at 3.00 a.m. (pg/ml)
Healthy people (20-40 y.o.)		63.4±9.2
Patients (60-74 y.o.), treated with polyvitamins	Initial level	32.4±6.8 *
	After correction	28.1±5.9 *
Patients (60-74 y.o.), treated with preparation of the pineal gland	Initial level	24.2±5.1 *
	After correction	59.0±12.6

* - $p<0.05$ as compared to the healthy people

Khavinson V. Peptidergic regulation of ageing (2009)

The influence of Epithalamin on telomere length

Age of patients, years	Normal limits of telomere length (b.p.)	Administration of Epithalamin	
		Initial value	After administration
5-10	13.88-15.89	-	-
25-30	11.78-13.78	-	-
45-50	9.67-11.68	**10.53±0.97**	**11.97±1.32***
60-65	8.09-10.10	**9.32±0.82**	**10.83±1.12***
75-80	6.51-8.52	-	-
90-95	4.93-6.94	-	-

* - $p<0.05$ as compared to the initial values

Tsuji A. et al. Forensic Science International. (2002)
Bekaert S. et al. Anticancer Research. (2005)

Pineal Gland Preparations

Epithalamin (Epinorm) – peptide complex with molecular weight 1000-5000 Da extracted from cattle pineal gland. The preparation is produced in flacons (ampoules) by 10 mg for intramuscular administration.
Course of treatment takes 5-10 days (1 injection daily).

Epitalon – Ala-Glu-Asp-Gly tetrapeptide. The preparation is produced in ampoules by 100 µg for intramuscular administration. Course of treatment takes 10 days (1 injection daily).

Endoluten – peptide complex with molecular weight 1000-5000 Da extracted from cattle pineal gland. The preparation is produced in capsules by 10 mg for oral administration.
Course of treatment takes 10-20 days (by 2 capsules daily).

The influence of Pineal Gland Preparations on telomere length in patients' blood cells

Age, years (norm)	Investigation	Telomere length (b.p.)		
		Peptide Preparations		
		Epithalamin	Epitalon	Endoluten
60-65 (8.09-10.10)	Initial value	9.32 ± 0.82 (n=25)	9.61 ± 0.93 (n=19)	9.43 ± 1.12 (n=21)
	After treatment	**10.83 ± 1.12** *	**10.72 ± 1.21** *	**10.62 ± 1.32** *
75-80 (6.51-8.52)	Initial value	7.33 ± 0.81 (n=21)	7.51 ± 0.91 (n=17)	7.63 ± 0.98 (n=18)
	After treatment	**8.73 ± 0.78** *	**8.91 ± 1.11** *	**8.66 ± 1.21** *

* - $p<0.05$ as compared to the initial indices

The influence Pineal Gland Preparations on the melatonin metabolite 6-OHMS excretion

Age, years	Investigation	6-OHMS excretion (ng/h)		
		Peptide Preparation		
		Epithalamin	Epitalon	Endoluten
60-65	Initial value	410 ± 38 (n=21)	445 ± 43 (n=19)	428 ± 47 (n=17)
	After treatment	**933 ± 86** *	**915 ± 97** *	**820 ± 92** *
75-80	Initial value	363 ± 41 (n=18)	371 ± 35 (n=22)	348 ± 43 (n=21)
	After treatment	**690 ± 63** *	**615 ± 71** *	**580 ± 62** *

* - $p<0.05$ as compared to the initial indices

normal limits for people aged 30-39– 1020-1900 ng/h

Enhancement of life resource in the elderly people after application of peptides

INDICES	CHANGES (after treatment with peptides)	Intensity of the changes (*)
Physical performance	Enhancement	1.8 – 1.9-fold
Fatigability in case of physical activity	Decrease	2.3 – 2.5-fold
Short memory	Improvement	by 56%
ARD and flu frequency	Decrease	2.4-fold
T-lymphocytes function	Enhancement	by 24-43%
Total antioxidant activity	Enhancement	by 53%
Melatonin in the blood	Enhancement	2.4-fold
Telomeres length	Enhancement	14-16%
Bone tissue density	Enhancement	In 73-83% of the patients
Survival rate 15years of observation	Enhancement	by 67%

* - $p<0.05$ as compared to the control

Korkushko O. et.al. (2002, 2006)

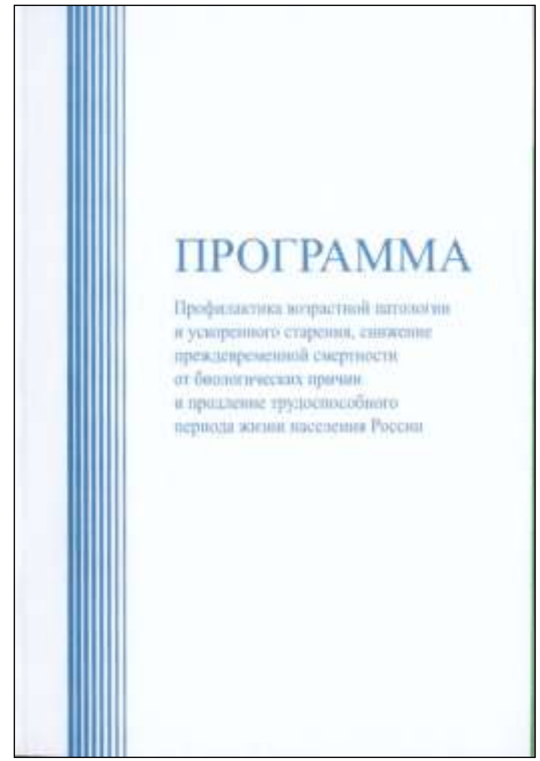
Enhancement of human vital resource

The application of peptide bioregulators contributed to significant **decrease in ageing rate in patients** (aged 40-55 years) exposed to harmful factors and **increase in survival rate** of elderly patients (observation period - 15 years), which is evidenced by:

1. Improved physical capacity
2. Reduced ageing rate of cardiovascular system
3. Normalized metabolism
4. Improved brain function
5. Increased resistance to viral diseases

Programme

**«Prevention of
age-related pathology and
expanding healthy working life»**



**Application of peptide bioregulators to
restore the functions of the body**

Expected results of the programme

Medical significance

- Increase in working capacity
- Reducing the rate of ageing
- Normalization of metabolism
- Brain function improvment
- Reduction in general morbidity
- Reduction of infectious diseases during epidemics

Social significance

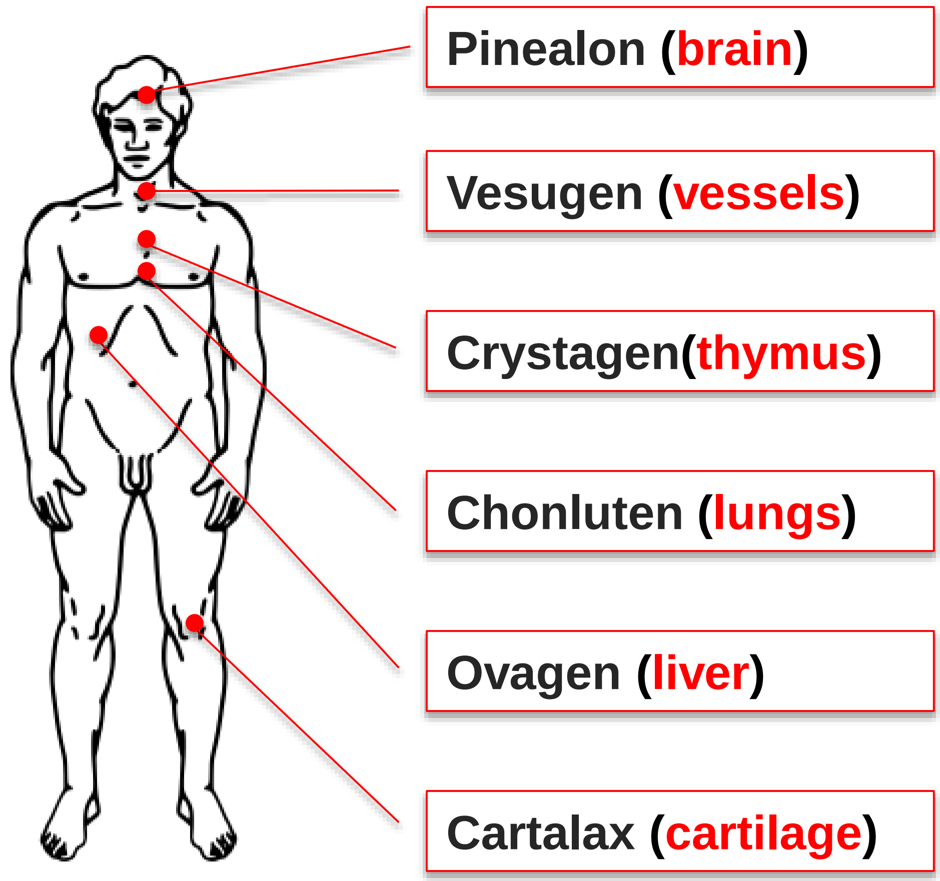
- Reducing the incidence of work-related diseases
- Slowdown the accelerated ageing of population
- Improvement of quality of life and extending working life
- Improvement of economic efficiency of workforce

Complex of the main peptides to enhance the resource and prevent age-related disorders

The scheme of treatment:

1. **Pinealon, Vesugen – 10 days**
2. **Crystagen, Chonluten – 10 days**
3. **Ovagen, Cartalax – 10 days**

Total 30 days



Health assessment is conducted before the application and repeated in 4 months.

This is followed by mathematical processing, statistics and recommendations. Recommended 2 courses each year.

Enhancement of vital resource of Russian Olympic team in rhythmic gymnastics

left to right: **A. Shumilova (coach), D. Kondakova, A. Zaripova (coach), J. Lukonina, Prof. V.Khavinson, E.Kanaeva, V. Schtelbaums (coach), I. Viner-Usmanova (main coach of team, honored coach of Russia), O. Buyanova (coach), D. Dmitrieva**

Peptide application areas

- **Preparations**
  - Natural preparations
  - Synthetic preparations
- **Biologically active supplements**
  - Natural
  - Synthetic
- **Cosmetology**
  - Preventive cosmetology
  - Medical cosmetology
- **Veterinary**
  - Animal husbandry
  - Aviculture
  - Domestic animals
- **Nutrition**
  - Functional foods and beverages
  - Sportsmen products

Conclusion

Theoretical, experimental and clinical investigations have shown the role of signal small peptides in epigenetic regulation of gene expression, protein synthesis, life resource and life span increase.

Researchers of the St. Petersburg Institute of Bioregulation and Gerontology

Left to right - Professors:
Ariev A., Baranovsky A., Anisimov V., Khavinson V., Kozina L., Chalisova N., Kvetnaia T., Trofimova S., Kheifits V., Morozov V., Baluzek M., Malinin V., Shataeva L., Kozlov K., Kvetnoy I., Ryzhak G.

Prof. Khavinson and the team of the Laboratory of Biogerontology of the St. Petersburg Institute of Bioregulation and Gerontology

Left to right Khalimov R., Prof. Khavinson V., Prof. Kvetnaia T., Basharina V.;
Dr. Tarnovskaya S., Plotnikova E., Dr. Linkova N.

The institutions where the main studies were conducted

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St. Petersburg Institute of Bioregulation and Gerontology
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Institute of Gerontology of National Academy of Medical Science of Ukraine
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Prof. O. Korkushko

The institutions where some of the studies were conducted

National Institute on Ageing (Baltimore, the USA)

Italian National Research Center on Ageing (Ancona, Italy)

Institute of Anatomy, Ludwig-Maximilians-University of Munich
(Munich, Germany)

Prince Felipe Research Center (Valencia, Spain)

University of Antwerp, Department of Biomedical Sciences
(Antwerp, Belgium)